Linear Algebra Notes

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These notes are based on two courses: *Introduction to Linear Algebra I (Math4018)* & *Introduction to Linear Algebra II (Math4022)* taught by Prof. Kowk-Wing Tsoi at the Department of Mathematics, NTU. The structure follows the material prepared by Prof. Tsoi, with some adjustments and simplifications to aid exercises and comprehension. The details and complete arguments are documented in my handwritten notes.

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Contents

1 Fields and Vector Spaces

Fields generalize the concepts of scalars. A set F is called a field if we can define addition and multiplication operations to satisfy 11 axioms.

Exercise 1.1 .
Write down the 11 axioms (5+5+1). These rules ensure we can add, subtract, multiply and divide by non-zero element in the field.

Example 1.1 .
Some non-examples of fields:

1. *$\mathbb{Z}$ is not a field (division).*
2. *$M_n(\mathbb{R})$ is not a field (commutativity under multiplication).*

A set V is called a vector space over a field F if the space endowed with two operations satisfies 10 axioms.

Exercise 1.2 .
Write down the 10 axioms (5+5). These rules help us determine whether a given space is a vector space.

Example 1.2 .
Let F be a field ($\mathbb{R}, \mathbb{C}, \mathbb{Q}, F_2$). $F^n = \{(x_1, x_2, \cdots x_n)\}$ is a vector space over F. In general, if two fields $F_1 \subseteq F_2$, then F_2 is a vector space over F_1.

Exercise 1.3 .
Prove for each $\mathbf{v} \in V$, its additive inverse $-\mathbf{v}$ is unique.

> *Proof.*
> Suppose $\exists\, \mathbf{v}' \neq \mathbf{v}''$ be two additive inverse of $\mathbf{v}$. Use $\mathbf{0}$ to relate $\mathbf{v}' = \mathbf{v}''$. ■

Use the axioms to derive the related property:

Property 1.1 .
Let V be a vector space over F. Let $\mathbf{v} \in V$ and $a \in F$.

1. $0 \cdot \mathbf{v} = \mathbf{0}$
2. $a \cdot \mathbf{0} = \mathbf{0}$
3. $(-a)\mathbf{v} = -(a\mathbf{v})$
4. $a(-\mathbf{v}) = -(a\mathbf{v})$

Proof.

The first two: let LHS be a vector $\mathbf{w}$, and use $\mathbf{w} + \mathbf{w} = \mathbf{w}$ to show $\mathbf{w} = \mathbf{0}$. The last two: use the uniqueness of additive inverse and the result of 1, 2. ■

Definition 1.1 .

Let V be a vector space over F. $W \subseteq V$ is a subspace of V if W is also a vector space over F with addition and scalar multiplication inherited from V.

Exercise 1.4 .

Write down the sufficient and necessarily condition of a subspace.

Theorem 1.1 (Subspace of $\mathbb{R}^n$).

Let A be a m by n matrix. Then the solution space of a homogeneous linear system of equations:

$$W = \{\mathbf{v} \in \mathbb{R}^n : A\mathbf{v} = \mathbf{0}\}$$

forms a subspace of $\mathbb{R}^n$.

Proof.

We can prove W is a subspace directly using the above theorem. Alternatively, consider a linear transformation $f : \mathbb{R}^n \to \mathbb{R}^m$, the matrix A is exactly the matrix representing f. By construction, W is the kernal of f. $ker(f)$ is a subspace of domain $\mathbb{R}^n$ (We will prove this in the future). ■

Theorem 1.2 .

Fix a positive integer n. The subset of trace-free matrices:

$$W = \{A \in M_n(F) : tr(A) = 0\}$$

forms a subspace of $M_n(F)$.

Proof.

This proof is similar to the previous one and left as an exercise. ■

Example 1.3 .

Differentiable function space is a subspace of continuous function space. This is an easy exercise.

Theorem 1.3 .

Intersection of subspaces is also a subspace.

Proof.
Left as an exercise. ∎

Example 1.4 .
Consider a vector space $M_n(F)$, W_1 be a symmetric matrices, and W_2 be a upper-triangular matrices. The intersection $W_1 \cap W_2$ is a diagonal matrices, which is a subspace of the $n-$square matrices vector space.

Lastly, we introduce some concepts earlier.

Property 1.2 .
Reminder:

1. *Union of two subspaces is not necessarily a subspace.*
2. *Intersection of two subspaces is a subspace.*
3. *The sum of two subspaces is the smallest subspace containing both.*
4. *Spanning set is a subspace.*

2 Basis and Dimension

Let V be a (finite-dimension) vector space over a field F. Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ be a finite subset of V.

Definition 2.1 (Linear Combination).
Any vector of the form

$$a_1\mathbf{v_1} + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n \ \ where\ a_i \in F$$

is called a linear combination of S.

Definition 2.2 (Linear Independence).

$$a_1\mathbf{v_1} + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n = \mathbf{0} \ \ implies\ \ a_i = 0,\ i = 1, 2, \ldots, n$$

The only way to express the zero vector as a linear combination of S is to take all scalars to zeros.

S is linear dependent if it is not linearly independent.

Theorem 2.1 .
S is linearly dependent iff either $\mathbf{v}_1 = 0$ or for some r, $\mathbf{v}_r$ is a linear combination of the preceding vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{r-1}$.

> *Proof.*
> ($\Rightarrow$) Let r be the largest integer such that $a_r \neq 0$.
> ($\Leftarrow$) There exists a linear combination and show that the scalars are not all zeors. ∎

Definition 2.3 (Linear Span).
The set of all linear combinations of S is:

$$span(S) = \{a_1\mathbf{v_1} + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n : a_i \in F\}.$$

Note that $\text{span}(\emptyset) = \{\mathbf{0}\}$.

Theorem 2.2 .
$span(S)$ is a subspace of V.

Definition 2.4 (Spanning / Generating Set).
We say that S spans V / S is a spanning/generating set of V if $span(S) = V$.

For every $\mathbf{v} \in V$ can be expressed as a linear combination of S.

Definition 2.5 (Basis).
A subset S of V is a basis of V if

1. *S is linearly independent.*
2. *S spans V.*

If S is a finite set, then V is finite-dimensional. Otherwise, V is infinite-dimensional. Additionally, a basis is not unique.

Remark .
If S spans V and $\mathbf{w} \in S$, then $\{\mathbf{w}\} \cup S$ is not linearly independent. Conversely, if $\mathbf{w} \notin S$, then $\{\mathbf{w}\} \cup S$ is still linearly independent.

Theorem 2.3 .
Suppose S is a basis of V. Then for every $\mathbf{v} \in V$, there exists a unique collection of scalars $a_1, a_2, \ldots, a_n$ such that

$$\mathbf{v} = a_1\mathbf{v_1} + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n.$$

Proof.
When we want to show the uniqueness, we could prove by contradiction. That is, fix a vector which could be expressed as two different linear combinations of S. ■

Definition 2.6 (Coordinates of vector).
Fix an order of elements in a basis S. We call $(a_1, a_2, \ldots a_n)$ is the coordinates of $\mathbf{v}$ w.r.t the basis S.

The coordinates are denoted by $[\mathbf{v}]_S = (a_1, a_2, \ldots, a_n)$, and the same vector could have different coordinates w.r.t different bases or different orders of the same basis.

Definition 2.7 (Dimension).
Suppose V is finite-dimensional vector space. The dimension of V is the size of a /any basis of V.

$$\dim_F(V) = |S|,$$

where S is a basis of V, $|\cdot|$ represents the size / cardinality of a set.

Note that $\text{span}(\emptyset) = \{\mathbf{0}\}$. The empty set spans zero vector space, and the empty set forms a basis of $\{\mathbf{0}\}$, $\dim_F(\{\mathbf{0}\}) = |\emptyset| = 0$. So zero vector space is the only zero-dimensional vector space.

Theorem 2.4 (Exchange / Replacement Theorem).
Suppose $S_1 = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a spanning set of V and $S_2 = \{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_m\}$ is linearly independent. Then $m \leq n$.

Theorem 2.5 .
Let V be a vector space over F and W be a subspace of V. Then $dim_F(W) \leq dim_F(V)$.

> *Proof.*
> If V is infinite-dimensional, there is nothing to prove. Now suppose V is finite-dimensional. Claim: W must be finite-dimensional. Suppose, by contradiction, there exists a subspace with an infinite basis B_W. B_W is linearly independent in V. Let B_V be a basis of V, which is a finite-dimensional spanning set in V. By replacement theorem, which is a contradiction. Note the above argument uses the axiom of choice. ■

Theorem 2.6 .
If both S_1 and S_2 are both bases of V, then $|S_1| = |S_2|$.

> *Proof.*
> Recall the definition of basis. Apply the exchange theorem to S_1 and S_2 from the 'both' sides. ■

Theorem 2.7 .
Let V be a vector space of dimension n and $S \subseteq V$ be a subset.

1. *If $|S| > n$, then S is linearly dependent.*
2. *If $|S| < n$, then S does not span V.*

> *Proof.*
>
> 1. Suppose, by contradiction, that S is linearly independent. Let B be any basis of V. We have $|B| = n$. By the exchange theorem, $|S| \leq |B| = n$, which is a contradiction. Thus S is linearly dependent.
> 2. Suppose, by contradiction, that S spans V. Let B be any basis of V. We have $|B| = n$. B is linearly independent and S spans V, by the exchange theorem, $|B| = n \leq |S|$, this contradicts $|S| < n$.
>
> ■

The proof from the above theorem provides a refined version of the exchange theorem:

Theorem 2.8 (Refined Exchange / Replacement Theorem).
Let T be any linearly independent subset of V. Let B be any basis of V. Let S be any spanning set of V. Then $|T| \leq |B| \leq |S|$, where $|B| = \dim_F(V)$.

Theorem 2.9 .
Let V and W be two finite-dimensional v.s. If $W \subseteq V$ and $\dim_F(W) = \dim_F(V)$, then $V = W$.

Proof.
It suffices to show that $W \supseteq V$. Suppose by contradiction,...Let S be a basis of W. Then we have $W = span(S)$ and $\mathbf{v} \notin span(S)$. Construct the larger size of linearly independent set in V. Apply the replacement theorem, then we have a contradiction. ∎

Theorem 2.10 .
Let V be a v.s. of dimension n. Then any n linearly independent subset of forms a basis of V.

Proof.
Easy proof. Let $W = span(S)$ and show that $W = V$. Recall any spanning set is a subspace of V. ∎

Exercise 2.1 .
The above theorem shows that a maximal linearly independent set forms a basis of V. In duality, a minimal spanning set also forms a basis of V. Prove the theorem.

Theorem 2.11 .
Let $S = \{\mathbf{v}_1, \mathbf{v}_2 \cdots \mathbf{v}_n, \mathbf{w}\}$. If $\mathbf{w}$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$, then $span(S) = span(S \setminus \{\mathbf{w}\})$.

Apply (2.1) iteratively, we have the following sifting method.

Theorem 2.12 (Sifting method).
Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$. There exists $S' \subseteq S$ s.t. S' is linearly independent and $span(S') = span(S)$.

Theorem 2.13 (Basis Extension Theorem).
Let S be a linearly independent subset of V. Then S can be extended into a basis of V.

Proof.
Let B be any basis of V. Consider $S \cup B$. Apply the sifting method to $S \cup B$. Question: Does S remain in the sifted set? ■

Theorem 2.14 (Duality of the Basis Extension Theorem).
Given S spans V, then there exists a subset $S' \subseteq S$ forms a basis of V.

Now let us prove replacement theorem

Proof.
Let $S_1 = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ be a spanning set of V and $S_2 = \{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_m\}$ be linearly independent. Idea is that insert each $\mathbf{w}_i$ in front of S_1 and sift. Think why $\{w_i\} \cup S$ must be linearly dependent, then we get the inequality.

■

Here we introduce sum of spaces as a subspace.

Exercise 2.2 .
Let W_1 and W_2 be two subspaces of a vector space V.

1. *Write down the definition of sum of spaces.*
2. *Prove this is smallest subspace containing W_1 and W_2.*
3. *Write down the definition of the direct sum of spaces.*
4. *Write down the dimension formula when the subspace is a direct sum of two spaces.*

Theorem 2.15 (Dimension Formula for Sum of Subspaces).
Let V_1 and V_2 be two subspaces of a finite-dimensional vector space V. Then

$$\dim_F(V_1 + V_2) = \dim_F(V_1) + \dim_F(V_2) - \dim_F(V_1 \cap V_2).$$

Proof.
Note that $W_1 \cap W_2 \subseteq W_1, W_2 \subseteq W_1 + W_2$. Start with the smallest set, apply basis extension theorem w.r.t W_1 and W_2. Check the union of the three basis, S, forms the basis in $W1 + W2$, i.e. S is L.I. and spans $W_1 + W_2$. ■

3 Linear Transformation

Let $f : V \to W$ be a linear transformation, and V is finite dimensional.

Definition 3.1 (linear function/transformation/map/homomorphism).
Check the function f: (i) Distribution on addition (ii) Scalars can be taken out of f.

Example 3.1 .
Verify the following examples are linear maps.

1. *Differentiation function: $D : \mathbb{R}[x]_{\leq n} \to \mathbb{R}[x]_{\leq n-1}$*
2. *Trace: $tr : M_n(F) \to F$*
3. *Fix a continuous function $g \in C(\mathbb{R})$. $S_g : C(\mathbb{R}) \to C(\mathbb{R})$ defined by $S_g(f(x)) = f(g(x))$*

Theorem 3.1 .
If $f : V \to W$ is a linear map. Then $f(\mathbf{0}_V) = \mathbf{0}_W$.

Alternatively, one can verify linearity via a contrapositive argument.

> **Remark** .
> *Note the following difference:*
>
> 1. $C(\mathbb{R}) \to C(\mathbb{R}) : f(x) \to f(x-1)$ *is about replacing the domain with some function.*
> 2. $\mathbb{R}^2 \to \mathbb{R}^2 : \{(x,y) : y = f(x)\} \to \{(x,y) : y = f(x-1)\}$ *is about shifting the graph of function $f(x)$ to 1 unit to the right.*

Theorem 3.2 .
Let $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ be a basis of V, then f is determined by $\{f(\mathbf{v}_1), f(\mathbf{v}_2), \ldots, f(\mathbf{v}_n)\}$, i.e.,

1. *For every $\mathbf{v} \in V$, $f(\mathbf{v})$ is a linear combination of $\{f(\mathbf{v}_1), f(\mathbf{v}_2), \ldots, f(\mathbf{v}_n)\}$. Actually, this shows $im(f) = span(\{f(\mathbf{v}_1), f(\mathbf{v}_2), \ldots, f(\mathbf{v}_n)\})$.*
2. *Suppose the linear map $g : V \to W$ satisfies $g(\mathbf{v}_i) = f(\mathbf{v}_i)\ \forall i = 1, 2, \cdots, n$. Then $g(\mathbf{v}) = f(\mathbf{v})$ for all $\mathbf{v} \in V$.*

Corollary .
If f is an injection ($ker(f) = 0$), then $\{f(\mathbf{v}_1), f(\mathbf{v}_2), \ldots, f(\mathbf{v}_n)\}$ forms a basis in $im(f)$.

The definitions of kernal and image are left as an exercise.

Theorem 3.3 .
Let $dim(V) = n$ and $dim(W) = k$

1. *$ker(f)$ is a subspace of V.*

2. *$dim(ker(f)) = nullity(f) = 0 \Leftrightarrow ker(f) = \{\mathbf{0}_w\} \Leftrightarrow f$ is injective.*

3. *$im(f)$ is a subspace of W.*

4. *$dim(im(f)) = rank(f) = k \Leftrightarrow im(f) = W \Leftrightarrow f$ is surjective.*

Theorem 3.4 (Rank-Nullity Theorem).

$$nullity(f) + rank(f) = dim_F(V)$$

Proof.
Start with the smallest set.
Since $ker(f) \subseteq V$, let $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ be a basis of $ker(f)$ and extend it to a basis of V, denoted by $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n, \mathbf{w}_1, \mathbf{w}_2, \cdots, \mathbf{w}_k\}$.
Show that $\{f(\mathbf{w}_1), f(\mathbf{w}_2), \ldots, f(\mathbf{w}_k)\}$ forms a basis of $im(f)$. ∎

Theorem 3.5 .
If $f : V \to W$ is a linear bijection, then f^{-1} is also linear.

Note that if f is bijective, then the inverse function of f (f^{-1}) exists.

Definition 3.2 (Linear Isomorphism).
Let V and W be two finite dimension vector spaces over a field F.

1. *$f : V \to W$ is a (linear) isomorphism from V onto W ($V \cong W$) if f is a bijective linear map.*

2. *V and W are said to be isomorphic as vector spaces if there exists a isomorphism $f : V \to W$.*

An trivial but important example is that $id_V : V \to V$ is a isomorphism. So every vector space is isomorphic to itself.

Definition 3.3 (Homeomorphism).
Let (X, d_X) and (Y, d_Y) be two metric spaces.

1. *$f : X \to Y$ is a homeomorphism ($X \cong Y$) if f is bijection, f is continuous and f^{-1} is also continuous.*

2. *X and Y are said to be homeomorphic as vector spaces if there exists a homeomorphism $f : X \to Y$.*

Theorem 3.6 .
Fix S as a basis of V, the coordinate mapping $\phi : V_S \to F^n$ defined by $\mathbf{v} \to [\mathbf{v}]_S$, is an isomorphism.

Theorem 3.7 .
$V \cong W \Leftrightarrow dim(V) = dim(F)$.

Proof.
($\Leftarrow$): $\phi_S : V \to F^n$ and $\phi'_S : W \to F^n$ are isomorphism. So $\phi_{S'}^{-1} \circ \phi_S$ is still an isomorphism. ∎

Definition 3.4 .
Let a linear map $f : V \to W$. Fix B_1 and B_2 as bases of V and W, where $dim(V) = n$ and $dim(W) = m$ The matrix is called the matrix representing f w.r.t the bases B_1 and B_2. We denote it as $[f]_{B_1}^{B_2} = (a_{ij})_{i,j}$. Equivalently, given any vector $\mathbf{v}_j \in V$, we have $f(\mathbf{v}_j) = \sum_{i=1}^{m} a_{ij}\mathbf{w}_i$.

Draw the fundamental diagram. The bridge between vector space and actual vector world is linear isomorphism.

Theorem 3.8 (Linear maps and matrix).

$$[f]_{B_1}^{B_2} \cdot [\mathbf{v}]_{B_1} = [f(\mathbf{v})]_{B_2}$$

Corollary .
Rank and nullity are invariant under different choices of bases. $\mathbf{v} \in ker(f) \Leftrightarrow f(\mathbf{v}) = \mathbf{0} \Leftrightarrow [f(\mathbf{v})]_{B_2} = [f]_{B_1}^{B_2} \cdot [\mathbf{v}]_{B_1} = \mathbf{0} \Leftrightarrow [\mathbf{v}]_{B_1} \in ker([f]_{B_1}^{B_2})$. *Note that the choice of bases (B_1, B_2) is arbitrary. By dimension formula, we deduce that rank and nullity are preserved.*

Theorem 3.9 (Composition and matrix multiplication).

$$[g \circ f]_{B_1}^{B_3} = [g]_{B_2}^{B_3} \cdot [f]_{B_1}^{B_2}$$

Similarly, we can show the connection regarding sum and scaling (linear homomorphism, exercise), inverse function / matrix (later).

Theorem 3.10 (Rank inequality).

Let $f : U \to V$ and $g : V \to W$ be two linear maps. Linear map version:

$$rank(g \circ f) \leq \min\{rank(f), rank(g)\}.$$

Matrix version:

$$rank(AB) \leq \min\{rank(A), rank(B)\}.$$

Theorem 3.11 .
Let $A \in M_{m\times n}(F)$ and $B \in M_{n\times m}(F)$. If $BA = I_n$ and $AB = I_m$, then $n = m = rank(A)$.

This theorem shows that only square matrices can possibly be invertible.

Definition 3.5 .
Let A be a n-square matrix. If there exists B such that $AB = BA = I_n$, then we say A is invertible/non-singular. And B is called the inverse (matrix) of $A(A^{-1})$.

Theorem 3.12 .

$$([f]_{B_1}^{B_2})^{-1} = [f^{-1}]_{B_2}^{B_1}.$$

Check $[id_V]_{B_1}^{B_1} = I_n$. Use the definition of inverse matrix.

Exercise 3.1 .
Row / column operations:

1. *Write down the three types of row operations and define the corresponding elementary row matrix.*

2. *Write down the three types of column operations and define the corresponding elementary column matrix*

Theorem 3.13 .
Let A be a $m \times n$ matrix.

1. *Performing a row operation on A is equivalent to left-multiplying an elementary row matrix to A.*

2. *Performing a column operation on A is equivalent to right-multiplying an elementary column matrix to A.*

Theorem 3.14 .
Let A be a $m \times n$ matrix.

1. *If Q is an invertible $m \times m$ matrix, then $rank(QA) = rank(A)$.*

2. *If P is an invertible $n \times n$ matrix, then $rank(AP) = rank(A)$.*

Proof.
Hint: rank inequality. ∎

The theorem shows that row / column operations preserves rank.

Exercise 3.2 . 1. *Define the row echelon form (REF) and column echelon form (CEF).*

2. *Prove elementary row matrix and elementary column matrix are invertible (Apply operations reversely).*

Theorem 3.15 .
Let A be a $m \times n$ matrix.

1. *A can be transformed into a REF by a finite number of row operations. Equivalently, there exists an invertible $m \times m$ matrix Q such that QA is in REF.*

2. *A can be transformed into a CEF by a finite number of column operations. Equivalently, there exists an invertible $n \times n$ matrix P such that AP is in CEF.*

Proof.
An algorithm helps us transform any matrix into echelon form. This is suitable for calculations with concrete examples. Invertiable matrix is the product of elementary matrices. ∎

Theorem 3.16 .
Let A be a matrix with m columns $\{\mathbf{c}_1, \mathbf{c}_2 \cdots \mathbf{c}_m\}$. Then

1. *$im(A) = span(\{\mathbf{c}_1, \mathbf{c}_2 \cdots \mathbf{c}_m\})$.*

2. *$rank(A)$ equals the maximal number of linearly independent columns of A.*

3. *$rank(A)$ equals the number of non-zero columns when A is transformed into a CEF.*

Proof.

1. Matrix vector multiplication.

2. By sifting method, the sifted subset forms a basis of $im(A)$.

3. Sifting method is equivalent to tranform A into CEF.

∎

Definition 3.6 (Smith Normal Form).
A $m \times n$ matrix is said to be in its Smith normal form if the matrix is both in its REF and CEF. Additionally, all non-zero entires of the matrix are 1.

Theorem 3.17 (QAP Theorem).
Let A be a $m \times n$ matrix. There exsits invertible matrices $Q \in M_m(F)$ and $P \in M_n(F)$ s.t. QAP is in its Smith normal form.

Theorem 3.18 .
Let A be a $m \times n$ matrix. Then $rank(A) = rank(A^\top)$.

> *Proof.*
> $rank(A) = rank(QAP) = rank(I_r) = r = rank(P^\top A^\top Q^\top) = rank(A^\top)$. ■

Theorem 3.19 .
TFAE:

1. *the rank of A ($dim(im(A))$).*
2. *the maximal number of linearly independent columns of A.*
3. *the number of non-zero columns when A is transformed into a CEF.*
4. *the rank of $A^\top$.*
5. *the maximal number of linearly independent rows of A.*
6. *the number of non-zero rows when A is transformed into a REF.*

> *Proof.*
> $rank(A) = rank(A^\top)$
> = maximal # of linearly independent columns of $A^\top$
> = maximal # of linearly independent rows of A
> = # of non-zero columns in CEF of $A^\top$ = # of non-zero rows in REF of A. ■

Theorem 3.20 .
Every invertible matrix is a product of elementary matrices.

With the above theorem, we can easily find the inverse of an invertible matrix A.

Theorem 3.21 .
Let A be a n-square matrix. If we conduct row operations to transform the argumented

matrix:

$$(A \mid I_n) \xrightarrow{\text{row operation}} (I_n \mid B)$$

Then $B = A^{-1}$.

> *Proof.*
> Left multiplying the inverse matrix of A is equivalent to Perform row operations since each invertible matrix is a product of elementary matrices. ■

Exercise 3.3 .
Write down the definition of homogeneous and inhomogeneous linear system of equations. Record the coefficient and argumented matrix.

Theorem 3.22 .
The solution of a homogenous system $(A \mid \mathbf{0})$ coincides $ker(A)$. Particularly, $dim(ker(A)) =$ (# of columns (domain)) $- rank(A)$.

Theorem 3.23 .
Let $A \in M_{m\times n}(F)$. If $m < n$, then the homogenous system $(A \mid \mathbf{0})$ has a non-trivial / non-zero solution.

> *Proof.*
> Intuitively, there are more variables than equations.
> $im(A) \subseteq F^m \to rank(A) < m$. By rank-nullity theorem, $dim(ker(A)) = n - rank(A) \geq n - m \geq 1 (\because n > m)$. ■

Theorem 3.24 .
Let $(A \mid \mathbf{b})$ be a general system of equations. The system is consistent (has at least one solution) iff $rank(A) = rank(A \mid \mathbf{b})$.

Theorem 3.25 (Principle of linearity).
Suppose $\mathbf{v}$ be a solution to the system of equation $(A \mid \mathbf{b})$, then the general solution of $(A \mid \mathbf{b})$ is

$$\mathbf{v} + ker(A) = \{\mathbf{v} + \mathbf{u} : \mathbf{u} \in ker(A)\}.$$

Corollary .
The system $(A \mid \mathbf{b})$ has a unique solution iff

1. $rank(A) = rank(A \mid \mathbf{b})$.

2. $ker(A) = \{\mathbf{0}\}$.

Theorem 3.26 (TFAE).
Let $T : V \to W$ be a linear transformation such that $dim(V) = dim(W)$. Having fixed bases, T can be represened by a square matrix A. TFAE:

1. *T is injective.*
2. *$ker(T) = \{\mathbf{0}\}$.*
3. *$nullity(T) = 0$.*
4. *$rank(T) = dim(V) = dim(W)$.*
5. *T is surjective.*
6. *$im(T) = W$.*
7. *T is an isomorphism.*
8. *A is invertible (A^{-1} exists).*
9. *$ker(A) = \{\mathbf{0}\} \to nullity(A) = 0$.*
10. *A is full rank ($rank(A) = rank(T)$).*
11. *All of rows of A are linearly independent.*
12. *All of columns of A are linearly independent.*
13. *The system of equations $A\mathbf{x} = \mathbf{v}$ has a unique solution, where $\mathbf{x} = A^{-1}\mathbf{v}$ ($\because rank(A) = rank(A \mid \mathbf{b})$ and $ker(A) = 0$).*
14. *$det(A) = det(A^\top) \neq 0$.*

Definition 3.7 (Change of coordinate matrix).
A matrix representing $id_V : V \to V$ w.r.t two bases of V. Then the matrix from S_1 basis to S_2 basis is denoted by $[id_V]_{S_1}^{S_2}$.

Definition 3.8 .
A linear transformation $T : V \to V$ is called a linear operator on V or an endomorphism on V.

Theorem 3.27 .
Let $T : V \to V$ be a linear operator and $Q = [id_V]_{S_1}^{S_2}$ be a change of coordinate matrix. Then:

$$[T]_{S_1}^{S_1} = Q^{-1} \cdot [T]_{S_2}^{S_2} \cdot Q.$$

Proof.
Draw the diagram to derive the equation. ■

Application: Diagonalization through $B = Q^{-1}AQ$. That is, choose a good basis (Q) appropriately s.t. B is as simple as possible (ideally a diagonal matrix).

Theorem 3.28 .
Let $T : V \to V$ be a linear operator on a finite-dimensional vector space V. Suppose S and S' are two bases of V. Then:

$$det([T]_S^S) = det([T]_{S'}^{S'}).$$

Definition 3.9 .
Let $T : V \to V$ be a linear operator on a finite-dimensional vector space V. Define the determinant of T by $det(T) := det([T]_S^S)$ for any choice of a basis S on V.

4 Determinants

Exercise 4.1 .
Let $A = (a_{ij})_{i,j} \in M_n(F)$*.*

1. *Write down the definition of determinants* $det(A)$ *by Laplace expansion (Consider* $n = 1$ *and* $n \geq 2$*).*

2. *Write down the definition of* $(i, j)-$*minor and* $(i, j)-$*cofactor.*

Remark .
Let $det : M_n(F) \to F$*.*

1. *det is not linear.*

2. *det is a multi-linear function.*

Example 4.1 .
Let A *be an upper triangular matrix. Then* $det(A) = \prod_{i=1}^{n} a_{ii}$*.*

> *Proof.*
> Prove by induction.
>
> 1. When $n = 1, det(A_1) = 1$.
>
> 2. Suppose $n = k$ holds. Then for $n = k + 1$, $det(A_{k+1}) = a_{11}M_{11}$. Note that $M_{1,j} = 0 \ \forall j \neq 1$.
>
> 3. So for a matrix A_k, we extend a larger matrix A_{k+1} by the left row and upper column.
>
> ■

Theorem 4.1 (Determinants of elementary matrices).
For elementary matrices, we have:

- $det(E^{I}_{p,q}) = -1$
- $det(E^{II}_{\lambda,p}) = \lambda$
- $det(E^{III}_{\mu,p,q}) = 1$
- $det(E) = det(E^{\top}) \neq 0$

Theorem 4.2 .
Let $A \in M_n(F)$:

1. *Exchangeing two rows of A multiplies $det(A)$ by 1.*
2. *Multiplying a row of A by a non-zero $\lambda \in F$ multiplies $det(A)$ by λ.*
3. *Adding a multiple of a row to another row does not change $det(A)$.*

> *Proof.*
> This proof is very hard.
>
> - For (1), we first prove swapping two rows and generalize to swap any two rows. Express the determinant as a linear combination of some $N_{p,q}$, which are minors deleting $1, 2$ rows and p, q columns.
> - For (2) and (3), it suffices to show $det : M_n(F) \to F$ is row-wise linear, or a multi-linear function. That is, determinants is linear on each row with other rows fixed.
>
> ■

Theorem 4.3 .
Let $A \in M_n(F)$.

1. *If $det(A)$ contains a zero row, then $det(A) = 0$.*
2. *Let $\lambda \in F$. Then $det(\lambda A) = \lambda^n det(A)$.*

Exercise 4.2 .
Compute $det(-A)$.

Theorem 4.4 .
A is not invertible if and only if $det(A) = 0$.

> *Proof.*
> $(\Rightarrow)$
>
> 1. By theorem (3.26), A is not invertible $\Leftrightarrow$ $REF(A)$ contains (at least) one zero row $\Leftrightarrow$ $det(REF(A)) = 0$.
> 2. Note $det(REF(A)) = det(E_1 E_2 \cdots E_k A) = det(E_1)det(E_2) \cdots det(E_k)det(A)$, where determinants of elementary matrices are non-zero. So $det(A) = 0$.

($\Leftarrow$)

1. Suppose by contradiction, then $det(A)det(A^{-1}) = det(I)$.
2. Logically equivalent to show that A is invertible then $det(A) = 0$. Since every invertible matrix is a product of elementary matrices, whose determinant are non-zeros.

$\blacksquare$

Remark .
The above theroem is logically equivalent to the (4.6).

Theorem 4.5 .
Let $A, B \in M_n(F)$, $det(AB) = det(A)det(B)$.

Proof.
If A is invertible, then it is an easy proof. If A is not invertible, we have $det(A) = 0$. Wish $det(AB) = det(A) = 0$. Since $rank(AB) \leq rank(A) < n$, we have $det(AB) = 0$. $\blacksquare$

Theorem 4.6 .
Let $A \in M_n(F)$.

1. *A is invertible* $\Longleftrightarrow det(A) \neq 0$.
2. *If A is invertible, then* $det(A^{-1}) = \frac{1}{det(A)}$

Theorem 4.7 .
$det(A^\top) = det(A)$.

Proof.
If A is invertible, then we can prove through elementary matrices. If A is not invertible, then we compute $rank(A^\top)$ by QAP and TFAE. $\blacksquare$

This implies column operations on determinants is workable.

Exercise 4.3 .
Write down the general form of Laplace expansion and prove it.

Definition 4.1 (Adjugate Matrix).
Let $A = (a_{ij} \in M_n(F))$. *The adjugate matrix of A is defined by* $adj(A) := \left((-1)^{i+j}M_{ij}\right)^\top$.

Theorem 4.8 .
Let $A \in M_n(F)$.Then we have:

1. $A \cdot adj(A) = det(A) \cdot I_n$.

2. *If A is invertible, then* $A^{-1} = \frac{1}{det(A)} \cdot adj(A)$.

> *Proof.*
> Refer to the handwritten notes. ■

Theorem 4.9 (Cramer's rule).
Let $A \in M_n(F)$ be an invertible matrix. Then the system of equations $A\mathbf{x} = \mathbf{b}$ has a unique solution (TFAE). Moreover, if $\mathbf{x} = (x_1, x_2, \cdots, x_n)^\top$, then $x_i = \frac{det(A_i)}{det(A)}$, where A_i is the matrix obtained by replace the $i-th$ column of A by the column vector $\mathbf{b}$.

> *Proof.*
> $\mathbf{x} = A^{-1}\mathbf{b} = \frac{1}{det(A)} adj(A)\mathbf{b}$. Spell the expression out. ■

Block Matrix

Exercise 4.4 .
Write down the definition of 2×2 block matrix. Show the multiplication of two block matrices.

Theorem 4.10 (Block inversion formula).
Suppose:

1. ***A** and **D** are invertible square matrices.*

2. ***A** and **D** are possibly different sizes.*

$$\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{O} & \mathbf{D} \end{pmatrix}^{-1} = \begin{pmatrix} \mathbf{A}^{-1} & -\mathbf{A}^{-1}\mathbf{B}\mathbf{D}^{-1} \\ \mathbf{O} & \mathbf{D}^{-1} \end{pmatrix}.$$

In particular, by determinant formula, $det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{O} & \mathbf{D} \end{pmatrix} = det(\mathbf{A})det(\mathbf{D}) \neq 0.$

Theorem 4.11 (Block determinant formula).
Suppose:

1. ***A** and **D** are square matrices.*

2. ***A** and **D** are possibly different sizes.*

$$det\begin{pmatrix}\mathbf{A} & \mathbf{B}\\ \mathbf{O} & \mathbf{D}\end{pmatrix} = det(\mathbf{A})det(\mathbf{D}).$$

Proof.
Both formulas need some observations about multiplication of matrices. For the second one, we wish to transform

$$\begin{pmatrix}\mathbf{I} & \mathbf{B}\\ \mathbf{O} & \mathbf{I}\end{pmatrix} \Rightarrow \begin{pmatrix}\mathbf{A} & \mathbf{B}\\ \mathbf{O} & \mathbf{D}\end{pmatrix}.$$

Then apply the follwoing formula. ■

Theorem 4.12 .
Let n be a positive integer and A be a square matrix. Then:

$$det\begin{pmatrix}\mathbf{I_n} & \mathbf{O}\\ \mathbf{O} & \mathbf{A}\end{pmatrix} = det\begin{pmatrix}\mathbf{A} & \mathbf{O}\\ \mathbf{O} & \mathbf{I_n}\end{pmatrix} = det(A).$$

Proof.
Prove by induction. ■

Exercise 4.5 .
Let $\mathbf{A}$ and $\mathbf{B}$ be square matrices of the same size, prove

$$det\begin{pmatrix}\mathbf{A} & \mathbf{B}\\ \mathbf{B} & \mathbf{A}\end{pmatrix} = det(\mathbf{A}+\mathbf{B})det(\mathbf{A}-\mathbf{B}).$$

Theorem 4.13 (Schur's factorization).
Suppose

1. *$\mathbf{A}$ and $\mathbf{D}$ are square matrices.*
2. *$\mathbf{A}$ and $\mathbf{D}$ are possibly different sizes.*
3. *$\mathbf{A}$ is invertible.*

$$det\begin{pmatrix}\mathbf{A} & \mathbf{B}\\ \mathbf{C} & \mathbf{D}\end{pmatrix} = det(\mathbf{A})det(\mathbf{D}-\mathbf{C}\mathbf{A}^{-1}\mathbf{B}).$$

Proof.
Observation: To kill $\mathbf{C}$, do row operations with a block of rows at the same time. Consider left multiplying a "block" elementary matrix. ■

5 Diagonalization

Definition 5.1 .
We say A and B of $M_n(F)$ are similar if there exists an invertible $Q \in M_n(F)$ s.t. $B = Q^{-1}AQ$.

Theorem 5.1 (Sufficient condition for similarity).
If $A \sim B$ and $A, B \in M_n(\mathbb{F})$, then

1. $rank(A) = rank(B)$.
2. $nullity(A) = nullity(B)$.
3. $det(A) = det(B)$.
4. $tr(A) = tr(B)$.
5. $p_A(x) = p_B(x)$: *the same chracteristic polynoimal.*
6. $m_A(x) = m_B(x)$: *the same minimal polynomial.*

If we further assume that F is algebratically closed (e.g. $\mathbb{C}$), then they have the same:

1. *Multi-set of eigenvalues (counted with algebraic multiplicities).*
2. *algebraic multiplicity of each eigenvalue.*
3. *geometric multiplicity of each eigenvalue.*

Proof.

1. By rank inequality, multiplying an invertible matrix preserves the rank of the original matrix.
2. By rank-nullity theorem.
3. Easy proof.
4. First prove $tr(AB) = tr(BA)$, where $A \in M_{m\times n}(F)$ and $B \in M_{n\times m}(F)$.
5. Derive $det(A - xI) = det(B - xI)$.
6. First prove the polynomial set satisfied by A and the one satisfied by B are equal. Then the monic polynomial of minimal degree satisfied by the corresponding matrix (i.e. minimal polynomial) is the same.

If we assume $A, B \in M_n(\mathbb{C})$, by the fundamental theorem of algebra,

1. Same chracteristic polynomial implies same set of eigenvalues.
2. Similar argument for algebraic multiplicites of eigenvalues.
3. Show $nullity(A - \lambda I_n) = nullity(B - \lambda I_n)$.

■

Remark . 1. *Multi-set of eigenvalues, which means the repetition of an eigenvalue matters. We will introduce the concept of spectrum later, and the statement is equivalent to say two similar matrix have the same spectrum.*

2. *$A \sim B \Leftrightarrow A$ and B represent the same linear operator under different bases.*

Definition 5.2 (Diagonalizable operators and matrices). 1. *A linear operator $T : V \to V$ is diagonalizable if $\exists$ a basis S of V s.t. the matrix $[T]_S^S$ is diagonal.*

2. *A matrix $A \in M_n(F)$ is diagonalizable if A is similar to a diagonal matrix, i.e., $\exists$ an invertible matrix Q s.t. $Q^{-1}AQ$ is diagonal.*

Definition 5.3 .
Let $T : V \to V$ be a linear operator.

1. *$\lambda \in F$ is an eigenvalue of T if $T(\mathbf{v}) = \lambda \mathbf{v}$ for some non-zero $\mathbf{v} \in V$.*
2. *$E_\lambda(T) = ker(A - \lambda \cdot id)$ is called the eigenspace w.r.t the eigenvalue λ.*
3. *Any non-zero $\mathbf{v} \in E_\lambda(T)$ is called an eigenvector of T w.r.t the eigenvalue λ. So any eigenvector satisfies $T(\mathbf{v}) = \lambda \mathbf{v}$.*

Theorem 5.2 .
If $\lambda_1, \lambda_2, \cdots \lambda_n$ be distinct eigenvalues of T and $\mathbf{v}_1, \mathbf{v}_2 \cdots \mathbf{v}_n$ be the corresponding eigenvectors, then $\{\mathbf{v}_1, \mathbf{v}_2 \cdots \mathbf{v}_n\}$ is linearly independent.

Proof.
Prove by induction. Suppose it holds for $n = k - 1$. Then consider $n = k$. Suppose $a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots a_k\mathbf{v}_k = 0$ (Equation 1). Apply T on both sides and use the identity $T(\mathbf{v}_i) = \lambda_i \mathbf{v}_i \, \forall i = 1, 2, \cdots k$ (Equation 2). Note that $\lambda_i \neq \lambda_j \, \forall i \neq j$. Reduce the $k - th$ term to obtain a linear combination of $\{\mathbf{v}_1, \mathbf{v}_2 \cdots \mathbf{v}_{k-1}\}$. By induction hypothesis and distinctness of eigenvalues, we have $a_1 = a_2 = \cdots a_{k-1} = 0$. Simplify the equation 1 to get $a_k = 0$. By

the principle of mathematical induction, the proof is completed. ■

Definition 5.4 (Chracteristic polynomial).
Two cases:

1. *Let $A \in M_n(F)$. The chracteristic polynoimal of A is defined by $p_A(x) = det(A - xI_n)$.*
2. *Let $T : V \to V$ be a linear operator. The chracteristic polynoimal of T is defined by $p_T(x) = det([T]_s^s - xI_n)$ (For any basis s of V).*

Theorem 5.3 .
Eigenvalues are the roots of the chracteristic polynomial of A. That is, λ is an eigenvalue of A iff $p_A(\lambda) = det(A - \lambda I_n) = 0$.

Remark .
The above theorem demonstrates how to find eigenvalues practically.

Theorem 5.4 (Diagonalizability Criterion I: Eigen-basis).
$A \in M_n(F)$ is diagonalizable iff $\exists$ a basis $\mathbf{v}_1, \mathbf{v}_2 \cdots \mathbf{v}_n$ in F^n s.t. each v_i is an eigenvector of A w.r.t. some eigenvalue λ_i.

Proof.
This is the definition of diagonalizability itself. Associate A with a linear operator $f_A : F^n \to F^n$. Let B be the standard basis.

1. Think why $A = [f_A]_B^B$.
2. f_A is diagonalizable iff $\exists$ a basis S of F^n s.t. $[f_A]_S^S$ is diagonal. $\Leftrightarrow$ A is diagonalizable iff $\exists$ an invertible $Q = [id_V]_S^B$.
3. Think: Q is essentially formed by putting each eignevectors in S as columns.
4. Think: Why S is a set of eigenvectors? Why the diagonal entries in $[f_A]_S^S$ are the corresponding eigenvalues? ($f_A(\mathbf{v}_i) = A\mathbf{v}_i = \lambda \mathbf{v}_i$).

■

Theorem 5.5 .
If $A \in M_n(F)$ has n distinct eigenvalues, then A is diagonalizable.

Proof.

1. Distinct eigenvalues shows that ...?

2. By replacement theorem, we find exactly a set of eigen-basis, hence A is diagonalizable.

■

Definition 5.5 .
Let $A \in M_n(\mathbb{C})$. By the fundamental theorem of algebra, $p_A(x)$ can be factorized as $(-1)^n(x-\lambda_1)^{r_1}(x-\lambda_2)^{r_2}\cdots(x-\lambda_k)^{r_k}$, where $\lambda_1, \lambda_2, \cdots \lambda_k$ are distinct. For each eigenvalue λ_i,

1. *Algebraic multiplicity of λ_i: # of copies of $(x-\lambda_i)$ in $p_A(x)$, denoted by $a(\lambda_i) = r_i$.*

2. *Geometric multiplicity of λ_i, denoted by $g(\lambda_i) = nullity(A - \lambda_i I_n)$, which measures the dimension of the eigenspace of λ_i.*

Theorem 5.6 .
Let $A \in M_n(\mathbb{C})$. For each λ of A:

$$1 \le g(\lambda) \le a(\lambda).$$

Proof.

1. $1 \le g(\lambda) : det(A - \lambda I_n) = 0$ and use TFAE.

2. $g(\lambda) \le a(\lambda)$: Take a basis of $(A - \lambda I_n)$ (assume dimension equals p). By basis extension theorem, we can obtain a basis S of $\mathbb{C}^n$. Apply associated linear operator T and get a "quasi-diagonla" matrix. Note that $p_{[T]_S^S}(x) = (-1)^n(x-\lambda)^p \times r(x)$. $r(x)$ may or may not contain $(x-\lambda)$.

■

Theorem 5.7 (Diagonalizability Criterion II: Multiplicity).
Let $A \in M_n(\mathbb{C})$. TFAE:

1. *A is diagonalizable.*

2. *$\exists$ a basis $\mathbf{v}_1, \mathbf{v}_2 \cdots \mathbf{v}_n$ in $\mathbb{C}^n$ s.t. each v_i is an eigenvector of A w.r.t. some eigenvalue λ_i.*

3. *For each λ of A, we have $g(\lambda) = a(\lambda)$.*

Proof.
It suffices to show:

1. (1) implies (3): Let D be a diagonal matrix, $A \sim D$ implies they have the same geometric multiplicities.
2. (3) implies (2): For each eigenspace, choose S_i to be a basis of $E_{\lambda_i}(A)$. Claim that $S = S_1 \cup S_2 \cup \cdots \cup S_k$ forms an eigen basis of A. Detail: for each λ_i, we may choose more than one eigenvector, why is the whole set S linearly independent?

■

Exercise 5.1 .
Take a polynomial $f \in F[x]$. For $A \in M_n(F)$, define $f(A)$.

Theorem 5.8 (Cayley-Hamilton Theorem).
Let $A \in M_n(F)$ and its chracteristic polynoimal $p_A(x)$. Then $p_A(A) = \mathbf{O}$ (the zero matrix).

Proof.
Use $(A - xI)adj(A - xI) = det(A - xI)I = p_A(x)I$. Compare coefficients.

■

Definition 5.6 (Minimal Polynomial).
Let $A \in M_n(\mathbb{C})$. The minimal polynoimal is the monic polynoimal of minimal degree s.t. $m_A(A) = \mathbf{O}$.

Monic means the leading coefficient is 1.

Theorem 5.9 .
Let $A \in M_n(\mathbb{C})$, $p_A(x)$ and $m_A(x)$ be its chracteristic polynomial and minial polynomial respectively.

1. *For any $f \in \mathbb{C}[x]$ satisfying $f(A) = \mathbf{O}$, then $f(x)$ is divisible by $m_A(x)$/ $m_A(x)$ divides $f(x)$.*
2. *$p_A(x)$ is divisible by $m_A(x)$/ $m_A(x)$ divides $p_A(x)$.*
3. *Each eigenvalue λ of A is a root of $m_A(x)$/ $m_A(\lambda) = 0$.*

Proof.

1. Conduct division of $f(x)$ by $m_A(x)$. Aim to argue the remainder $r(x)$ must be zero polynomial. Suppose not, then $r(x)$ is a samller degree polynomial satisfying $r(A) = 0$, which yields a contradiction.

2. By Cayley-Hamilton and the property (1).

3. Use (5.10), we have: $m_A(A)\mathbf{v} = m_A(\lambda)\mathbf{v}$. This forces $m_A(\lambda) = 0$.

∎

Theorem 5.10 .
Let f be a non-constant polynomial. If λ is an eigenvalue of A, then $f(\lambda)$ is an eigenvalue of $f(A)$.

Proof.
Key observation: $A^k\mathbf{v} = \lambda^k\mathbf{v}$ for any $k \in \mathbb{N}$. ∎

Lemma .
Every matrix $A \in M_n(\mathbb{C})$ has a unique minimal polynomial.

Proof.
Use the property (1). ∎

Lemma (Candidate of minial polynomial).
Let $A \in M_n(\mathbb{C})$ and $\lambda_1, \lambda_2, \cdots \lambda_k$ be all distinct eigenvalues of A. If $p_A(x) = (-1)^n \cdot (x-\lambda_1)^{r_1}(x-\lambda_2)^{r_2}\cdots(x-\lambda_k)^{r_k}$. Then $m_A(x) = (x-\lambda_1)^{s_1}(x-\lambda_2)^{s_2}\cdots(x-\lambda_k)^{s_k}$, where $1 \leq s_i \leq r_i$ for each $i = 1, 2, \cdots, k$.

Proof.
Property (2) and (3) imply this lemma. ∎

Theorem 5.11 (Diagonalizability Criterion III: Minimal Polynomial).
Let $A \in M_n(\mathbb{C})$. TFAE:

1. *A is diagonalizable.*

2. *$\exists$ a basis $\mathbf{v}_1, \mathbf{v}_2 \cdots \mathbf{v}_n$ in $\mathbb{C}^n$ s.t. each v_i is an eigenvector of A w.r.t. some eigenvalue λ_i.*

3. *For each λ of A, we have $g(\lambda) = a(\lambda)$.*

4. *The minial polynoimal $m_A(x)$ has no repeated factor.*

Proof.

1. (2) implies (4): Suppose (2) is true. Let $p_A(x) = (-1)^n \cdot (x-\lambda_1)^{r_1}(x-\lambda_2)^{r_2}\cdots(x-\lambda_k)^{r_k}$ and $q(x) = (x-\lambda_1)(x-\lambda_2)\cdots(x-\lambda_k)$, where $\lambda_i \neq \lambda_j \forall i \neq j$. Aim to show $q(x) = m_A(x)$. By property (3) in (5.9), we have $q(x) \mid m_A(x)$. Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \cdots \mathbf{v}_n\}$ be a basis of $\mathbb{C}^n$ and eigenvectors of A. For each $\mathbf{v} \in S$, $A\mathbf{v} = \lambda\mathbf{v}$ for some $\lambda = \lambda_i$. By (5.10), $q(A)\mathbf{v} = q(\lambda)\mathbf{v}$ and $q(\lambda) = 0$. Note that $S \subseteq ker(q(A)) \to |S| \leq nullity(q(A)) \leftrightarrow rank(q(A)) = 0 \leftrightarrow q(A) = \mathbf{O}$. So by property (1) in (5.9), $m_A(x) \mid q(A)$.

2. Suppose $m_A(x) = (x-\lambda_1)(x-\lambda_2)\cdots(x-\lambda_k)$, where $\lambda_i \neq \lambda_j \forall i \neq j$. By definition, $m_A(A) = \mathbf{O}$. Use (1), we have:

$$nullity(\prod_{i=1}^{k}(A-\lambda_i)) \leq \sum_{i=1}^{k} nullity(A-\lambda_i I) \leq \sum_{i=1}^{k} g(\lambda_i) = n.$$

However, $nullity(m_A(A)) = nullity(\mathbf{O}) = n$. This forces the equality.

■

Proposition 1 .
Let $B, C \in M_n(F)$, then $nullity(BC) \leq nullity(B) + nullity(C)$.

Proof.
First, show that $ker(C) \subseteq ker(BC)$. Extend a basis of $ker(C)$, $\{\mathbf{v}_1, \mathbf{v}_2, \cdots \mathbf{v}_n\}$ into a basis of $ker(BC)$, $\{\mathbf{v}_1, \mathbf{v}_2, \cdots \mathbf{v}_n, \mathbf{w}_1, \mathbf{w}_2, \cdots, \mathbf{w}_p\}$.
Next, show $\{C\mathbf{w}_1, C\mathbf{w}_2, \cdots, C\mathbf{w}_p\} \subseteq ker(B)$. In particular, $\{C\mathbf{w}_1, C\mathbf{w}_2, \cdots, C\mathbf{w}_p\}$ is linearly independent. (Show the linear combination lies in $ker(C)$, and use the linear independence of the basis in $ker(BC)$). ■

Theorem 5.12 (Schur Triangulation).
Every square matrix $A \in M_n(\mathbb{C})$ is similar to an upper triangular matrix.

Proof.
Prove by induction. Left as an exercise. ■

Definition 5.7 (Spectrum).
The spectrum of a linear operator T is the multi-set contains the all eigenvalues of T, denoted by $spec(T) = \{\lambda_1, \lambda_2, \cdots \lambda_n\}$. Note that these elements might repeat.

Theorem 5.13 (Spectral mapping theorem).
Let $A \in M_n(\mathbb{C})$ and $f \in \mathbb{C}[x]$ be a non-constant polynomial.

1. *If λ is an eigenvalue of A, then $f(\lambda)$ is an eigenvalue of $f(A)$.*

2. *If μ is an eigenvalue of $f(A)$, then $\exists$ an eigenvalue λ of A s.t. $\mu = f(\lambda)$.*

That is, if $spec(A) = \{\lambda_1, \lambda_2, \cdots \lambda_n\}$, then $spec(f(A)) = \{f(\lambda_1), f(\lambda_2), \cdots f(\lambda_n)\}$. So spectrum mapping is a bijection between multi-sets.

> *Proof.*
> Key result is that diagonal entries of upper triangular matrix is eigenvalues. Use Schur triangulation to deduce the spectrum mapping. ■

Application of Diagonalization

Definition 5.8 (T-invariant subspace).
Let $T : V \to V$ be a linear operator. A subspace $W \subseteq V$ is said to be a T-invariant subspace *if $T(W) \subseteq W$. Moreover, the restriction of T on W is defined by $T|_W : W \to W$, where $T|_W(\mathbf{w}) = T(\mathbf{w}) \, \forall \mathbf{w} \in W$.*

Theorem 5.14 .
Let $W \subseteq V$ be an T-invariant subspace and let $T|_W : W \to W$ be the restriction. Then $m_{T|_W}(x)$ divdides $m_T(x)$.

> *Proof.*
> Key observation: $m_T(T) = \mathbf{0}$ implies $m_T(T|_W) = \mathbf{0}$. Use the property (1) in (5.9). ■

Exercise 5.2 .
Verify that

1. *$ker(T|_W) = ker(T) \cap W$.*

2. *$im(T|_W) = \{T(\mathbf{w}) : \mathbf{w} \in W\}$.*

Theorem 5.15 (Simultaneous Eigenvector).
Let $T_1, T_2 : V \to V$ be two commutable linear operators of a complex finite-dimensional V, then there exists $\mathbf{v}$ s.t. $\mathbf{v}$ is simultaneously an eigenvector of both T_1 and T_2.

Proof.
Let $E_\lambda(T_1)$ be an eigenspace of T_1.

1. Claim that $E_\lambda(T_1)$ is T_2 invariant.

2. Consider the restriction $T_2|_{E_\lambda(T_1)} : E_\lambda(T_1) \to E_\lambda(T_1)$. Take an eigenvector $\mathbf{v}$ of $T_2|_{E_\lambda(T_1)}$ w.r.t. some eigenvalue λ'. This vector is simultaneously an eigenvector of $E_\lambda(T_1)$.

■

Theorem 5.16 (Simultaneous Diagonalizability).
Let $T_1, T_2 : V \to V$ be two commutable and diagonalizable linear operators of a complex finite-dimensional V, then there exists a simultaneous eigen-basis S s.t. $[T_1]_S^S$ and $[T_2]_S^S$ are both diagonal and hence simultaneously diagonalizable.

Proof.
This is just an stronger verison of the previous theorem. ■

Definition 5.9 (Exponential of Matrix).
Let $A \in M_n(\mathbb{C})$. The exponential of A is defined by:

$$e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!}.$$

Theorem 5.17 .
If $D = \begin{pmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{pmatrix}$ is a diagonal matrix, then $e^D = \begin{pmatrix} e^{d_1} & 0 & \cdots & 0 \\ 0 & e^{d_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & e^{d_n} \end{pmatrix}$

Theorem 5.18 .
Let $A, B \in M_n(\mathbb{C})$, then:

1. *e^A converges absolutely.*

2. *If $AB = BA$, then $e^{A+B} = e^A e^B$.*

Proof.
The above theorems are stated without proof, as they may require knowledge of normed spaces, which is beyond the current scope. ■

Theorem 5.19 .
Let $A \in M_n(\mathbb{C})$*.*

1. *If* A B*, then* e^A e^B*.*
2. *Consider a function* $f : \mathbb{C} \to M_n(\mathbb{C})$ *defined by* $f(t) = e^{At}$*, then*

$$f'(t) = \frac{d}{dt}e^{At} = Ae^{At}.$$

Corollary .
Give a system of differential equations:

$$\frac{d}{dt}\mathbf{v}(t) = A\mathbf{v}(t),$$

where $A \in M_n(\mathbb{C})$*, then the solution is given by:*

$$\mathbf{v}(t) = e^{At}\mathbf{v}(0).$$

Remark .
To compute e^{At}*, we need to adress higher powers of* A*, then diagonalization or Cayley-Hamilton is useful here.*

Exercise 5.3 .
Give a system of difference equations:

$$\mathbf{v}_n = A\mathbf{v}_{n-1},$$

where $A \in M_n(\mathbb{C})$*, then the solution is given by:*

$$\mathbf{v}_n = A^n\mathbf{v}_0.$$

6 Linear Duality

Let V and W be two finite-dimensional vector spaces over a field $\mathcal{F}$ s.t. $dim(V) = n$ and $dim(W) = m$. Recall our *dream diagram*:

$$\begin{tikzcd}
V \arrow[r, "\mathcal{L}(V,W)"] \arrow[d, "\phi_S"'] & W \arrow[d, "\phi_{S'}"] \\
\mathcal{F}^n \arrow[r, "\mathcal{M}_{m\times n}(\mathcal{F})"'] & \mathcal{F}^m
\end{tikzcd}$$

Fix S as a basis of V, the coordinate mapping $\phi : V \to F^n$ defined by $\mathbf{v} \mapsto [\mathbf{v}]_S$ is an isomorphism. So does the coordinate mapping $\phi_{S'}$.

Now consider the vector space of all linear functions from V to W, denoted by $\mathcal{L}(V, W) = \{f : V \to W : f \text{ is linear}\}$, then we have:

Theorem 6.1 .
Fix bases S and S' of V and W respectively, then there exists an isomorphism of vector spaces:

$$\Phi_S^{S'} : \mathcal{L}(V, W) \to \mathcal{M}_{m\times n}(\mathcal{F}),$$

defined by $f \mapsto [f]_S^{S'}$.

Corollary .
If both V, W are finite dimensional over $\mathcal{F}$, then

$$dim_{\mathcal{F}}(\mathcal{L}(V, W)) = dim_{\mathcal{F}}(V)dim_{\mathcal{F}}(W).$$

Next, we apply the above result to the special case when $W = \mathcal{F}$.

Definition 6.1 (Dual Space).
Let V be a vector space over a field $\mathcal{F}$. The dual space of V, denoted by V^*, is defined by:

$$V^* = \mathcal{L}(V, \mathcal{F}) = \{f : V \to \mathcal{F} : f \text{ is linear}\}.$$

Take any element $f \in V^*$, we call $f : V \to \mathcal{F}$ is linear functional on V.

Theorem 6.2 .
Fix a basis S of V and $B = \{1\}$ be a standard basis of $\mathcal{F}$, then there exists an isomorphism of vector spaces:

$$\Phi_S^B : \mathcal{L}(V, \mathcal{F}) \to \mathcal{M}_{1\times n}(\mathcal{F}),$$

defined by $f \mapsto [f]_S^B$.

Corollary .
If both V is finite dimensional over $\mathcal{F}$, then

$$dim_{\mathcal{F}}(V^*) = dim_{\mathcal{F}}(V).$$

Definition 6.2 (Dual Basis).
Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \cdots \mathbf{v}_n\}$ be a basis of V. $S^* = \{\mathbf{v}_1^*, \mathbf{v}_2^*, \cdots \mathbf{v}_n^*\}$ is said to be the dual basis of S in V^* if $\phi_S^B(\mathbf{v}_k^*) = e_k$ for each $k = 1, 2, \cdots n$.

Lemma .
Fix any basis $S = \{\mathbf{v}_1, \mathbf{v}_2, \cdots \mathbf{v}_n\}$ be a basis of V, then its dual basis $S^* = \{\mathbf{v}_1^*, \mathbf{v}_2^*, \cdots \mathbf{v}_n^*\}$ satisfies, for each i:

$$\begin{cases} \mathbf{v}_i^*(\mathbf{v}_j) = 1, \text{ if } i = j, \\ \mathbf{v}_i^*(\mathbf{v}_j) = 0, \text{ if } i \neq j. \end{cases}$$

Proof.
For $i = 1$, consider $\mathbf{v}_1^* : V \to \mathcal{F}$. By definition, $\Phi_S^B(\mathbf{v}_1^*) = e_1$, i.e., $[v_1^*]_S^B = (1, 0, \cdots, 0)$. This implies that $\mathbf{v}_1^*(\mathbf{v}_1) = 1 \cdot 1$ and $\mathbf{v}_1^*(\mathbf{v}_j) = 0 \cdot 1$ for $j \neq 1$. Now repeat the above argument for $i = 2, 3, \cdots n$. ∎

Definition 6.3 (Dual linear maps).
Let $T : V \to W$ be a linear map btw two vector spaces V and W. The dual of T is a linear map btw the dual spaces of V and W:

$$T^* : W^* \to V^* \text{ given by } w^* \mapsto w^* \circ T.$$

Theorem 6.3 .
Let $T : V \to W$ be a linear map btw two finite-dimensional vector spaces V and W. Fix bases S and S' of V and W respectively, then

$$[T^*]_{S'^*}^{S^*} = ([T]_S^{S'})^T.$$

Proof.
Let $S = \{\mathbf{v}_1, \mathbf{v}_2 \cdots \mathbf{v}_n\}$ be a basis of V. Let $S' = \{\mathbf{w}_1, \mathbf{w}_2 \cdots \mathbf{w}_m\}$ be a basis of W. Suppose $[T]_S^{S'} = A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$, we want to show $[T^*]_{S'^*}^{S^*} = A^\top$.
Let $\mathbf{v} = \sum_{k=1}^n c_k \mathbf{v}_k$. Consider $T^*(\mathbf{w}_1^*)(\mathbf{v}) = \mathbf{w}_1^* \circ T(\mathbf{v}) =$

$\mathbf{w}_1^*(\sum_{k=1}^n c_k T(\mathbf{v}_k)) = \mathbf{w}_1^*(\sum_{k=1}^n c_k \sum_{j=1}^m a_{jk}\mathbf{w}_j) = \sum_{k=1}^n c_k a_{1k} \cdot 1$. Note that $c_k = v_k^*(\mathbf{v})$, so we get $T^*(\mathbf{w}_1^*) = b_1^* \circ T(\mathbf{v}) = \sum_{k=1}^n a_{1k} v_k^*(\mathbf{v})$. ■

Definition 6.4 (Annihilator).
Let V be a vector space and $U \subseteq V$ be a subspace. The annihilator of U is a subspace of V^* defined by:

$$U^0 = \{f \in V^* : f(\mathbf{u}) = 0 \; \forall \mathbf{u} \in U\}.$$

Theorem 6.4 (Linear Duality Theorem (I)).
Let V be a finite dimensional vector space and $U \subseteq V$, then:

$$dim(V) = dim(U) + dim(U^0).$$

Proof.
Let $\{\mathbf{u}_1, \mathbf{u}_2, \cdots \mathbf{u}_k\}$ be a basis of U. By basis extension theorem, we can extend it into $\{\mathbf{u}_1, \mathbf{u}_2, \cdots \mathbf{u}_k, \mathbf{u}_{k+1}, \cdots \mathbf{u}_n\}$. So $\{\mathbf{u}_1^*, \mathbf{u}_2^*, \cdots \mathbf{u}_k^*, \mathbf{u}_{k+1}^*, \cdots \mathbf{u}_n^*\}$ is the dual basis in V^*. Next, we want to show $\{\mathbf{u}_{k+1}^*, \cdots \mathbf{u}_n^*\}$ forms a basis of U^0.

1. For each $k+1 \leq j \leq n$, $\mathbf{u}_j^*(\mathbf{u}_i) = 0$ for $1 \leq i \leq k$. So $\mathbf{u}_j^* \in U^0$.

2. Let $\theta \in U^0 \subseteq V^*$, so $\theta = \sum_{i=1}^n a_i \mathbf{u}_i^*$, and we wish $a_1 = a_2 = \cdots a_k = 0$. Let $\mathbf{u} \in U$, so $\mathbf{u} = \sum_{i=1}^k \mathbf{u}_i$. By definition, $\theta(\mathbf{u_j}) = 0$ forces $\sum_{i=1}^k a_i \mathbf{u}_i^*(\mathbf{u}_j) = 0$ for $1 \leq j \leq k$.

3. Linear independence is trivial.

By the claim, we have $dim(U^0) = n - k = dim(V) - dim(U)$. ■

Theorem 6.5 (Linear Duality Theorem (II)).
Let $T : V \to W$ be a linear map btw finite dimensional vector spaces. Then we have:

1. $ker(T^*) = (im(T))^0$.

2. $rank(T) = rank(T^*)$.

3. $im(T^*) = (ker(T))^0$.

Proof.
Left as an exercise. ■

Corollary .
(6.5) *implies that*

1. T^* *is injective iff* T *is surjective.*

2. $rank(A) = rank(A^\top)$.

3. T^* *is surjective iff* T *is injective.*

Proof.

1. $im(T) = W \leftrightarrow dim((im(T)^0)) = 0 = dim(ker(T^*))$.

2. Recall theorem (6).

3. $ker(T) = \{\mathbf{0}\} \leftrightarrow dim((ker(T))^0) = dim(im(T^*)) = dim(V) = dim(V^*)$

■

Definition 6.5 (Double Dual).
The dual of a dual space is defined by:

$$(V^*)^* = \{g : V^* \to F : g \text{ is linear}\}.$$

Theorem 6.6 (Linear Duality Theorem (III)).
Let $ev : V \to (V^*)^*$ *defined by* $\mathbf{v} \mapsto ev(\mathbf{v})$*. That is, for each* $\mathbf{v} \in V$, $ev(\mathbf{v}) : V^* \to \mathcal{F}$ *is a linear functional on* V^**. So the ev function satisfies that for* $f \in V^*$, $ev(\mathbf{v})(f) = f(\mathbf{v})$*. Under this setting, we have:*

1. *ev is an injective linear map.*

2. *If* V *is finite dimensional, then ev is an isomorphism btw* V *and* $(V^*)^*$*.*

Proof.

1. Take $\mathbf{v} \in ker(ev)$, we have $ev(\mathbf{v}) = \mathbf{0}$. That is, $ev(\mathbf{v})(f) = f(\mathbf{v}) = 0$ for $f \in V^*$. Since f is aribitrary in the dual space V^*, this forces $\mathbf{v} = \mathbf{0}$ and $ker(ev) = \{\mathbf{0}\}$. This proves the injectiveness.

2. Note that $dim(V) = dim(V^*) = dim((V^*)^*)$, $ev{:}V \to (V^*)^*$ is injective linear map from 1, by (3.26), ev is also a surjective map and hence an isomorphism.

■

Remark .

If V is infinite dimensional, then ev *might not be surjective in general, which is beyond the current scope.*

7 Jordan Canonical Form

Definition 7.1 (Jordan Block).
A Jordan block $J_{\lambda,k}$ of eigenvealue λ and index/size k is a $k \times k$ upper triangular matrix such that$\cdots$

Theorem 7.1 .
Let $J = J_{\lambda,k} \in M_k(\mathbb{C})$, then:

1. $p_J(x) = (\lambda - x)^k$.

2. $a_J(\lambda) = k$, *and* $g_J(\lambda) = 1$.

3. *For* $1 \leq i \leq k$, $\text{rank}((J - \lambda I_k)^i) = k - i$ *and* $\text{nullity}((J - \lambda I_k)^i) = i$.

4. $\text{nullity}((J - \lambda I_k)^i) - \text{nullity}((J - \lambda I_k)^{i-1}) = \begin{cases} 1, & \text{if } 1 \leq i \leq k \\ 0, & \text{if } i \geq k+1 \end{cases}.$

5. $m_J(x) = (x - \lambda)^k$.

Proof.
1,2 is trivial by inspection. For 3 and 4, let $N_k = J - \lambda I_k$. The key observation is that the diagonal of $1'$s is shifted by one unit to the right after multiplying N_k itself. For 5, since $N_k^i \neq \mathbf{O}$ for $i < k$, the minimal polynomial of J must be the characteristic polynomial of J. ∎

Remark .
2 and 5 shows that J is diagonalizable only when J is of size 1.

Definition 7.2 .
Direct sum of matrices Let $A \in M_m(F)$, $B \in M_n(F)$, then the direct sum $A \oplus B$ is a block diagonal matrix in $M_{m+n}(F)$ such that

$$A \oplus B = \begin{pmatrix} A & \mathbf{O} \\ \mathbf{O} & B. \end{pmatrix}$$

Exercise 7.1 .
Write down the definition of a direct sum of two subspaces. They are two different concepts.

Theorem 7.2 .
Let $A \in M_m(F)$, $B \in M_n(F)$, then:

1. $\text{rank}(A \oplus B) = \text{rank}(A) + \text{rank}(B)$

2. $\text{nullity}(A \oplus B) = \text{nullity}(A) + \text{nullity}(B)$

3. $p_{A\oplus B}(x) = p_A(x) \cdot p_B(x)$

4. $a_{A\oplus B}(\lambda) = a_A(\lambda) + a_B(\lambda)$

5. $g_{A\oplus B}(\lambda) = g_A(\lambda) + g_B(\lambda)$

6. $m_{A\oplus B}(x) = lcm\{m_A(x), m_B(x)\}$

Proof.
1–5 are left as an exercise. For 6, observe that for any polynomial $f(x) \in F[x]$, one has $f(A \oplus B) = f(A) \oplus f(B)$. Let $m_{A\oplus B} = f(x)$, $f(A \oplus B) = \mathbf{0}$ by the definition of minimal polynomial, which forces $f(x)$ is a common multiple of $m_A(x)$ and $m_B(x)$. For the "least" part, we provide two methods. Suppose $g(x)$ is another common multiple, we want to show f divides g. Since $m_A(x) \mid g(x)$ and $m_B(x) \mid g(x)$, $g(A \oplus B) = \mathbf{0}$ impies $f \mid g$.

Alternatively, conduct the division $g(x) = f(x) \cdot q(x) + r(x)$. It suffices to show $r(x) \equiv 0$. Suppose, by contradiction, $deg(r) < deg(f)$. However,$r(A) = r(B) = \mathbf{0}$ violates the minimality. ∎

Definition 7.3 (JCF).
A matrix J is in its Jordan Canonical Form if it is a direct sum of Jordan blocks:

$$J = J_{\lambda_1,k_1} \oplus J_{\lambda_2,k_2} \oplus \cdots \oplus J_{\lambda_n,k_n}.$$

Corollary . 1. *Any diagonal matrix is exactly in its JCF$\cdots$*

2. *Write down the $p_J(x)$ and $m_J(x)$ in terms of $p_{J_{\lambda_i,k_i}}(x)$ and $m_{J_{\lambda_i,k_i}}(x)$.*

Theorem 7.3 .
Let J in its JCF and fix $\lambda \in Spec(J)$, then:

1. *# of Jordan blocks of eigenvalue $\lambda = g_J(\lambda)$.*

2. *# of Jordan blocks of eigenvalue λ of index(size) at least i* $= \text{nullity}((J - \lambda I_k)^i) - \text{nullity}((J - \lambda I_k)^{i-1})$.

Proof.
Let $J = J_{\lambda_1,k_1} \oplus J_{\lambda_2,k_2} \oplus \cdots \oplus J_{\lambda_n,k_n}$. Suppose $\lambda_1 = \lambda_2 = \cdots \lambda_l = \lambda$, $l \le n$, $\lambda \notin \{\lambda_{l+1}, \lambda_{l+2}, \cdots \lambda_n\}$. Let $N_k = J - \lambda I_k$, recall (3) and (4) in theorem (7.1),

1. $g_J(\lambda) = \sum_{i=1}^{n} \text{nullity}(N_{\lambda_i,k_i}) = \underbrace{1+1+\cdots+1}_{l \text{ times}} + \underbrace{0+0+\cdots+0}_{\lambda\neq\lambda_i, i\geq l} = l$, which equals # of Jordan blocks of eigenvalue λ.

2. Because $N_k = N_{k_1} \oplus N_{k_2} \oplus \cdots \oplus N_{k_l} \oplus T$, where T is some direct sum of Jordan blocks of eigenvalue that is not λ. Fix i, $\text{nullity}(N_k^i) - \text{nullity}(N_k^{i-1}) = \sum_{j=1}^{l} \left(\text{nullity}(N_{k_j}^i) - \text{nullity}(N_{k_j}^{i-1})\right) = \#\{1 \le j \le l : i \le k_j\}$, which equals # of Jordan blocks of eigenvalue λ of index(size) at least i.

■

Combine the definition of JCF and the theroem (7), we can define the JCF of a general matrix $A \in M_n(\mathbb{C})$.

Definition 7.4 (JCF of a general matrix).
Let $A \in M_n(\mathbb{C})$, the JCF of A is a matrix of J_A s.t.

1. *J_A is a direct sum of Jordan blocks.*

2. *# of Jordan blocks of eigenvalue $\lambda = g_A(\lambda)$.*

3. *# of Jordan blocks of eigenvalue λ of index(size) at least i* $= \text{nullity}\left((A - \lambda I_k)^i\right) - \text{nullity}\left((A - \lambda I_k)^{i-1}\right)$.

Remark .
How to find the JCF of A?

1. *Find spectrum of A via $p_A(x)$.*

2. *Find $g_A(\lambda)$ to determine # of Jordan blocks.*

3. *Find $r_i = \text{nullity}\left((A - \lambda I_k)^i\right) - \text{nullity}\left((A - \lambda I_k)^{i-1}\right)$ to determine individual size of a Jordan block.*

Let $T : V \to V$ be a linear operator over a finite-dimensional complex vector space V. In the context of diagonalization, we aim to find an eigenbasis η such that $[T]_\eta^\eta$ is a diagonal matrix. Likewise, our goal now is to find a basis γ such that

$[T]_\gamma^\gamma$ is in its JCF. We first look at a part of matrix corresponding to an eigenvalue λ. Suppose there exists such a part of basis $\beta = \{\mathbf{v}_1, \mathbf{v}_2, \cdots \mathbf{v}_k\}$, and we put each elements by columns to form a matrix P, then we have:

$$[T]_B^B P = P[T]_\beta^\beta$$

$$T([\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k]) = [\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k] \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda & 1 \\ 0 & 0 & \cdots & 0 & \lambda \end{pmatrix},$$

where B is the standard basis. Comparing the columns on both sides yields the following system of equations:

$$\begin{aligned} T(\mathbf{v}_1) &= \lambda \mathbf{v}_1 & \qquad & (T - \lambda\,\mathrm{id})(\mathbf{v}_1) = \mathbf{0} \\ T(\mathbf{v}_2) &= \mathbf{v}_1 + \lambda \mathbf{v}_2 & & (T - \lambda\,\mathrm{id})(\mathbf{v}_2) = \mathbf{v}_1 \\ T(\mathbf{v}_3) &= \mathbf{v}_2 + \lambda \mathbf{v}_3 & \Longleftrightarrow & (T - \lambda\,\mathrm{id})(\mathbf{v}_3) = \mathbf{v}_2 \\ &\vdots & & \vdots \\ T(\mathbf{v}_k) &= \mathbf{v}_{k-1} + \lambda \mathbf{v}_k & & (T - \lambda\,\mathrm{id})(\mathbf{v}_k) = \mathbf{v}_{k-1} \end{aligned}$$

Observe that:

1. $\mathbf{v}_1$ is the only eigenvector among β.
2. Given $\mathbf{v}_k$, we can generate a chain of β by setting $\Big\{\mathbf{v}_k, (T - \lambda\,\mathrm{id})(\mathbf{v}_k), (T - \lambda\,\mathrm{id})(\mathbf{v}_{k-1}), \cdots (T - \lambda\,\mathrm{id})(\mathbf{v}_2)\Big\}$, or equivalently, $\Big\{\mathbf{v}_k, (T - \lambda\,\mathrm{id})(\mathbf{v}_k), (T - \lambda\,\mathrm{id})^2(\mathbf{v}_k), \cdots, (T - \lambda\,\mathrm{id})^{k-1}(\mathbf{v}_k)\Big\}$.
3. $(T - \lambda\,\mathrm{id})^k(\mathbf{v}_k) = (T - \lambda\,\mathrm{id})(\mathbf{v}_1) = \mathbf{0}$, so $\mathbf{v}_k \in \ker(T - \lambda\,\mathrm{id})^k \setminus \ker(T - \lambda\,\mathrm{id})^{k-1} (\because \mathbf{v}_1 \neq \mathbf{0})$.

These motivates the definition of Jordan chain:

Definition 7.5 (Jordan Chain).
Let $T : V \to V$ be a linear operator and $\lambda \in Spec(T)$. We say a set of non-zero vectors $\beta = \{\mathbf{v}_1, \mathbf{v}_2, \cdots \mathbf{v}_k\}$ is a Jordan chain of T corresponding to the eigenvalue λ of length k if ..., where $\mathbf{v}_k$ is the end vector and $\mathbf{v}_1$ is the initial vector.

Definition 7.6 (Generalized eigenspace).
Let $T : V \to V$ be a linear operator and λ be an eigenvalue of T. Define

$$K_\lambda \equiv \bigcup_{i \geq 1} \ker(T - \lambda)^i$$

to be the generalized eigenspace of T w.r.t eigenvalue λ. Any non-zero vector in K_λ is called a generalized eigenvector.

Theorem 7.4 .
Suppose m is the smallest positive integer such that $\ker(T-\lambda I)^m = \ker(T-\lambda I)^{m+1}$. Then

1. *we have a strict inclusions of subspaces:*
$$\{\mathbf{0}\} \subset \ker(T-\lambda I) \subset \ker(T-\lambda I)^2 \subset \cdots \subset \ker(T-\lambda I)^m.$$
2. $K_\lambda = \ker(T-\lambda I)^m$.
3. *Let $p \le m$ and $\mathbf{v} \in \ker(T-\lambda I)^p \setminus \ker(T-\lambda I)^{p-1}$.*
The set $\{\mathbf{v}, (T-\lambda I)(\mathbf{v}), \cdots, (T-\lambda I)^{p-1}(\mathbf{v})\}$ forms a Jordan chain of T w.r.t eigenvalue λ of length p.

Theorem 7.5 .
Let λ be an eigenvalue of T. If $\beta_1, \beta_2, \cdots \beta_q$ are Jordan chains of λ such that their initial vectors are linearly independent, then $\gamma_\lambda = \beta_1 \cup \beta_2 \cup \cdots \cup \beta_q$ is linearly independent.

Proof.
Let $|\gamma_\lambda| = n$, we aim to do induction on the size of union of Jordan chains, which are linearly independent. If $n = 1$, there are nothing to prove. Suppose $n < k$ is true, then for $n = k$, let $W = span(\gamma_\lambda)$, we want to show $dim(W) = k$. Consider a restrction map $T - \lambda I|_W : W \to W$ and a set $\gamma'_\lambda = \beta'_1 \cup \beta'_2 \cup \cdots \beta'_q$ obtained by removing the end vector of each Jordan chain. By induction hypothesis, γ'_λ is linearly independent. Note that γ_λ spans W, $(T-\lambda I)(\gamma_\lambda) = \gamma'_\lambda$ spans $(T-\lambda)(W)$, i.e. $im(T-\lambda|_W)$. So γ'_λ forms a basis of $im(T-\lambda I_W)$ and $\text{rank}(T-\lambda I) = k-q$. Hence

$$\underbrace{k}_{|\gamma_\lambda|} \ge dim(W) = \text{rank}(T-\lambda|_W) + \text{nullity}(T-\lambda I) \ge k-q+q = k,$$

where $\ker(T-\lambda I|_W) = \ker(T-\lambda I) \cap W$ has at least q linearly independent eigenvector (initial vectors). This forces $dim(W) = k$. ■

Theorem 7.6 .
There exists a basis γ_λ of generalized eigenspace K_λ such that γ_λ is formed by a union of Jordan chains of the eigenvalue λ.

Proof.
We prove by the induction on the dimension of K_λ. Let $n = dim(K_\lambda)$, $n = 1$, there are nothing to prove. Now suppose $n < k$ such that K_λ admits a basis formed by a union of Jordan chains, then for $n = k$, consider a restriction $T - \lambda I|_{K_\lambda} : K_\lambda \to K_\lambda$. The proof strategy goes as following:

1. $ker(T - \lambda I) \supset \{\mathbf{0}\}$, by rank-nullity, $im(T - \lambda I|_{K_\lambda}) \subset K_\lambda$.

2. Claim: Let $u = im(T - \lambda I|_{K_\lambda})$, u is the generalized eigenspace of $T|_u$ w.r.t eigenvalue λ. From this claim and induction hypothesis, there exist Jordan chains $\beta_1, \beta_2, \cdots, \beta_q$ such that their union, γ, forms a basis of u. So $dim(u) = \text{rank}(T - \lambda I|_{K_\lambda}) = |\gamma|$.

3. For each i, take the initial vector w_i from β_i. By basis extension theorem, the set $\{w_1, w_2, \cdots, w_q, u_1, u_2, \cdots, u_s\}$ forms a basis in $\ker(T - \lambda I)$. Since $dim(K_\lambda) = \text{nullity}(T - \lambda I|_{K_\lambda}) + \text{rank}(T - \lambda I|_{K_\lambda}) = \text{nullity}(T - \lambda I) + dim(u) = q + s + |\gamma|$.

4. On the other hand, $\beta_i \subseteq im(T - \lambda I|_{K_\lambda}) \subseteq im(T - \lambda I)$, for each i, there exists a pre-image v_i of the end vector of β_i. Let $\tilde{\beta_i} = \beta_i \cup \{v_i\}$ and the set $\tilde{\gamma} = \tilde{\beta_1} \cup \tilde{\beta_2} \cup \cdots \cup \tilde{\beta_q} \cup \{u_1\} \cup \{u_2\} \cup \cdots \cup \{u_s\}$. The initial vectors are linearly independent, so $\tilde{\gamma}$ is linearly independent, and $|\tilde{\gamma}| = |\gamma| + q + s$. Hence, $\tilde{\gamma}$ forms a maximal linearly independent set in K_λ and hence a basis.

■

Theorem 7.7 .
Let λ and μ be distinct eigenvalues of $T : V \to V$.

1. *K_λ is a $T-$invariant subspace.*

2. *The restriction of $T - \mu I$ to K_λ is invertible.*

Proof.
For 1, use T and I can commute. Given that $T - \mu I|_{K_\lambda}$ is well-defined by 1, we want to show this restriction is bijective. By TFAE, taking any vector $\mathbf{v} \in \text{Ker}(T - \mu I) \cap K_\lambda$, we hope $\mathbf{v} = \mathbf{0}$. Suppose it does not. Since $\mathbf{v} \in K_\lambda$, there exists a smallest integer p s.t. $(T - \lambda I)^p \mathbf{v} = \mathbf{0}$. Let $\mathbf{y} = (T - \lambda I)^{p-1}\mathbf{v}$ such that $T\mathbf{y} = \lambda\mathbf{y}$. On the other hand, $\mathbf{v} \in \text{Ker}(T - \mu I)$, we can show that

$\mathbf{y} \in \text{Ker}(T - \mu I)$, using commutativity btw T and I. Then $T\mathbf{y} = \mu\mathbf{y}$, which yields a contradiction. ∎

Theorem 7.8 .
Let a be a algebraic multiplicity of the eigenvalue λ of T, then we have $K_\lambda = \text{Ker}(T - \lambda I)^a$.

Proof.
$(\supseteq)$ is clear, for $(\subseteq)$, take $\mathbf{v} \in K_\lambda$, write $p_T(x) = (x-\lambda)^a(x-\mu_1)^{b_1}(x-\mu_2)^{b_2}\cdots(x-\mu_k)^{b_k}$, where $\{\lambda, \mu_1, \mu_2, \cdots, \mu_k\}$ are distinct. By Cayley-Hamilton, one has: $p_T(T)(\mathbf{v}) = (T-\lambda I)^a(T-\mu_1 I)^{b_1}(T-\mu_2 I)^{b_2}\cdots(T-\mu_k I)^{b_k}(\mathbf{v}) = \mathbf{O}(\mathbf{v}) = \mathbf{0}$. Consider the map $(T-\mu_1 I)^{b_1}(T-\mu_2 I)^{b_2}\cdots(T-\mu_k I)^{b_k}$ restricted to K_λ, which is invertible by theorem (7.7), then $(T-\lambda I)^a(\mathbf{v}) = \mathbf{0}$. This completes the proof. ∎

Remark .
Recall theorem (7.4), the above theorem asserts that $m \leq a$.

Theorem 7.9 .
Let λ and μ be distinct eigenvalues of T, then $K_\lambda \cap K_\mu = \{\mathbf{0}\}$.

Proof.
Take $\mathbf{v} \in K_\lambda \cap K_\mu$ and $p_T(x) = (x-\lambda)^a(x-\mu)^b q(x)$. Since $gcd\{(x-\lambda)^a, (x-\mu)^b\} = 1$, by Euclidean algorithm, there exists f, g s.t. $f(x)(x-\lambda)^a + g(x)(x-\mu)^b = 1$. Plug $x = T$ and apply the equation to $\mathbf{v}$, we must have $\mathbf{v} = \mathbf{0}$ by theorem (7.8) ∎

Theorem 7.10 (Primary Decomposition Theorem).
Let V be a finite-dimensional vector space over $\mathbb{C}$ and $T : V \to V$ be a linear operator. Let $\lambda_1, \lambda_2 \cdots, \lambda_k$ be distinct eigenvalues of T, then:

$$V = K_{\lambda_1} \oplus K_{\lambda_2} \oplus \cdots K_{\lambda_k}.$$

Proof.
Let $U = K_{\lambda_1} \oplus K_{\lambda_2} \oplus \cdots K_{\lambda_k}$. We hope $dim(U) = dim(V)$. (Direct) sum of space is a subspace, $U \subseteq V$ implies $dim(U) < dim(V)$. Next, $dim(U) = \sum_{i=1}^k dim(K_{\lambda_i}) \geq \sum_{i=1}^k \text{nullity}(T-\lambda_i I)^{a_i} \geq \text{nullity}(\prod_{i=1}^k (T-\lambda_i I)^{a_i}) = \text{nullity}(\mathbf{0}) = k$. Combine $U \subseteq V$ and $dim(U) = dim(V)$, we conclude

$U = V$. ■

Theorem 7.11 (JCF Theorem I).
Let V be a finite dimensional vector space over complex field, $T : V \to V$ be any linear operator, and the corresponding matrix representing T w.r.t standard basis, $[T]_B^B$, then there exists a basis γ of V formed by a union of Jordan chains, which is called a Jordan basis. Moreover, $[T]_B^B$ similar to its JCF, $[T]_\gamma^\gamma$.

Proof.
By primary decomposition theorem (7.10) and theorem (7.6), there exists such union of Jordan chains and forms a maximal linearly independent set and hence a basis. ■

Theorem 7.12 (JCF Theorem II & III).
Let $A, B \in M_n(\mathbf{C})$, and up to a permutation of Jordan blocks, then:

1. *J_A is uniquely determined by A.*
2. *$A \sim B$ if and only if $J_A = J_B$.*

Proof.
1 is trivial by definition. For 2, we focus on $(\Rightarrow)$. By similarity, $Spec(A) = Spec(B)$ and $(A - \lambda I)^i \sim (B - \lambda I)^i$, which implies $\text{nullity}(A - \lambda I)^i = \text{nullity}(B - \lambda I)^i$ for each integer i. Hence, they have the same JCF. ■

8 Inner Product Space

8.1 Finite Dimensional Inner Product Spaces

Definition 8.1 (Inner Product Space).
Let V be a vector space over $\mathbb{C}$ and $\langle -,- \rangle : V \times V \to \mathbb{C}$ be a function. We say the pair $(V, \langle -,- \rangle)$ is a complex inner product space if it satisfies:

1. *(Linearity in 1st component) For any $u, v, w \in V$ and $\alpha, \beta \in \mathbb{C}$,:*
$$\langle \alpha u + \beta v, w \rangle = \alpha \langle u, w \rangle + \beta \langle v, w \rangle.$$
2. *(Conjugate Symmetry) For any $u, v \in V$:*
$$\langle u, v \rangle = \overline{\langle v, u \rangle}.$$
3. *(Non-negativity) For any $u \in V$:*
$$\langle u, u \rangle \geq 0.$$
4. *(Non-degeneracy) For any $u \in V$:*
$$\|u\| = \sqrt{\langle u, u \rangle} = 0$$
iff $u = 0$.

And $\langle -,- \rangle$ is called an inner product on V.

Remark .
Dot product is the standard inner product on $\mathbb{C}^n$.

Definition 8.2 .
Let $(V, \langle -,- \rangle)$ be a complex inner product space.

1. *The norm of $x \in V$ is defined as $\|x\| = \sqrt{\langle x, x \rangle}$.*
2. *x, $y \in V$ are orthogonal if $\langle x, y \rangle = 0$.*

Theorem 8.1 .
Let $(V, \langle -,- \rangle)$ be a complex inner product space, for $u, v, w \in V$ and $\alpha, \beta \in \mathbb{C}$.

1. *Almost linear in 2nd comp: $\langle u, \alpha v + \beta w \rangle = \overline{\alpha} \langle u, v \rangle + \overline{\beta} \langle u, w \rangle$.*
2. $||\alpha u|| = |\alpha| ||u||$.
3. *$\langle u, v \rangle = 0$ for any $v \in V \Leftrightarrow u \equiv 0$.*
4. *$\langle u, w \rangle = \langle v, w \rangle$ for any $w \in V \Leftrightarrow u = v$.*

Proof.
Left as an exercise. ■

Theorem 8.2 .
Let $(V, \langle -, - \rangle)$ be a complex inner product space, for $u, v \in V$.

- *Cauchy-Schwarz:* $|\langle u, v\rangle| \leq ||u|| \cdot ||v||$.
- *Triangle:* $||u + v|| \leq ||u|| + ||v||$.

Proof.
For Cauchy-Schwarz, if $v = 0$, then it is trivial. Assume $v \neq 0$, inspired by the projection vector in $\mathbb{R}^2$, consider $z = u - \frac{\langle u,v\rangle}{\|v\|^2}v$. First show z, v are orthogonal, and by non-negativity, $\langle z, z\rangle \geq 0$. After some expansion, we have

$$\|u\|^2 - \frac{\langle u, v\rangle}{\|v\|^2}\langle v, u\rangle \geq 0,$$

rearrage and take the square root, then we are done.

For triangle inequality,

$$\begin{aligned} ||u+v||^2 &= \langle u+v, u+v\rangle = \langle u,u\rangle + \langle u,v\rangle + \langle v,u\rangle + \langle v,v\rangle \\ &= \langle u,u\rangle + 2Re(\langle u,v\rangle) + \langle v,v\rangle \\ &\leq ||u||^2 + 2|\langle u,v\rangle| + ||v||^2 \\ &\leq ||u||^2 + 2||u||||v|| + ||v||^2 = (||u|| + ||v||)^2. \end{aligned}$$

Hence $||u + v|| \leq ||u|| + ||v||$. ■

Theorem 8.3 .
Let $(V, \langle -, - \rangle)$ be a complex inner product space. For any $u, v \in V$, we have:

- ***Parallelogram Identity*** $\|u+v\|^2 + \|u-v\|^2 = 2(\|u\|^2 + \|v\|^2)$.
- ***Generalized Pythagorean Theorem*** *If $u_1, u_2, \ldots, u_n$ are mutually orthogonal (i.e., $\langle u_i, u_j\rangle = 0$ for all $i \neq j$), then*

$$\left\| \sum_{i=1}^{n} u_i \right\|^2 = \sum_{i=1}^{n} \|u_i\|^2.$$

- ***Polarization Identity***

$$\langle u, v\rangle = \frac{1}{4}\left(\|u+v\|^2 - \|u-v\|^2 + i\|u+iv\|^2 - i\|u-iv\|^2\right).$$

Proof.
Left as an exercise. ∎

Theorem 8.4 (Jordan-von Neumann).
Let V be a vector space equipped with a norm $\|\cdot\|$. This norm arises from an inner product (i.e., there exists an inner product on V such that $\|v\| = \sqrt{\langle v, v\rangle}$ for all $v \in V$) if and only if the norm satisfies the parallelogram identity:

$$\|u+v\|^2 + \|u-v\|^2 = 2(\|u\|^2 + \|v\|^2) \quad \forall u, v \in V.$$

Proof.
Beyond the scope of this note. ∎

Definition 8.3 (Orthogonal and Orthonormal Sets).
*Let $(V, \langle -, -\rangle)$ be an inner product space. A subset $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ of V is called an **orthogonal set** if every pair of distinct vectors is orthogonal, i.e.,*

$$\langle \mathbf{v}_i, \mathbf{v}_j\rangle = 0 \quad \text{for all } i \neq j.$$

*Furthermore, S is called an **orthonormal set** if it is an orthogonal set and every vector in S is a unit vector. Compactly, we can write the condition for an orthonormal set using the Kronecker delta δ_{ij}:*

$$\langle \mathbf{v}_i, \mathbf{v}_j\rangle = \delta_{ij} = \begin{cases} 1, & \text{if } i = j \\ 0, & \text{if } i \neq j \end{cases}$$

Theorem 8.5 .
Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ be an orthogonal set of non-zero vectors in an inner product space V. Then S is linearly independent.

Theorem 8.6 .
Let $(V, \langle -, -\rangle)$ be a finite-dimensional complex inner product space. If $S = \{u_1, u_2, \cdots, u_k\}$ is an orthogonal basis of V, then for any $v \in V$,

$$v = \sum_{i=1}^{k} \alpha_i u_i, \quad \text{where } \alpha_i = \frac{\langle v, u_i\rangle}{\|u_i\|^2}.$$

In particular, if S is an ONB, then $\alpha_i = \langle v, u_i\rangle$.

Theorem 8.7 .
Let $(V, \langle -, -\rangle)$ be a finite-dimensional complex inner product space. Let $\beta = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ be an orthonormal basis (ONB) for V. For any $\mathbf{v}, \mathbf{w} \in V$, let $[\mathbf{v}]_\beta$ and $[\mathbf{w}]_\beta$ be their coordinate vectors relative to β. Then the inner product of $\mathbf{v}$ and $\mathbf{w}$ is equal to the standard

dot product of their coordinate vectors:

$$\langle \mathbf{v}, \mathbf{w} \rangle_V = \langle [\mathbf{v}]_\beta, [\mathbf{w}]_\beta \rangle_{\text{dot}}$$

Theorem 8.8 (Gram-Schmidt Process).
Let $(V, \langle -, - \rangle)$ be an inner product space. Given a linearly independent set $S = \{v_1, v_2, \cdots, v_l\}$ in V, if

1. $w_1 = v_1$,
2. $w_k = v_k - \sum_{j=1}^{k-1} \frac{\langle v_k, w_j \rangle}{\|w_j\|^2} w_j$ *for* $2 \le k \le l$.

Then:

1. $S' = \{w_1, w_2, \cdots, w_k\}$ *is an orthogonal set of non-zero vectors.*
2. $\text{Span}(S) = \text{Span}(S')$.

Proof.
We shall prove by induction on $l = |S|$. For $l = 1$, there is nothing to prove. Suppose the claim is true for $l = k - 1$, then for $l = k$, we need to check:

1. $w_k \neq 0$.
2. $\langle w_k, w_i \rangle = 0$ for $1 \le i \le l$.
3. Take

   $$w_k \in \text{Span}(\{w_1, w_2, \cdots, w_{k-1}, v_k\}) = \text{Span}(\{v_1, v_2, \cdots, v_{k-1}, v_k\}),$$

   so $\text{Span}(S') \subseteq \text{Span}(S)$. Note that $\text{Span}(S)$ is given to be linearly independent and $\text{Span}(S')$ is linearly independent by orthogonality of non-zero vectors, so both sets have the same dimension. Hence $\text{Span}(S') = \text{Span}(S)$.

■

Definition 8.4 (Orthogonal Projection).
Let $(V, \langle -, - \rangle)$ be a finite-dimensional inner product space and E be a subspace of V. Let $S = \{u_1, u_2, \cdots, u_k\}$ be an ONB of E. The projection operator onto E is defined as:

$$P_E(v) = \sum_{j=1}^{k} \langle v, u_j \rangle u_j.$$

Remark .

1. *The projection operator $P_E : V \to E \subseteq V$ is a linear opeator.*
2. *P_E is an idempotent operator, i.e. $P_E^2 v = P_E v \quad \forall v \in V$.*

Theorem 8.9 .
Let $(V, \langle -, - \rangle)$ be a finite-dimensional inner product space and $P_E : E \to V$ be the projection onto a subspace E. Then:

1. *$P_E(v) \in E$.*
2. *$v - P_E(v)$ is orthogonal to every vector in E.*

Proof.
For 1, it is clear by definition.
For 2, it suffices to check $\langle v - P_E(v), u_j \rangle = 0 \quad \forall j = 1, 2, \cdots, k$. ∎

Theorem 8.10 (Minimal Distance Property).
Let $(V, \langle -, - \rangle)$ be a finite-dimensional inner product space and E be a subspace of V. Fix any $v \in V$, then:

1. *Among all vectors in E,*
$$\|P_E(v) - v\| \leq \|z - v\| \quad \forall z \in E.$$
2. *If the equality holds, then $z = P_E(v)$.*

Proof.
For 1, observe that $\langle z - P_E(v), v - P_E(v) \rangle = 0$, then we can use Pythoagorean theorem.
For 2, the equality holds means $\|z - P_E(v)\| = 0$. By non-degeneracy, $z = P_E(v)$. ∎

Definition 8.5 (Orthogonal Complement).
Let $(V, \langle -, - \rangle)$ be an inner product space and E be a subset of V. The orthogonal complement of E is defined as:

$$E^{\perp} = \{v \in V : \langle v, u \rangle\} = 0 \quad \forall u \in E.$$

Remark .

1. *E is not neccessarily a subspace.*

2. *$E^{\perp}$ is a subspace of V.*

Theorem 8.11 .
Let $(V, \langle -,- \rangle)$ be a finite-dimensional inner product space and E be a subspace of V.

1. *$v - P_E(v) \in E^{\perp}$.*

2. $\mathrm{Ker}(P_E) = E^{\perp}$.

Proof.
For 1, it is followed by $v - P_E(v)$ is orthogonal to any vector in E.
For 2, take $v \in \mathrm{Ker}(P_E)$, then $v - P_E(v) = v \in E^{\perp}$. Conversely, take $v \in E^{\perp}$, then $P_E(v) = \underbrace{v - P_E(v)}_{\in E^{\perp}} + v = P_E(v) \in E^{\perp}$.
By non-degeneracy, $P_E(v) = 0$. ■

Theorem 8.12 (Orthogonal Decomposition).
Let $(V, \langle -,- \rangle)$ be a finite-dimensional inner product space and E be a subspace of V.

1. *$V = E \oplus E^{\perp}$*

2. $\dim(V) = \dim(E) + \dim(E^{\perp})$..

Proof.
For 1, observe that $v = P_E(v) + (v - P_E(v))$ for any $v \in V$. To show E and $e = E^{\perp}$ has trivial intersection, we take $v \in E \cap E^{\perp}$, then $\langle v, v \rangle = 0$. By non-degeneracy, $v = 0$.
For 2, it follows from 1 directly. ■

Theorem 8.13 .
Let $A \in M_{m \times n}(\mathbb{R})$, $x \in \mathbb{R}^n$ $y \in \mathbb{R}^m$.

1. *$\langle Ax, y \rangle_{\mathrm{dot}} = \langle x, A^{\top} y \rangle_{\mathrm{dot}}$.*

2. $\mathrm{rank}(A^{\top} A) = \mathrm{rank}(A)$.

3. *If rank$(A) = n$, then $A^{\top} A$ is inveritble.*

Proof.
For 1 and 3, it is an easy exercise.
For 2, we first show $\ker(A^{\top} A) = \ker(A)$.

For any $v \in \ker(A^\top A)$, $\langle A^\top Av, v\rangle = 0$. By 1, $\langle Av, Av\rangle = 0$, so $Av = 0$ by degeneracy. The converse is trivial. In addition, both A and $A^\top A$ shares the same dimension of domain (The matrics have the same number of columns). By rank-nullity, $\operatorname{rank}(A^\top A) = \operatorname{rank}(A)$. ∎

Question (Best Solution Problem).
Given a system of equations $X\beta = y$ to be inconsistent, where $X \in M_{m\times n}(\mathbb{R})$, $y \in \mathbb{R}^m$. Find $\beta_0 \in \mathbb{R}^n$ that minimizes $\|X\beta - y\|$, i.e.

$$\|X\beta_0 - y\| \le \|X\beta - y\|$$

for any $\beta \in \mathbb{R}^n$.

Remark .
The norm is induced by the usual dot product on real vector space.

To solve the best solution problem, we employ the following method:

Theorem 8.14 (Least Square Methods).
Let $X \in M_{m\times n}(\mathbb{R})$, $y \in \mathbb{R}^m$. Suppose $\beta = \beta_0 \in \mathbb{R}^n$ is a minimizer of $\|y - X\beta\|$, then:

1. *β_0 satisfies $X^\top X\beta = X^\top y$;*
2. *If $\operatorname{rank}(X) = n$, then $\beta_0 = (X^\top X)^{-1}X^\top y$.*

Proof.
For 1, let $W = \operatorname{Im}(X) = \{X\beta : \beta \in \mathbb{R}^n\}$. By the minimal distance property,

$$\begin{aligned}\|P_W(y) - y\| &\le \|w - y\| \ \forall w \in W\\ \Rightarrow \|P_W(y) - y\| &\le \|X\beta - y\| \ \forall \beta \in \mathbb{R}^n.\end{aligned}$$

So it suffices to find β_0 such that $P_W(y) = X\beta_0$. This is possible since $P_W(y) \in W$. Let $P_W(y) = X\beta_0$ for some $\beta_0 \in \mathbb{R}^n$. Note that

$$\langle P_W(y) - y, w\rangle = \langle X\beta_0 - y, X\beta\rangle = \langle X^\top(X\beta_0 - y), \beta\rangle = 0 \quad \forall \beta \in \mathbb{R}^n,$$

then we have $X^\top(X\beta_0 - y) = 0$.

For 2, it is an easy exercise. ∎

Question (Minimal Solution Problem).
Given a system of equations $Ax = b$ to be infinitely many solutions, where $A \in M_{m\times n}(\mathbb{R})$, $b \in \mathbb{R}^m$. Find $u \in \mathbb{R}^n$ that s.t. $\|u\| \le \|x\|$ for all solutions x to the system.

Theorem 8.15 .
Let $A \in M_{m\times n}(\mathbb{R})$, then $\operatorname{Ker}(A) = (\operatorname{Im}(A^\top))^\perp$.

> *Proof.*
> Left as an exercise. ■

Theorem 8.16 .
Given a system of equations $Ax = b$ to be infinitely many solutions, where $A \in M_{m\times n}(\mathbb{R})$, $b \in \mathbb{R}^m$.

1. *There exists a unique minimal solution u to the system and $u \in \operatorname{Im}(A^\top)$.*
2. *Let v be a solution to $AA^\top v = b$, then $A^\top v = u$.*

> *Proof.*
> For 1, for any solution x_0 to the system, we can write a general solution $u = x_0 - v$ for some $v \in \operatorname{Ker}(A)$. It suffices to find v such that $\|x_0 - v\| \leq \|x_0 - w\|$ for all $w \in \operatorname{Ker}(A)$. Consider the projection $v = P_{\operatorname{Ker}(A)}(x_0)$. By the minimal distance property, v is unique, so $u = x_0 - v \in (\operatorname{Ker}(A))^\perp = \operatorname{Im}(A^\top)$ is also unique.
>
> For 2, suppose $AA^\top v = b$ and $Au = b$ by construction, then $A^\top v - u \in \operatorname{Ker}(A) \cap \operatorname{Im}(A^\top) = \{0\}$. Hence $A^\top v = u$. ■

Remark .
Why $AA^\top v = b$ is consistent? Note that $\operatorname{Ker}(A^\top) = \operatorname{Ker}(AA^\top)$. *Use* $\operatorname{Ker}(A) = \operatorname{Im}(A^\top)^\perp$ *and take orthogonal complement on both sides, one has:* $\operatorname{Im}(AA^\top) = \operatorname{Im}(A)$, *then we are done.*

Theorem 8.17 .
Let $A \in M_n(\mathbb{C})$ and $v, w \in \mathbb{C}^n$, then

$$\langle Av, w\rangle = \langle v, \overline{A^\top} w\rangle.$$

Definition 8.6 (Adjoint Operator).
Let $A \in M_n(\mathbb{C})$. The adjoint of A is defined by its conjugate transpose, denoted by $A^* = \overline{A^\top}$.

In general, the adjoint of a linear operator T is an operator $T^* : V \to V$ if

$$\langle Tv, w\rangle = \langle v, T^* w\rangle \ \forall\ v, w \in V$$

Theorem 8.18 .
For each linear operator, its adjoint operator exists and is unique.

Proof.
For existence, we first fix an ONB β, and define $[T^*]_\beta^\beta = \overline{([T]_\beta^\beta)^\top}$. Use theorem (8.7), we check that

$$\langle Tv, w\rangle = \langle [Tv]_\beta, [w]_\beta\rangle = \langle [v]_\beta, [T^*]_\beta^\beta [w]_\beta\rangle = \langle v, T^*w\rangle.$$

For uniqueness, we left as an exercise. ∎

Theorem 8.19 .
Let $T, U : V \to V$ be two linear operators on an inner product space V. We have:

1. $(T+U)^* = T^* + U^*$.
2. $(cT)^* = \bar{c}T^*$ *for any* $c \in \mathbb{C}$.
3. $(TU)^* = U^*T^*$.
4. $(T^*)^* = T$.

Theorem 8.20 .
Let S be an ONB of V, and S' be another basis of V. TFAE:

1. *S' is also an ONB of V.*
2. *The change of coordinate matrix U from S' to S satisfies $U^*U = U^*U = I_n$ (i.e. U is unitary).*

Definition 8.7 .
Let $A \in M_n(\mathbb{C})$,

1. *A is normal if $AA^* = A^*A$.*
2. *A is unitary if $AA^* = A^*A = I_n$ (i.e. $A^* = A^{-1}$).*
3. *A is Hermitian/self-adjoint if $A = A^*$.*

Theorem 8.21 .
For a finite-dimensional inner product space V, the set of unitary operators forms a group. That is,

$$U(V) = \{T : V \to V : T \text{ is unitary}\} \text{ is}$$

1. *closed under composition,*
2. *closed under inversion.*

Theorem 8.22 .
Let $T \in U(V)$ and $\lambda \in \mathrm{Spec}(T)$. Then $|\lambda| = 1$, i.e.

$$\lambda = e^{i\theta} = \cos\theta + i\sin\theta \ \forall \theta \in \mathbb{R}.$$

Proof.
Compute $\langle Tv, Tv\rangle = \langle v, T^*Tv\rangle = \langle v, v\rangle$ by unitarity of T. On the other hand, $\langle Tv, Tv\rangle = \langle \lambda v, \lambda v\rangle = \lambda\overline{\lambda}\,\|v\|$. Compare them and $v \neq 0$, we obtain $|\lambda| = 1$. ∎

Lemma .
Let T be a linear operator on a finite-dimensional complex inner product space $(V, \langle -, -\rangle)$. If $\langle Tv, Tw\rangle = \langle v, w\rangle$ $\forall v, w \in V$, then T is unitary.

Proof.
Suppose $\langle Tv, Tw\rangle = \langle v, w\rangle \quad \forall v, w \in V$. Then

$$\begin{aligned}&\langle v, T^*Tw\rangle = \langle v, w\rangle\\&\langle v, (T^*T - I)w\rangle = 0 \quad \forall v, w \in V.\end{aligned}$$

Since this holds for every $v \in V$,

$$\begin{aligned}&(T^*T - I)w = 0 \quad \forall w \in V\\&T^*T = I.\end{aligned}$$

∎

Theorem 8.23 .
Let $(V, \langle -, -\rangle)$ be a finite-dimensional inner product space and $T : V \to V$ be a linear operator. Then T is a unitary operator if and only if T is an isometry. That is,

$$T \in U(V) \iff \|Tv\| = \|v\| \quad \forall v \in V.$$

Proof.
$(\Rightarrow)$ Suppose T is unitary, $T^*T = I$. Then for any $v \in V$,

$$\|Tv\|^2 = \langle Tv, Tv\rangle = \langle v, T^*Tv\rangle = \langle v, v\rangle = \|v\|^2.$$

$(\Leftarrow)$ Assume

$$\|Tv\| = \|v\| \qquad \forall v \in V.$$

For any $x, y \in V$, by the polarization identity,

$$\begin{aligned}\langle x, y\rangle = \frac{1}{4}\Big(&\|x + y\|^2 - \|x - y\|^2\Big)\\ + \frac{i}{4}\Big(&\|x + iy\|^2 - \|x - iy\|^2\Big).\end{aligned}$$

Using the hypothesis,

$$\begin{aligned}\langle x, y\rangle &= \frac{1}{4}\Big(\|T(x+y)\|^2 - \|T(x-y)\|^2\Big)\\ &\quad + \frac{i}{4}\Big(\|T(x+iy)\|^2 - \|T(x-iy)\|^2\Big).\end{aligned}$$

Since T is linear,

$$\begin{aligned}\langle x, y\rangle &= \frac{1}{4}\Big(\|Tx+Ty\|^2 - \|Tx-Ty\|^2\Big)\\ &\quad + \frac{i}{4}\Big(\|Tx+iTy\|^2 - \|Tx-iTy\|^2\Big)\\ &= \langle Tx, Ty\rangle.\end{aligned}$$

Therefore

$$\langle Tx, Ty\rangle = \langle x, y\rangle \qquad \forall x, y \in V.$$

By the previous lemma, T is unitary. ■

Theorem 8.24 .
Let $A \in M_n(\mathbb{C})$. The following are equivalent.

1. $AA^* = A^*A = I_n$.
2. *The rows of A form an orthonormal basis for $\mathbb{C}^n$.*
3. *The columns of A form an orthonormal basis for $\mathbb{C}^n$.*

Proof.

Let $A = \begin{pmatrix} — & r_1 & — \\ — & r_2 & — \\ & \vdots & \\ — & r_n & — \end{pmatrix}$. Observe that:

$$AA^* = \begin{pmatrix} \langle r_1, r_1\rangle_{\text{dot}} & \langle r_1, r_2\rangle_{\text{dot}} & \cdots \\ \vdots & \ddots & \vdots \\ \cdots & \cdots & \langle r_n, r_n\rangle_{\text{dot}} \end{pmatrix}.$$

(1) $\Leftrightarrow$ (2): $AA^* = I_n \Leftrightarrow \langle r_i, r_j\rangle_{\text{dot}} = \delta_{ij} \Leftrightarrow \{r_1, r_2, \ldots, r_n\}$ forms an ONB. Similarly, replace rows by columns and AA^* by A^*A in the above argument to show (1) $\Leftrightarrow$ (3). ■

Theorem 8.25 .
If T is a Hermitian operator on V, then all eigenvalues of T are real.

Proof.
Let T be a Hermitian operator. Take any eigenvalue $\lambda \in \mathrm{Spec}(T)$, so $Tv = \lambda v$ for some $v \neq 0$. Consider:

$$\langle Tv, v\rangle = \langle \lambda v, v\rangle = \lambda\langle v, v\rangle$$

On the other hand, since T is Hermitian ($T = T^*$):

$$\langle Tv, v\rangle = \langle v, T^* v\rangle = \langle v, Tv\rangle = \langle v, \lambda v\rangle = \bar{\lambda}\langle v, v\rangle$$

Hence, $\lambda\langle v, v\rangle = \bar{\lambda}\langle v, v\rangle$. Since $v \neq 0$, we have $\langle v, v\rangle \neq 0$, which implies:

$$\lambda = \bar{\lambda} \Leftrightarrow \lambda \in \mathbb{R}.$$

■

Theorem 8.26 .
Let T be a Hermitian operator on V. Then any eigenvectors that correspond to different eigenvalues of T are orthogonal.

Proof.
WTS: If $Tu = \lambda u$ and $Tw = \mu w$, where $\lambda \neq \mu$, then $\langle u, w\rangle = 0$. Consider:

$$\langle Tu, w\rangle = \langle \lambda u, w\rangle = \lambda\langle u, w\rangle$$

Also:

$$\langle u, T^* w\rangle = \langle u, Tw\rangle = \langle u, \mu w\rangle = \bar{\mu}\langle u, w\rangle = \mu\langle u, w\rangle$$

(Since T is Hermitian, its eigenvalues are real, so $\bar{\mu} = \mu \in \mathbb{R}$).
Thus, $\lambda\langle u, w\rangle = \mu\langle u, w\rangle$, which implies:

$$(\lambda - \mu)\langle u, w\rangle = 0$$

Since $\lambda \neq \mu$, we must have $\langle u, w\rangle = 0$. ■

Proposition 2 .
Let $(V, \langle -, -\rangle)$ be a complex finite-dimensional inner product space and $T : V \to V$ be a normal operator. Then:

1. $\|T(v)\| = \|T^*(v)\| \ \forall v \in V$.

2. *If λ is an eigenvalue of T, then $\overline{\lambda}$ is an eigenvalue of T^*.*

Proof.
Fix any $v \in V$. Since T is normal, $TT^* = T^*T$,

$$\begin{aligned}\|Tv\|^2 &= \langle Tv, Tv\rangle = \langle v, T^*Tv\rangle = \langle v, TT^*v\rangle \\ &= \langle T^*v, T^*v\rangle = \|T^*v\|^2.\end{aligned}$$

Next, let $\lambda \in \mathrm{Spec}(T)$, then there exists $v \neq 0$ such that $Tv = \lambda v \Rightarrow (T - \lambda I)v = 0$. One can verify that $T - \lambda I$ is normal.
By the first part, $\|(T-\lambda I)v\| = \|(T-\lambda I)^*v\|$. Since $(T-\lambda I)v = 0$, we obtain $\|(T-\lambda I)^*v\| = 0$. By non-degeneracy, $(T^* - \overline{\lambda}I)v = 0$. Since $v \neq 0$, $\overline{\lambda} \in \mathrm{Spec}(T^*)$. ■

Theorem 8.27 .
Let T be a normal operator on V. Then any eigenvectors that correspond to different eigenvalues of T are orthogonal.

Proof.
We can apply the same argument as in the proof of theorem (8.26), combining with the previous proposition. ■

Remark .
This is the generalization of theorem (8.26).

Definition 8.8 .
Let $A \in M_n(\mathbb{R})$. Then

1. *A is called **orthogonal** if $AA^t = A^tA = I_n$ (orthogonal = real unitary).*

2. *A is called **symmetric** if $A^t = A$ (symmetric = real Hermitian).*

Remark .
For convenience, we often write

$$O(n) \overset{def}{=} \{A \in M_n(\mathbb{R}) : A^tA = AA^t = I_n\}$$

to be the set of n by n orthogonal matrices. This is also a 'group' (closed under multiplication, inversion).

Theorem 8.28 .
If $A \in M_n(\mathbb{R})$ is an orthogonal matrix, then

1. *all the eigenvalues λ of A satisfy $|\lambda| = 1$;*

2. $\det(A) = \pm 1$;

3. *the columns/rows of A form an orthonormal basis of $\mathbb{R}^n$.*

Remark .
Although A is real, it does not imply its eigenvalues are real (e.g., rotation matrix).

Proof.
Proof of 1 & 3: They follow directly from properties of unitary matrices.
Proof of 2: By def, if A is orthogonal, $AA^t = I_n$.

$$\begin{aligned} &\det(AA^t) = \det(I_n) \\ \Rightarrow &\det(A)\det(A^t) = 1 \\ \Rightarrow &\det(A) = \pm 1. \end{aligned}$$

■

Theorem 8.29 .
If $A \in M_n(\mathbb{R})$ is a symmetric matrix, then

1. *all the eigenvalues of A are real;*

2. *any eigenvectors that correspond to different eigenvalues of A are orthogonal (with respect to the real dot product).*

Proof.
Symmetric = real Hermitian. (As A is real, $A^* = \overline{A^t} = A^t = A$). These matrices enjoy all properties of Hermitian matrices. ■

8.2 Spectral Theorem

Definition 8.9 .

1. *$A \in M_n(\mathbb{C})$ is called **unitarily diagonalizable** if there exists an unitary matrix U ($U^* = U^{-1}$) such that U^*AU is a diagonal matrix.*

2. *$A \in M_n(\mathbb{R})$ is called **orthogonally diagonalizable** if there exists an orthogonal matrix P ($P^t = P^{-1}$) such that P^tAP is a diagonal matrix.*

Theorem 8.30 (Refined Schur Triangulation).
Let $T : V \to V$ be a linear operator. Then there exists an orthonormal basis β of V such that the matrix $[T]_\beta^\beta$ is an upper triangular matrix.

Proof.
By Schur Triangulation, $\exists$ a basis $\beta' = \{v_1, v_2, \ldots, v_n\}$ of V s.t. $[T]_{\beta'}^{\beta'}$ is upper triangular. Apply Gram-Schmidt and normalize, we obtain an ONB $\beta = \{u_1, u_2, \ldots, u_n\}$.
We want to show $T(u_k) \in \text{span}\{u_1, u_2, \ldots, u_k\}$ for each $k = 1, 2, \ldots, n$. We proceed by induction on k. $k = 1$ is trivial. Now assume the claim is true up to $k-1$, by Gram-Schmidt, $u_k - v_k \in \text{span}\{u_1, u_2, \ldots, u_{k-1}\}$. So $T(u_k) - T(v_k) \in \text{span}\{T(u_1), T(u_2), \ldots, T(u_{k-1})\} \subseteq \text{span}\{u_1, u_2, \ldots, u_{k-1}\}$ by induction hypothesis. Note that $T(v_k) \in \text{span}\{v_1, v_2, \ldots, v_k\} = \text{span}\{u_1, u_2, \ldots, u_k\}$. Hence, $T(u_k) \in \text{span}\{u_1, u_2, \ldots, u_k\}$ for each $k = 1, 2, \ldots, n$. ∎

Theorem 8.31 (Hermitian Spectral Theorem).
If $A \in M_n(\mathbb{C})$ is Hermitian, then A is unitarily diagonalizable.

Proof.
By refined Schur's Triangulation, $\exists U \in U(n)$ s.t. $U^*AU = D$ is upper triangular. We should observe that D is also Hermitian, and hence diagonal. ∎

Theorem 8.32 .
Let $A \in M_n(\mathbb{C})$ be an upper triangular matrix. Then A is normal if and only if A is diagonal.

Proof.
Let A be upper triangular.
$(\Leftarrow)$ Trivial.
$(\Rightarrow)$ Suppose A is normal as well, $A^*A = AA^*$. Let $A = (a_{ij})$. Compare the $(1,1)$ entries of AA^* and A^*A:

$$(AA^*)_{11} = |a_{11}|^2 + |a_{12}|^2 + \cdots + |a_{1n}|^2$$

$$(A^*A)_{11} = |a_{11}|^2$$

As A is normal, $AA^* = A^*A$, which forces:

$$|a_{12}|^2 + |a_{13}|^2 + \cdots + |a_{1n}|^2 = 0 \Rightarrow a_{1j} = 0 \text{ for } j \geq 2.$$

Repeat this argument to other diagonal entries of AA^* and A^*A. We can deduce $a_{ij} = 0$ whenever $i \neq j$. So A is diagonal. ∎

Theorem 8.33 ($\mathbb{C}$omplex Spectral Theorem).
$A \in M_n(\mathbb{C})$ is unitarily diagonalizable if and only if A is normal.

Remark .
Equivalently, in terms of 'operator language', the Complex Spectral Theorem states that $T : V \to V$ is a normal operator if and only if there exists an orthonormal basis β such that $[T]_\beta^\beta$ is diagonal.

> *Proof.*
> ($\Rightarrow$): Let A be normal. By refined Schur triang., $\exists U \in U(n)$ s.t. $U^*AU = D$ is upper triangular. Observe that D is normal, by theorem (8.32) D must be diagonal.
> ($\Leftarrow$): Exercise by direct compute. ∎

Theorem 8.34 ($\mathbb{R}$eal Spectral Theorem).
Let $A \in M_n(\mathbb{R})$. The following are equivalent.

1. *$A \in M_n(\mathbb{R})$ is symmetric. (i.e. $A^t = A$)*
2. *There exists $P \in O(n)$ such that P^tAP is diagonal (or so-called 'orthogonally diagonalizable').*

> *Proof.*
> (1) $\Rightarrow$ (2): Let A be symmetric. $A^* = \overline{A^t} = A^t = A$. So A is Hermitian (hence normal). By Complex Spectral Theorem, $\exists U \in U(n)$ such that $U^*AU = D$ is diagonal. Recall the columns of U consist of eigenvectors of A. As A is symmetric, all its eigenvalues are real. As $A \in M_n(\mathbb{R})$, the eigenvectors (solutions to $Ax = \lambda x$) can be chosen to be real. Hence, we can choose U to be a "real" matrix. Thus, U is orthogonal (real unitary), $U \in O(n)$.
>
> (2) $\Rightarrow$ (1): Suppose A is orthogonally diagonalizable. $\exists P \in O(n)$ such that $P^tAP = D$ is diagonal, where $P^t = P^{-1}$. Note that $A = PDP^t$, $A^t = (PDP^t)^t = (P^t)^tD^tP^t = PDP^t = A$ since D is diagonal. A is symmetric. ∎

Proposition 3 .
If $A \in M_n(\mathbb{C})$ is normal, then A and A^* share the same eigenvectors.

Proof.
Given A is normal, by Spectral Theorem, $\exists\, U$ unitary such that $U^*AU = D$ is diagonal. Since the columns of U forms an eigen-basis (and ONB as well) of A by taking the conjugate transpose on both sides, we have $U^*A^*U = \overline{D}$, A^* can be diagonalized by the same matrix U. Hence, the columns of U are eigenvectors of A^* as well.
Explicitly, fix every eigenvector v of A with eigenvalues λ, write $Av = \lambda v$, since $(A - \lambda I)$ is also normal, by proposition (2), $\|(A - \lambda I)v\| = \|(A - \lambda I)^* v\| = 0 \Rightarrow A^* v = \bar{\lambda} v$. ■

Theorem 8.35 (Riesz Representation (Finite-Dimensional Version)).
Let $(V, \langle -, - \rangle)$ be a finite-dimensional complex inner product space. For each linear functional $\varphi \in V^*$, there exists a unique $u \in V$ such that

$$\varphi(v) = \langle v, u \rangle \quad \text{for all } v \in V.$$

Remark .
In other words, there is a natural identification between the dual space V^* and V.

Proof.
Existence Fix $\{v_1, v_2, \ldots, v_n\}$ to be an ONB of V. Then for any $v \in V$:

$$\begin{aligned} v &= \sum_{i=1}^{n} \langle v, v_i \rangle v_i \\ \Rightarrow \varphi(v) &= \varphi\left(\sum_{i=1}^{n} \langle v, v_i \rangle v_i\right) = \sum_{i=1}^{n} \langle v, v_i \rangle \varphi(v_i) \end{aligned}$$

Let $u = \sum_{i=1}^{n} \overline{\varphi(v_i)} v_i$. Then $\langle v, u \rangle = \left\langle v, \sum_{i=1}^{n} \overline{\varphi(v_i)} v_i \right\rangle = \sum_{i=1}^{n} \varphi(v_i) \langle v, v_i \rangle = \varphi(v)$. We are done by taking u.
Uniqueness Suppose $\exists\, u_1, u_2$ s.t. for all $v \in V$:

$$\begin{aligned} \varphi(v) &= \langle v, u_1 \rangle = \langle v, u_2 \rangle \\ \Rightarrow &\langle v, u_1 - u_2 \rangle = 0 \quad \forall v \in V \end{aligned}$$

Take $v = u_1 - u_2$, by non-degeneracy, we obtain $u_1 = u_2$. ■

9 Singular Value Decomposition

9.1 SVD

Proposition 4 .
Let $A \in M_n(\mathbb{C})$. Consider the Hermitian matrix A^*A.

1. *A^*A is PSD.*

2. *There exists a PSD Hermitian matrix D such that $A^*A = D^2$.*

Proof.

1. For any $\mathbf{v} \in \mathbb{C}^n$,

$$\langle A^*A\mathbf{v}, \mathbf{v}\rangle = \langle A\mathbf{v}, A\mathbf{v}\rangle = \|A\mathbf{v}\|^2 \geq 0.$$

Thus, all eigenvalues $\lambda \geq 0$, meaning A^*A is PSD.

2. By the spectral theorem,

$$V^*A^*AV = \Lambda = \operatorname{diag}(\lambda_1, \lambda_2, \dots, \lambda_n).$$

Write

$$\Sigma = \operatorname{diag}(\mu_1, \mu_2, \dots, \mu_n), \qquad \mu_i = \sqrt{\lambda_i} \geq 0.$$

Then $\Sigma^2 = \Lambda$ and $\Sigma^* = \Sigma$. Therefore,

$$A^*A = V\Lambda V^* = V\Sigma^2V^* = (V\Sigma V^*)(V\Sigma V^*) = D^2.$$

Let

$$D = V\Sigma V^*.$$

Then D is Hermitian and PSD.

■

Theorem 9.1 (SVD: $\mathbb{C}$omplex version).
Let $A \in M_{m\times n}(\mathbb{C})$ be a matrix of rank k. There exist

1. *unitary matrices $U \in U(m)$ and $V \in U(n)$;*

2. *a diagonal matrix*

$$\Sigma = \begin{pmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_k \end{pmatrix}$$

with

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_k > 0,$$

such that A admits a factorization of the form

$$A = U \underbrace{\begin{pmatrix} \Sigma & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix}}_{\Sigma_0 \in M_{m\times n}(\mathbb{R})} V^*.$$

Proof.
The proof is constructive. Consider $A \in M_{m\times n}(\mathbb{C})$, Then $A^*A \in M_n(\mathbb{C})$ is a square matrix. Let $k = \text{rank}(A) = \text{rank}(A^*A)$ (why?), and $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_k > 0$ be the non-zero eigenvalues of A^*A.

Step 1. Diagonalize A^*A. As A^*A is Hermitian, and hence normal, by the Spectral Theorem, there exists $V \in U(n)$ such that

$$V^*(A^*A)V = \begin{pmatrix} \Sigma^2 & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix}.$$

Take $\sigma_i = \sqrt{\lambda_i}$, and write $V = (V_1 \mid V_2)$, where V_1 contains the first k columns.

Step 2. Compare the blocks of $(AV)^*(AV)$. Since

$$AV = (AV_1 \mid AV_2),$$

compute:

$$(AV)^*(AV) = \begin{pmatrix} (AV_1)^*(AV_1) & (AV_1)^*(AV_2) \\ (AV_2)^*(AV_1) & (AV_2)^*(AV_2) \end{pmatrix} = \begin{pmatrix} \Sigma^2 & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix}.$$

By comparing blocks, we have

$$(AV_1)^*(AV_1) = \Sigma^2 = \Sigma^*\Sigma$$

and

$$(AV_2)^*(AV_2) = \mathbf{O},$$

which implies $AV_2 = \mathbf{O}$ since $\ker((AV_2)^*(AV_2)) = \ker((AV_2))$.

Step 3. Construct U_1.
Let
$$U_1 = AV_1\Sigma^{-1} \in M_{m\times k}(\mathbb{C}).$$
Note that
$$U_1^* U_1 = \Sigma^{-1}V_1^* A^* AV_1\Sigma^{-1} = I_k,$$
so the columns of U_1 are orthonormal. By the Basis Extension Theorem and Gram-Schmidt, we can extend U_1 to a unitary matrix
$$U = (U_1 \mid U_2) \in U(m).$$

Step 4. Verify the decomposition.
Now compute:
$$U^*AV = \begin{pmatrix} U_1^* \\ U_2^* \end{pmatrix} A(V_1 \mid V_2) = \begin{pmatrix} U_1^*AV_1 & U_1^*AV_2 \\ U_2^*AV_1 & U_2^*AV_2 \end{pmatrix}.$$
Substitute
$$AV_1 = U_1\Sigma, \qquad AV_2 = \mathbf{O}.$$
Then
$$U^*AV = \begin{pmatrix} U_1^*U_1\Sigma & \mathbf{O} \\ U_2^*U_1\Sigma & \mathbf{O} \end{pmatrix} = \begin{pmatrix} \Sigma & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix}.$$
Thus
$$A = U\Sigma_0 V^*,$$
completing the SVD. ∎

Remark (Vector form of SVD).
Let $\{\mathbf{u}_1, \dots, \mathbf{u}_m\}$ be the columns of U and $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be the columns of V. The vector form of the SVD of A is the sum of rank-one matrices:
$$A = \sum_{i=1}^{k} \sigma_i \mathbf{u}_i \mathbf{v}_i^*.$$

Theorem 9.2 (Operator Form of SVD).
Let
$$T : W_1 \to W_2$$
be a linear map between two finite-dimensional complex inner product spaces. There exist

1. *an ONB* $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ *of* W_1 *and an ONB* $\{\mathbf{u}_1, \dots, \mathbf{u}_m\}$ *of* W_2*;*

2. *positive values*
$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_k > 0,$$

such that, where $k = \operatorname{rank}(T)$*,*

$$T(\mathbf{v}_i) = \begin{cases} \sigma_i \mathbf{u}_i, & \text{if } i \leq k, \\ \mathbf{0}, & \text{otherwise.} \end{cases}$$

Definition 9.1 (Singular values).
Let $A \in M_{m\times n}(\mathbb{C})$*. The values*

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_k > 0$$

that occur in the SVD of A *are called the singular values of* A*.*

Theorem 9.3 .
If A *admits an SVD*
$$A = U\Sigma_0 V^*,$$

then the diagonal entries of Σ*,*

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_k > 0,$$

are given by the positive square roots of the non-zero eigenvalues of the Hermitian matrix A^*A*. In particular, the values* σ_i *are uniquely determined by* A*.*

Proof.
Suppose A admits an SVD of the form

$$A = U\Sigma_0 V^*.$$

Then we compute A^*A:

$$\begin{aligned} A^*A &= \left(U \begin{pmatrix} \Sigma & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix} V^* \right)^* \left(U \begin{pmatrix} \Sigma & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix} V^* \right) \\ &= V \begin{pmatrix} \Sigma^* & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix} U^*U \begin{pmatrix} \Sigma & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix} V^* \\ &= V \begin{pmatrix} \Sigma^2 & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix} V^*. \end{aligned}$$

Since V is unitary, this means A^*A is unitarily similar to a diagonal matrix with entries σ_i^2. So σ_i^2 is an eigenvalue of A^*A, and σ_i is the positive square root of the eigenvalue of A^*A. ■

Theorem 9.4 (Polar Decomposition).
Let $A \in M_n(\mathbb{C})$. Then there exists a unitary matrix $L \in U(n)$ and a positive semi-definite (PSD) Hermitian matrix P such that

$$A = LP.$$

Moreover, if A is invertible, then the factorization is unique.

Proof.
Existence. By SVD,

$$A = U\Sigma V^*,$$

where $U, V \in U(n)$ and Σ is PSD diagonal. Hence

$$A = (UV^*)(V\Sigma V^*) = LP.$$

Here,

$$L = UV^*, \qquad P = V\Sigma V^*.$$

Then L is unitary, and P is Hermitian and PSD.

Uniqueness. Suppose there exist L_1, L_2 and P_1, P_2 such that

$$A = L_1P_1 = L_2P_2.$$

Then

$$A^*A = P_1^*L_1^*L_1P_1 = P_1^*P_1 = P_1^2$$

and

$$A^*A = P_2^*L_2^*L_2P_2 = P_2^*P_2 = P_2^2.$$

Since P_1, P_2 are Hermitian and PSD (actually PD if A is invertible), if $P_1^2 = P_2^2$, then $P_1 = P_2$ (Exercise), and hence $L_1 = L_2$. ■

Applications: Low-Rank Approximation

Theorem 9.5 (Von Neumann's Trace Inequality).
Let $A, B \in M_{m\times n}(\mathbb{R})$. Suppose $\operatorname{rank}(A) = k$ *and* $\operatorname{rank}(B) = k'$, *and let $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_k > 0$ and $\mu_1 \geq \mu_2 \geq \cdots \geq \mu_{k'} > 0$ be the singular values of A and B,*

respectively. Then:

$$|\operatorname{tr}(AB^\top)| \le \sum_{i=1}^{\min\{k,k'\}} \sigma_i \mu_i.$$

Proof.

Define $\displaystyle\sum_{i=1}^{\min\{k,k'\}} \sigma_i\mu_i = \sum_{i=1}^{N} \sigma_i\mu_i$, where $N > k, k'$.

Set $\sigma_i = 0$ if $i > k$, and $\mu_i = 0$ if $i > k'$. By SVD,

$$A = \sum_{i=1}^{N} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top, \quad B = \sum_{i=1}^{N} \sigma_i \mathbf{x}_i \mathbf{y}_i^\top,$$

where $\{\mathbf{u}_i\}_{i=1}^m$ and $\{\mathbf{x}_i\}_{i=1}^m$ are ONBs of $\mathbb{R}^m$, and $\{\mathbf{v}_i\}_{i=1}^n$ and $\{\mathbf{y}_i\}_{i=1}^n$ are ONBs of $\mathbb{R}^n$.

Define, for $\ell, s \le N$, $A_\ell = \displaystyle\sum_{i=1}^{\ell} \mathbf{u}_i \mathbf{v}_i^\top$, $B_s = \displaystyle\sum_{j=1}^{s} \mathbf{x}_j \mathbf{y}_j^\top$.

Von Neumann observes $A = \displaystyle\sum_{\ell=1}^{N} (\sigma_\ell - \sigma_{\ell+1}) A_\ell$ and $B = \displaystyle\sum_{s=1}^{N} (\mu_s - \mu_{s+1}) B_s$.

Then $\operatorname{tr}(AB^\top) = \displaystyle\sum_{\ell=1}^{N}\sum_{s=1}^{N} (\sigma_\ell - \sigma_{\ell+1})(\mu_s - \mu_{s+1}) \operatorname{tr}(A_\ell B_s^\top)$.

Next, we show that $\operatorname{tr}(A_\ell B_s^\top) \le \min\{\ell, s\}$ for any $\ell, s \le N$.

$$\begin{aligned}
|\operatorname{tr}(A_\ell B_s^\top)| &= \Big|\sum_{i=1}^{\ell}\sum_{j=1}^{s} \operatorname{tr}(\mathbf{u}_i \mathbf{v}_i^\top \mathbf{y}_j \mathbf{x}_j^\top)\Big| \\
&= \Big|\sum_{i=1}^{\ell}\sum_{j=1}^{s} \operatorname{tr}(\mathbf{v}_i^\top \mathbf{y}_j \mathbf{x}_j^\top \mathbf{u}_i)\Big| = \Big|\sum_{i=1}^{\ell}\sum_{j=1}^{s} \mathbf{v}_i^\top \mathbf{y}_j \mathbf{x}_j^\top \mathbf{u}_i\Big| \\
&= \Big|\sum_{i=1}^{\ell}\sum_{j=1}^{s} \langle \mathbf{y}_j \mathbf{x}_j^\top \mathbf{u}_i, \mathbf{v}_i\rangle\Big| = \Big|\sum_{i=1}^{\ell}\sum_{j=1}^{s} \langle \mathbf{v}_i, \mathbf{y}_j \mathbf{x}_j^\top \mathbf{u}_i\rangle\Big|.
\end{aligned}$$

Let $\mathbf{w}_i = \displaystyle\sum_{j=1}^{s} \mathbf{y}_j \mathbf{x}_j^\top \mathbf{u}_i = \sum_{j=1}^{s} \langle \mathbf{x}_j, \mathbf{u}_i\rangle \mathbf{y}_j$. Claim: $\|\mathbf{w}_i\| = 1$ for any $i \in \{1, 2, \cdots, \ell\}$. By Pythagorean theorem,

$$\begin{aligned}
\|\mathbf{w}_i\| &= \sqrt{\sum_{j=1}^{s} \left\|\langle \mathbf{x}_j, \mathbf{u}_i\rangle \mathbf{y}_j\right\|^2} \le \sqrt{\sum_{j=1}^{m} \left\|\langle \mathbf{x}_j, \mathbf{u}_i\rangle \mathbf{y}_j\right\|^2} = \sqrt{\sum_{j=1}^{m} |\langle \mathbf{x}_j, \mathbf{u}_i\rangle|^2 \left\|\mathbf{y}_j\right\|^2} \\
&= \sqrt{\sum_{j=1}^{m} |\langle \mathbf{x}_j, \mathbf{u}_i\rangle|^2} = \|\mathbf{u}_i\| = 1. \quad (\{y_j\} \text{ and } \{x_j\} \text{ are ONBs})
\end{aligned}$$

So

$$
\begin{aligned}
|\operatorname{tr}(A_\ell B_s^\top)| &= |\sum_{i=1}^{\ell}\sum_{j=1}^{s}\langle \mathbf{v}_i, \mathbf{y}_j\mathbf{x}_j^\top \mathbf{u}_i\rangle| \\
&= |\sum_{i=1}^{\ell}\langle \mathbf{v}_i, w_i\rangle| \le \sum_{i=1}^{\ell}\|\mathbf{v}_i\|\,\|\mathbf{w}_i\| = \sum_{i=1}^{\ell} 1\times 1 = \ell.
\end{aligned}
$$

By symmetry, we also have $|\operatorname{tr}(A_\ell B_s^\top)| \le s$.
Hence,

$$
\begin{aligned}
|\operatorname{tr}(AB^\top)| &= |\sum_{\ell=1}^{N}\sum_{s=1}^{N}(\sigma_\ell-\sigma_{\ell+1})(\mu_s-\mu_{s+1})\operatorname{tr}(A_\ell B_s^\top)| \\
&\le \sum_{\ell=1}^{N}\sum_{s=1}^{N}(\sigma_\ell-\sigma_{\ell+1})(\mu_s-\mu_{s+1})|\operatorname{tr}(A_\ell B_s^\top)| \\
&\le \sum_{\ell=1}^{N}\sum_{s=1}^{N}(\sigma_\ell-\sigma_{\ell+1})(\mu_s-\mu_{s+1})\min\{\ell,s\} \\
&\le \sum_{s=1}^{N}(\mu_s-\mu_{s+1})(\sum_{\ell=1}^{s}(\sigma_\ell-\sigma_{\ell+1})\ell + \sum_{\ell=s+1}^{N}(\sigma_\ell-\sigma_{\ell+1})s) \\
&\le \sum_{s=1}^{N}(\mu_s-\mu_{s+1})(\sum_{\ell=1}^{s}\sigma_\ell - s\sigma_{s+1} + s\sigma_{s+1}) \\
&\le \sum_{s=1}^{N}(\mu_s-\mu_{s+1})\sum_{\ell=1}^{s}\sigma_\ell = \sum_{s=1}^{N}\mu_s\sigma_s = \sum_{i=1}^{\min\{k,k'\}}\sigma_i\mu_i.
\end{aligned}
$$

This completes the proof. ■

Theorem 9.6 (Low-rank approximation).
Let $A \in M_{m\times n}(\mathbb{R})$ be of rank k with singular values

$$\sigma_1 \ge \sigma_2 \ge \cdots \ge \sigma_k > 0.$$

Fix a value $l < k$. For any matrix $B \in M_{m\times n}(\mathbb{R})$ of rank l, one has

$$\|A-B\|_F \ge \sqrt{\sigma_{l+1}^2+\sigma_{l+2}^2+\cdots+\sigma_k^2}.$$

The equality holds when

$$B = \sum_{i=1}^{l}\sigma_i\mathbf{u}_i\mathbf{v}_i^t,$$

which is the truncated SVD of A.

Proof.
Let $\mu_1 \geq \mu_2 \geq \cdots \mu_l > 0$ be the singular values of B.

$$\begin{aligned}
\|A - B\|_F^2 &= \operatorname{tr}((A-B)^\top(A-B)) = \operatorname{tr}((A-B)(A-B)^\top) \\
&= \operatorname{tr}(AA^\top) + \operatorname{tr}(BB^\top) - \operatorname{tr}(AB^\top) - \operatorname{tr}(BA^\top) \\
&= \sum_{i=1}^{k} \sigma_i^2 + \sum_{i=1}^{l} \mu_i^2 - 2\operatorname{tr}(AB^\top) \\
&\geq \sum_{i=1}^{k} \sigma_i^2 + \sum_{i=1}^{l} \mu_i^2 - 2\sum_{i=1}^{l} \sigma_i\mu_i \quad \text{(Von Neumann's Trace Inequality)} \\
&= \sum_{i=1}^{l} \sigma_i^2 + \sum_{i=1}^{l} \mu_i^2 - 2\sum_{i=1}^{l} \sigma_i\mu_i + \sum_{i=l+1}^{k} \sigma_i^2 \\
&= \sum_{i=1}^{l} (\sigma_i - \mu_i)^2 + \sum_{i=l+1}^{k} \sigma_i^2 \\
&\geq \sum_{i=l+1}^{k} \sigma_i^2.
\end{aligned}$$

The equality holds if $\sigma_i = \mu_i$ for $i = 1, 2, \cdots, l$. ■

9.2 Pseudoinverse

Definition 9.2 (Moore–Penrose inverse).
Let $T : V \to W$ be a linear map between two inner product spaces. Denote the inverse of

$$T|_{\operatorname{Ker}(T)^\perp} : \operatorname{Ker}(T)^\perp \to \operatorname{Im}(T)$$

simply by

$$T^{-1} : \operatorname{Im}(T) \to \operatorname{Ker}(T)^\perp.$$

The pseudoinverse, or Moore–Penrose inverse, of T is a linear map $T^\dagger : W \to V$ defined by

$$T^\dagger(w) = T^{-1}\big(P_{\operatorname{Im}(T)}(w)\big).$$

Theorem 9.7 .
Let $T : V \to W$ be a linear map between two inner product spaces. Then its pseudoinverse $T^\dagger$ satisfies:

1. *$T^\dagger T : V \to V$ coincides with $P_{\operatorname{Ker}(T)^\perp}$;*
2. *$TT^\dagger : W \to W$ coincides with $P_{\operatorname{Im}(T)}$.*

Proof.
By orthogonal decomposition, $V = \mathrm{Ker}(T) \oplus \mathrm{Ker}(T)^{\perp}$.
Let $\mathbf{v} \in V$, $\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2$, where $\mathbf{v}_1 \in \mathrm{Ker}(T)$, $\mathbf{v}_2 \in \mathrm{Ker}(T)^{\perp}$.
Thus

$$\begin{aligned} T^{\dagger}T(\mathbf{v}) &= T^{\dagger}(T(\mathbf{v}_2)) \\ &= T^{-1}\big(P_{\mathrm{Im}(T)}(T(\mathbf{v}_2))\big) = T^{-1}(T(\mathbf{v}_2)) \\ &= \mathbf{v}_2 = P_{\mathrm{Ker}(T)^{\perp}}(\mathbf{v}). \end{aligned}$$

Similarly, for $W = \mathrm{Im}(T) \oplus \mathrm{Im}(T)^{\perp}$, let $\mathbf{w} = \mathbf{w}_1 + \mathbf{w}_2$. Then

$$\begin{aligned} TT^{\dagger}(\mathbf{w}) &= T\Big(T^{-1}\big(P_{\mathrm{Im}(T)}(\mathbf{w}_1 + \mathbf{w}_2)\big)\Big) \\ &= T(T^{-1}(\mathbf{w}_1)) = \mathbf{w}_1 = P_{\mathrm{Im}(T)}(\mathbf{w}). \end{aligned}$$

∎

Definition 9.3 (Computational definition of pseudoinverse).
Let $A \in M_{m\times n}(\mathbb{C})$ have an SVD

$$A = U\begin{pmatrix} \Sigma & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix} V^*.$$

Then we define its pseudo-inverse to be

$$A^{\dagger} := V\begin{pmatrix} \Sigma^{-1} & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix} U^* \in M_{n\times m}(\mathbb{C}).$$

Theorem 9.8 (Penrose).
Let $A \in M_{m\times n}(\mathbb{C})$. Then its pseudoinverse satisfies the Penrose conditions:

1. $AA^{\dagger}A = A$.
2. $A^{\dagger}AA^{\dagger} = A^{\dagger}$.
3. $(AA^{\dagger})^* = AA^{\dagger}$.
4. $(A^{\dagger}A)^* = A^{\dagger}A$.

Proof.
Substitute the SVD forms

$$A = U\Sigma_0 V^*, \qquad A^{\dagger} = V\Sigma_0^{-1}U^*,$$

where
$$\Sigma_0 = \begin{pmatrix} \Sigma & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix}, \qquad \Sigma_0^{-1} = \begin{pmatrix} \Sigma^{-1} & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix},$$
and
$$\Sigma^{-1} = \operatorname{diag}(\sigma_1^{-1}, \dots, \sigma_k^{-1}).$$

Condition 1.

$$\begin{aligned} AA^\dagger A &= U\Sigma_0 V^* V \Sigma_0^{-1} U^* U \Sigma_0 V^* \\ &= U\Sigma_0 \Sigma_0^{-1} \Sigma_0 V^* \\ &= U\Sigma_0 V^* \\ &= A. \end{aligned}$$

Condition 2.

$$\begin{aligned} A^\dagger A A^\dagger &= V\Sigma_0^{-1} U^* U \Sigma_0 V^* V \Sigma_0^{-1} U^* \\ &= V\Sigma_0^{-1} \Sigma_0 \Sigma_0^{-1} U^* \\ &= V\Sigma_0^{-1} U^* \\ &= A^\dagger. \end{aligned}$$

Condition 3. Since

$$\begin{aligned} AA^\dagger &= U\Sigma_0 V^* V \Sigma_0^{-1} U^* \\ &= U\Sigma_0 \Sigma_0^{-1} U^* \\ &= U \begin{pmatrix} I_k & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix} U^*, \end{aligned}$$

and
$$\begin{pmatrix} I_k & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix}$$
is Hermitian, we get
$$(AA^\dagger)^* = AA^\dagger.$$

Condition 4. Since

$$\begin{aligned} A^\dagger A &= V\Sigma_0^{-1} U^* U \Sigma_0 V^* \\ &= V\Sigma_0^{-1} \Sigma_0 V^* \\ &= V \begin{pmatrix} I_k & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix} V^*, \end{aligned}$$

and

$$\begin{pmatrix} I_k & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix}$$

is Hermitian, we get

$$(A^\dagger A)^* = A^\dagger A.$$

Therefore, $A^\dagger$ satisfies all four Penrose conditions. ■

Theorem 9.9 (Penrose Uniqueness).
If $X \in M_{n\times m}(\mathbb{C})$ satisfies the four Penrose conditions, then

$$X = A^\dagger.$$

Proof.
Suppose both X_1 and X_2 satisfy the Penrose conditions. First observe the following two facts.
Observation 1. By $(AX)^* = AX$ and $XAX = X$, we have

$$X(AX)^* = XAX = X.$$

Since $(AX)^* = X^*A^*$, this gives

$$XX^*A^* = X.$$

Observation 2. By $AXA = A$ and $(XA)^* = XA$, we have

$$A(XA)^* = AXA = A.$$

Since $(XA)^* = A^*X^*$, this gives

$$AA^*X^* = A.$$

Taking adjoints gives

$$XAA^* = A^*.$$

Now

$$\begin{aligned}
X_1 &= X_1X_1^*A^* && \text{(Obs 1 for } X_1\text{)} \\
&= X_1X_1^*(AX_2A)^* && \text{(Condition 1 for } X_2\text{)} \\
&= X_1X_1^*A^*(AX_2)^* && \\
&= X_1X_1^*A^*(AX_2) && \text{(Condition 3 for } X_2\text{)}
\end{aligned}$$

$$
\begin{aligned}
&= X_1 A X_2 && \text{(Obs 1 for } X_1\text{)} \\
&= X_1 A A^* X_2^* X_2 && \text{(Obs 2 for } X_2\text{)} \\
&= A^* X_2^* X_2 && \text{(Obs 2 for } X_1\text{)} \\
&= (X_2 A)^* X_2 = X_2 A X_2 = X_2 && \text{(Condition 4 \& 2 for } X_2\text{).}
\end{aligned}
$$

■

Theorem 9.10 .
Theoretical definition and computational definition of pseudoinverse are equivalent.

Proof.
Let $T^\dagger : W \to V$ be the pseudoinverse of $T : V \to W$. Recall $T^\dagger T = P_{\ker(T)^\perp}$ and $TT^\dagger = P_{\mathrm{Im}(T)}$.
For $v \in V$, we have $TT^\dagger T(v) = P_{\mathrm{Im}(T)}(T(v)) = T(v)$. Hence

$$TT^\dagger T = T.$$

For $w \in W$, we have $T^\dagger TT^\dagger(w) = P_{\mathrm{Ker}(T)^\perp}(T^\dagger(w)) = T^\dagger(w)$,
since $T^\dagger(w) \in \ker(T)^\perp$. Hence

$$T^\dagger TT^\dagger = T^\dagger.$$

Let $\beta = \{v_1, \dots, v_k, v_{k+1}, \dots, v_n\}$ be an ONB of V, where

$$\mathrm{span}\{v_1, \dots, v_k\} = \ker(T)^\perp, \qquad \mathrm{span}\{v_{k+1}, \dots, v_n\} = \ker(T).$$

Then

$$\left[P_{\ker(T)^\perp}\right]_\beta^\beta = \begin{pmatrix} I_k & \mathbf{O} \\ \mathbf{O} & \mathbf{O}_{n-k} \end{pmatrix},$$

which is Hermitian.
Similarly, let $\gamma = \{w_1, \dots, w_k, w_{k+1}, \dots, w_m\}$ be an ONB of W, where

$$\mathrm{span}\{w_1, \dots, w_k\} = \mathrm{Im}(T), \qquad \mathrm{span}\{w_{k+1}, \dots, w_m\} = \mathrm{Im}(T)^\perp.$$

Then

$$\left[P_{\mathrm{Im}(T)}\right]_\gamma^\gamma = \begin{pmatrix} I_k & \mathbf{O} \\ \mathbf{O} & \mathbf{O}_{m-k} \end{pmatrix},$$

which is Hermitian.
Since orthogonal projections are Hermitian, we get

$$(T^\dagger T)^* = T^\dagger T, \qquad (TT^\dagger)^* = TT^\dagger.$$

Let $[T]_\beta^\gamma = A$, by Penrose conditions, $[T^\dagger]_\gamma^\beta = A^\dagger$. ∎

Solve System of Equations

Theorem 9.11 (Least Square Method).
Let $A \in M_{m\times n}(\mathbb{C})$ and fix $\mathbf{b} \in \mathbb{C}^m$. Then $\mathbf{x} = A^\dagger\mathbf{b}$ minimizes the norm $\|A\mathbf{x} - \mathbf{b}\|$. That is,

$$\|A(A^\dagger\mathbf{b}) - \mathbf{b}\| \leq \|A\mathbf{x} - \mathbf{b}\| \qquad \text{for all } \mathbf{x} \in \mathbb{C}^n.$$

Proof.
Recall that $AA^\dagger$ is the orthogonal projection onto $\mathrm{Im}(A)$. Thus

$$AA^\dagger\mathbf{b} = P_{\mathrm{Im}(A)}(\mathbf{b}).$$

By the minimal distance property of orthogonal projections, for any vector $\mathbf{y} \in \mathrm{Im}(A)$, we have

$$\|P_{\mathrm{Im}(A)}(\mathbf{b}) - \mathbf{b}\| \leq \|\mathbf{y} - \mathbf{b}\|.$$

Since any $\mathbf{y} \in \mathrm{Im}(A)$ can be written as

$$\mathbf{y} = A\mathbf{x}$$

for some $\mathbf{x} \in \mathbb{C}^n$, we get

$$\|A(A^\dagger\mathbf{b}) - \mathbf{b}\| \leq \|A\mathbf{x} - \mathbf{b}\|.$$

∎

Corollary (Linear Regression).
For the linear regression problem, finding $\mathbf{x} = (c_1, \ldots, c_n)^t$ that minimizes the sum of squared errors $\|A\mathbf{x} - \mathbf{b}\|^2$ is achieved by $\mathbf{x} = A^\dagger\mathbf{b}$.

Theorem 9.12 (Best Solution Problem).
Let $A \in M_{m\times n}(\mathbb{C})$ and fix $\mathbf{b} \in \mathbb{C}^m$. Suppose the system of equations $A\mathbf{x} = \mathbf{b}$ is

inconsistent. Then $\mathbf{x} = A^\dagger \mathbf{b}$ *gives the best approximation of a solution of minimal norm, i.e. it satisfies:*

1. $$\|A(A^\dagger \mathbf{b}) - \mathbf{b}\| \leq \|A\mathbf{x} - \mathbf{b}\| \qquad \textit{for all } \mathbf{x} \in \mathbb{C}^n.$$

   So $A^\dagger \mathbf{b}$ *is a best solution.*

2. *If* $\mathbf{z} \in \mathbb{C}^n$ *satisfies*
   $$\|A\mathbf{z} - \mathbf{b}\| = \|A(A^\dagger \mathbf{b}) - \mathbf{b}\|,$$
   then
   $$\|A^\dagger \mathbf{b}\| \leq \|\mathbf{z}\|.$$
   So $A^\dagger \mathbf{b}$ *has the minimal norm among all the best solutions.*

Proof.
(1) has been proved in theorem (9.11). For (2), suppose $\mathbf{z}$ is another best solution, $\|A\mathbf{z} - \mathbf{b}\| = \|A(A^\dagger \mathbf{b}) - \mathbf{b}\|$. Recall that if E is a subspace and $\mathbf{v} \in V$, then

$$\|P_E(\mathbf{v}) - \mathbf{v}\| \leq \|\mathbf{z} - \mathbf{v}\| \quad \forall \mathbf{z} \in E.$$

Indeed,

$$\mathbf{z} - \mathbf{v} = \underbrace{(P_E(\mathbf{v}) - \mathbf{v})}_{\in E^\perp} + \underbrace{(\mathbf{z} - P_E(\mathbf{v}))}_{\in E}.$$

Hence, by the Pythagorean theorem,

$$\|\mathbf{z} - \mathbf{v}\|^2 = \|P_E(\mathbf{v}) - \mathbf{v}\|^2 + \|\mathbf{z} - P_E(\mathbf{v})\|^2 \geq \|P_E(\mathbf{v}) - \mathbf{v}\|^2.$$

The equality holds when $\mathbf{z} = P_E(\mathbf{v})$.
Now,

$$AA^\dagger \mathbf{b} = P_{\mathrm{Im}(A)} \mathbf{b} \in \mathrm{Im}(A), \qquad A\mathbf{z} \in \mathrm{Im}(A).$$

Also,

$$\|A\mathbf{z} - \mathbf{b}\|^2 = \|P_{\mathrm{Im}(A)}\mathbf{b} - A\mathbf{z}\|^2 + \|P_{\mathrm{Im}(A)}\mathbf{b} - \mathbf{b}\|^2.$$

Since $\mathbf{z}$ is another best solution, $\|A\mathbf{z} - \mathbf{b}\| = \|A(A^\dagger \mathbf{b}) - \mathbf{b}\|$,

$$A\mathbf{z} = P_{\mathrm{Im}(A)}\mathbf{b} = A(A^\dagger \mathbf{b}).$$

Multiply by $A^\dagger$ and by the Penrose conditions,

$$\underbrace{A^\dagger A\mathbf{z}}_{P_{\ker(A)^\perp}(\mathbf{z})} = A^\dagger A(A^\dagger \mathbf{b}) = A^\dagger \mathbf{b}.$$

Thus $A^\dagger \mathbf{b}$ is the projection of $\mathbf{z}$ onto $\ker(A)^\perp$, we have $\|A^\dagger \mathbf{b}\| \leq \|\mathbf{z}\|$. ■

Theorem 9.13 (Minimal Solution Problem).
Suppose the system $A\mathbf{x} = \mathbf{b}$ is consistent. $\mathbf{x} = A^{\dagger}\mathbf{b}$ gives the solution of minimal norm. If $\mathbf{z} \in \mathbb{C}^n$ is any other solution, then

$$\|A^{\dagger}\mathbf{b}\| \leq \|\mathbf{z}\|.$$

Proof.
Given $A\mathbf{x} = \mathbf{b}$ has a solution, we have $\mathbf{b} \in \mathrm{Im}(A)$. Then $A(A^{\dagger}\mathbf{b}) = P_{\mathrm{Im}(A)}(\mathbf{b}) = \mathbf{b}$, so $\mathbf{x} = A^{\dagger}\mathbf{b}$ is a solution.
Suppose $\mathbf{z}$ satisfies $A\mathbf{z} = \mathbf{b}$. Multiply by $A^{\dagger}$: $A^{\dagger}A\mathbf{z} = A^{\dagger}\mathbf{b}$. Since $A^{\dagger}A = P_{\mathrm{Ker}(A)^{\perp}}$, this means $P_{\mathrm{Ker}(A)^{\perp}}(\mathbf{z}) = A^{\dagger}\mathbf{b}$. Because the projection of $\mathbf{z}$ is $A^{\dagger}\mathbf{b}$, its norm must be less than or equal to the norm of the original vector $\mathbf{z}$. Therefore,

$$\|A^{\dagger}\mathbf{b}\| \leq \|\mathbf{z}\|.$$

■

Remark .
The power of pseudoinverse is that the method works in 'rank-deficient' setting.

9.3 Conclusion: SVD knows everything

Theorem 9.14 .
Let $A \in M_{m\times n}(\mathbb{C})$, and suppose that a singular value decomposition of A is given by

$$A = U \begin{pmatrix} \Sigma & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix} V^*$$

with

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_k > 0.$$

Let

$$\{\mathbf{u}_1, \ldots, \mathbf{u}_m\}$$

be the columns of U and

$$\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$$

be the columns of V. Then:

(a) $\mathrm{rank}(A) = k$, $\mathrm{nullity}(A) = n - k$.

(b) A basis of $\mathrm{Ker}(A)$ *is given by* $\{\mathbf{v}_{k+1}, \ldots, \mathbf{v}_n\}$.

(c) A basis of $\mathrm{Im}(A)$ *is given by* $\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$.

(d) *A basis of* $\mathrm{Ker}(A^*)$ *is given by* $\{\mathbf{u}_{k+1}, \ldots, \mathbf{u}_m\}$.

(e) *A basis of* $\mathrm{Im}(A^*)$ *is given by* $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$.

(f) *The vector form of the SVD of A is*

$$A = \sum_{i=1}^{k} \sigma_i \mathbf{u}_i \mathbf{v}_i^*.$$

(g) *The pseudoinverse of A is*

$$A^\dagger = V \begin{pmatrix} \Sigma^{-1} & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix} U^*.$$

(h) *The vector form of the pseudoinverse is*

$$A^\dagger = \sum_{i=1}^{k} \sigma_i^{-1} \mathbf{v}_i \mathbf{u}_i^*.$$

(i) *The orthogonal projection onto* $\mathrm{Im}(A)$ *is*

$$AA^\dagger = \sum_{i=1}^{k} \mathbf{u}_i \mathbf{u}_i^*.$$

(j) *The orthogonal projection onto* $\mathrm{Im}(A^*)$ *is*

$$A^\dagger A = \sum_{i=1}^{k} \mathbf{v}_i \mathbf{v}_i^*.$$

(k) *The orthogonal projection onto* $\mathrm{Ker}(A)$ *is*

$$I_n - A^\dagger A = \sum_{i=k+1}^{n} \mathbf{v}_i \mathbf{v}_i^*.$$

(l) *The orthogonal projection onto* $\mathrm{Ker}(A^*)$ *is*

$$I_m - AA^\dagger = \sum_{i=k+1}^{m} \mathbf{u}_i \mathbf{u}_i^*.$$

(m) *For every* $\mathbf{x} \in \mathbb{C}^n$*, one has*

$$\|A\mathbf{x}\|^2 = \sum_{i=1}^{k} \sigma_i^2 \left| \langle \mathbf{x}, \mathbf{v}_i \rangle \right|^2.$$

(n) *The maximum value of* $\|A\mathbf{x}\|$ *subject to* $\mathbf{x} \in \mathbb{C}^n$ *with* $\|\mathbf{x}\| = 1$ *is* σ_1.

(o) *For each integer l with $1 \leq l \leq k$, the best approximation to A of rank l with respect to the Frobenius norm is*

$$A_l = \sum_{i=1}^{l} \sigma_i \mathbf{u}_i \mathbf{v}_i^*.$$

Moreover,

$$\|A - A_l\|_F = (\sigma_{l+1}^2 + \cdots + \sigma_k^2)^{1/2}.$$

(p) *If A is square and invertible, then $A^\dagger = A^{-1}$.*

(q) *If $A\mathbf{x} = \mathbf{b}$ is consistent, then $A^\dagger \mathbf{b}$ is the unique solution of minimal norm among all solutions.*

(r) *If $A\mathbf{x} = \mathbf{b}$ is inconsistent, then $A^\dagger \mathbf{b}$ is a least-squares solution, and indeed the unique least-squares solution of minimal norm.*

Proof.
Common facts. Since

$$A = U\Sigma_0 V^*$$

and U, V are invertible, we have

$$\operatorname{rank}(A) = \operatorname{rank}(\Sigma_0) = k.$$

Hence, by the rank-nullity theorem,

$$\operatorname{nullity}(A) = n - k.$$

Moreover, from

$$AV = U\Sigma_0,$$

we get

$$A\mathbf{v}_i = \begin{cases} \sigma_i \mathbf{u}_i, & i \leq k, \\ \mathbf{0}, & i > k. \end{cases}$$

Thus

$$\operatorname{Ker}(A) = \operatorname{span}\{\mathbf{v}_{k+1}, \ldots, \mathbf{v}_n\}, \qquad \operatorname{Im}(A) = \operatorname{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}.$$

Similarly, taking adjoints gives

$$A^* = V\Sigma_0^* U^*, \qquad A^* U = V\Sigma_0^*.$$

So

$$A^*\mathbf{u}_i = \begin{cases} \sigma_i \mathbf{v}_i, & i \le k, \\ \mathbf{0}, & i > k. \end{cases}$$

Therefore

$$\operatorname{Ker}(A^*) = \operatorname{span}\{\mathbf{u}_{k+1}, \dots, \mathbf{u}_m\}, \qquad \operatorname{Im}(A^*) = \operatorname{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}.$$

(i) Since $P_{\operatorname{Im}(A)} = AA^\dagger$, we have

$$\begin{aligned} P_{\operatorname{Im}(A)} &= AA^\dagger \\ &= \left(\sum_{i=1}^k \sigma_i \mathbf{u}_i \mathbf{v}_i^*\right)\left(\sum_{j=1}^k \sigma_j^{-1} \mathbf{v}_j \mathbf{u}_j^*\right) \\ &= \sum_{i=1}^k \mathbf{u}_i \mathbf{u}_i^*. \end{aligned}$$

(j) Since $P_{\operatorname{Im}(A^*)} = P_{\operatorname{Ker}(A)^\perp} = A^\dagger A$, we have

$$P_{\operatorname{Im}(A^*)} = A^\dagger A = \sum_{i=1}^k \mathbf{v}_i \mathbf{v}_i^*.$$

(k) For $\mathbf{v} \in V$, write

$$\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2, \qquad \mathbf{v}_1 \in \operatorname{Ker}(A), \quad \mathbf{v}_2 \in \operatorname{Ker}(A)^\perp.$$

Then

$$(I_n - A^\dagger A)\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2 - \mathbf{v}_2 = \mathbf{v}_1 = P_{\operatorname{Ker}(A)}(\mathbf{v}).$$

Hence

$$I_n - A^\dagger A = \sum_{i=1}^n \mathbf{v}_i \mathbf{v}_i^* - \sum_{i=1}^k \mathbf{v}_i \mathbf{v}_i^* = \sum_{i=k+1}^n \mathbf{v}_i \mathbf{v}_i^*.$$

(m) For $\mathbf{x} \in \mathbb{C}^n$,

$$\begin{aligned} A\mathbf{x} &= \left(\sum_{i=1}^k \sigma_i \mathbf{u}_i \mathbf{v}_i^*\right)\mathbf{x} \\ &= \sum_{i=1}^k \sigma_i (\mathbf{v}_i^* \mathbf{x}) \mathbf{u}_i \\ &= \sum_{i=1}^k \sigma_i \langle \mathbf{x}, \mathbf{v}_i \rangle \mathbf{u}_i. \end{aligned}$$

Since $\{\mathbf{u}_i\}$ is an ONB,

$$\|A\mathbf{x}\|^2 = \sum_{i=1}^k \sigma_i^2 \left|\langle \mathbf{x}, \mathbf{v}_i \rangle\right|^2.$$

(n) Suppose $\|\mathbf{x}\| = 1$. Since

$$\mathbf{x} = \sum_{i=1}^{n} \langle \mathbf{x}, \mathbf{v}_i \rangle \mathbf{v}_i, \qquad \|\mathbf{x}\|^2 = \sum_{i=1}^{n} |\langle \mathbf{x}, \mathbf{v}_i \rangle|^2 = 1,$$

we have

$$\begin{aligned} \|A\mathbf{x}\|^2 &= \sum_{i=1}^{k} \sigma_i^2 |\langle \mathbf{x}, \mathbf{v}_i \rangle|^2 \\ &\leq \sigma_1^2 \sum_{i=1}^{k} |\langle \mathbf{x}, \mathbf{v}_i \rangle|^2 \\ &\leq \sigma_1^2 \sum_{i=1}^{n} |\langle \mathbf{x}, \mathbf{v}_i \rangle|^2 \\ &= \sigma_1^2. \end{aligned}$$

Thus

$$\|A\mathbf{x}\| \leq \sigma_1.$$

Equality is attained when $\mathbf{x} = \mathbf{v}_1$, since

$$A\mathbf{v}_1 = \sigma_1 \mathbf{u}_1 \quad \Longrightarrow \quad \|A\mathbf{v}_1\|^2 = \sigma_1^2 \|\mathbf{u}_1\|^2 = \sigma_1^2.$$

Parts (f), (g), (h), (l), (o), (p), (q), and (r) were proved earlier or follow by similar arguments, so they are left as exercises. ∎

Remark .

For $A \in M_{m\times n}(\mathbb{R})$, A can be regarded as a bounded linear map from V to W. The operator norm is

$$\|A\|_{op} = \sup_{\|\mathbf{x}\|=1} \|A\mathbf{x}\| = \sigma_1.$$

10 Bilinear and Quadratic Forms

In this section, we assume any vector space V is finite-dimensional over $\mathbb{R}$.

10.1 Bilinear and Quadratic Forms

Definition 10.1 (Bilinear Form).
We say a function $\tau : V \times V \to \mathbb{R}$ is bilinear if it is linear in each component fixing the other one. i.e.(Exercise)

Remark .
Real inner product space is a special case of bilinear form with additonal assumptions (exercise).

Definition 10.2 .
Let $\beta = \{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ be a basis of V and $\tau : V \times V \to \mathbb{R}$ be a bilinear form. The matrix representing a bilinear form w.r.t β is given by a square matrix $\psi_\beta(\tau) = A = (A_{ij})$ by setting $A_{ij} = \tau(v_i, v_j)$.

Theorem 10.1 .
Let β be a basis of V and $\tau : V \times V \to \mathbb{R}$ be a bilinear form, for any $\mathbf{v}, \mathbf{w} \in V$, one has:

$$\tau(\mathbf{v}, \mathbf{w}) = [\mathbf{v}]_\beta^\top \psi_\beta(\tau)[\mathbf{w}]_\beta.$$

> *Proof.*
> It suffices to verify the above equality for any two basis vectors. ■

Proposition 5 .
Let $\tau : V \times V \to \mathbb{R}$ be a non-zero bilinear form, TFAE:

1. $\operatorname{rank}(\psi_\beta(\tau)) = 1$ *for any basis β of V.*
2. *There exist two linear functionals $f, g \in V^*$ such that $\tau(\mathbf{v}, \mathbf{w}) = f(\mathbf{v})g(\mathbf{w})$.*

Corollary .
Product of two linear functionals is a bilinear form of at most rank one.

Remark .
The converse is not true. Consider the real dot product and use (5) to disprove it.

Theorem 10.2 (Congruence).
Let β_1, β_2 be two bases of V and $\tau : V \times V \to \mathbb{R}$ be a bilinear form. Set $A = \psi_{\beta_1}(\tau)$ and $B = \psi_{\beta_2}(\tau)$, then there exists an invertible matrix P such that $A = P^\top BP$.

Proof.
(Sketch) Recall P is a change of coordinate matrix from β_1 to β_2, then for any $\mathbf{v} \in V$, $P[\mathbf{v}]_{\beta_1} = [\mathbf{v}]_{\beta_2}$. Use theorem (10.1) to deduce. ■

Definition 10.3 .
Two matrices $A, B \in M_n(\mathbb{R})$ are congruent if there exists an invertible matrix P such that
$$A = P^\top BP.$$

Remark .

1. *For linear operators, we say two matrices A and B are similar if...*
2. *For the congruence of bilinear forms, $P^\top$ is not necessarily P^{-1}.*

Definition 10.4 .
A bilinear form is called

1. *symmetric if $\tau(\mathbf{v}_1, \mathbf{v}_2) = \tau(\mathbf{v}_2, \mathbf{v}_1)$ for any $\mathbf{v}_1, \mathbf{v}_2 \in V$.*
2. *anti-symmetric if $\tau(\mathbf{v}_1, \mathbf{v}_2) = -\tau(\mathbf{v}_2, \mathbf{v}_1)$ for any $\mathbf{v}_1, \mathbf{v}_2 \in V$.*

Theorem 10.3 .
Fix a basis β of V

1. *τ is symmetric iff $\psi_\beta(\tau)$ is a symmetric matrix.*
2. *τ is anti-symmetric iff $\psi_\beta(\tau)$ is an anti-symmetric matrix.*

Proof.
Omit. ■

Theorem 10.4 .
Any bilinear form τ can be uniquely written as a sum $\tau_1 + \tau_2$, where τ_1 is symmetric and τ_2 is anti-symmetric.

Proof.
First observe that any bilinear form

$$\tau(\mathbf{v},\mathbf{w}) = \underbrace{\frac{\tau(\mathbf{v},\mathbf{w})+\tau(\mathbf{w},\mathbf{v})}{2}}_{\text{Symmetric}} + \underbrace{\frac{\tau(\mathbf{v},\mathbf{w})-\tau(\mathbf{w},\mathbf{v})}{2}}_{\text{Anti-symmetric}}.$$

For the uniqueness, suppose $\tau = \tau_1 + \tau_2 = \tau_1' + \tau_2'$, then $\tau_1 - \tau_1' = \tau_2' - \tau_2 \equiv \delta$. Note that δ is both symmetric and anti-symmetric, this forces $\delta \equiv 0$. ■

Definition 10.5 (Quadratic form).
A quadratic form $q : V \to \mathbb{R}$ is a function of the form

$$q(\mathbf{v}) = \tau(\mathbf{v},\mathbf{v})$$

for some bilinear form $\tau : V \times V \to \mathbb{R}$.

Theorem 10.5 .
There is a one-to-one correspondence between:

1. *Quadratic forms $q : V \to \mathbb{R}$.*
2. *Symmetric bilinear forms $\tau : V \times V \to \mathbb{R}$.*
3. *Symmetric matrices $A \in \mathrm{Sym}_n(\mathbb{R})$.*

Proof.
We know the bijection between symmetric bilinear forms and symmetric matrices via $\psi_\beta(\tau)$.
To show the bijection between (1) and (2), given a symmetric bilinear form τ, $q(\mathbf{v}) = \tau(\mathbf{v},\mathbf{v})$ is clearly a quadratic form.
Conversely, given any quadratic form q, we have $q(\mathbf{v}) = \tau(\mathbf{v},\mathbf{v})$ for some bilinear form τ. Note that $\tau(\mathbf{v},\mathbf{v}) = \tau_1(\mathbf{v},\mathbf{v}) + \underbrace{\tau_2(\mathbf{v},\mathbf{v})}_{0}$, where τ_1 is symmetric and τ_2 is anti-symmetric. Via the Polarization identity, we can recover τ_1 from q:

$$\tau_1(\mathbf{v},\mathbf{w}) = \frac{q(\mathbf{v}+\mathbf{w}) - q(\mathbf{v}) - q(\mathbf{w})}{2}.$$

This identity establishes a unique symmetric bilinear form for every quadratic form. ■

Remark .
If $V = \mathbb{R}^n$, the quadratic form coincides with a homogeneous polynomial of degree 2:

$$q(\mathbf{x}) = \mathbf{x}^\top A\mathbf{x} = \sum_{i=1}^{n} \alpha_{ii} x_i^2 + \sum_{i<j} 2\alpha_{ij} x_i x_j,$$

where $A \in \mathrm{Sym}_n(\mathbb{R})$.

Theorem 10.6 (Diagonalization of symmetric bilinear forms).
Let $(V, \langle -, - \rangle)$ be a finite-dimensional real inner product space. Then every symmetric bilinear form τ is diagonalizable.

Proof.
Fix $\mathbf{v} \in V$. Consider the linear functional $\tau : V \to \mathbb{R}$ defined by $\mathbf{w} \mapsto \tau(\mathbf{v}, \mathbf{w})\ \forall\, \mathbf{w} \in V$. By the Riesz Representation Theorem, there exists a unique vector $T(\mathbf{v}) \in V$ such that $\tau(\mathbf{v}, \mathbf{w}) = \langle T(\mathbf{v}), \mathbf{w} \rangle$ for all $\mathbf{w} \in V$.
This defines an operator $T : V \to V$. Because τ is bilinear, T is linear. Since τ is symmetric,

$$\langle T\mathbf{v}, \mathbf{w} \rangle = \tau(\mathbf{v}, \mathbf{w}) = \tau(\mathbf{w}, \mathbf{v}) = \langle T\mathbf{w}, \mathbf{v} \rangle = \langle \mathbf{v}, T\mathbf{w} \rangle,$$

so T is self-adjoint (Hermitian).
By the Real Spectral Theorem, there exists an ONB $\beta = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ of eigenvectors of T such that

$$\tau(\mathbf{v}_i, \mathbf{v}_j) = \left\langle T\mathbf{v}_i, \mathbf{v}_j \right\rangle = \left\langle \lambda_i \mathbf{v}_i, \mathbf{v}_j \right\rangle = \lambda_i \delta_{ij}.$$

Hence $\psi_\beta(\tau)$ is a diagonal matrix. ∎

Remark .
Spectral theorem gives an ONB that helps to diagonalize τ, but it does not rule out the possibility that there could be any other choice of basis. Indeed, the diagonalization of a bilinear form is not unique.

Theorem 10.7 (Diagonalization of Quadratic Forms).
Let $(V, \langle -, - \rangle)$ be an inner product space and $q : V \to \mathbb{R}$ be a quadratic form. Then there exists a basis $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ such that

$$q(\mathbf{v}) = \sum_{i=1}^{n} \alpha_i x_i^2,$$

where

$$\mathbf{v} = \sum_{i=1}^{n} x_i \mathbf{v}_i \qquad \text{and} \qquad \alpha_i \in \mathbb{R}.$$

Proof.
By the diagonalization of symmetric bilinear forms, there is a basis $\beta = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ such that $\tau(\mathbf{v}_i, \mathbf{v}_j) = \alpha_i \delta_{ij}$. Then

$$\begin{aligned} q(\mathbf{v}) &= \tau\left(\sum x_i \mathbf{v}_i, \sum x_j \mathbf{v}_j\right) \\ &= \sum_{i=1}^{n} x_i^2 \tau(\mathbf{v}_i, \mathbf{v}_i) = \sum_{i=1}^{n} \alpha_i x_i^2. \end{aligned}$$

■

Remark .
Diagonaliability of quadratic forms confirms that completing the square of any quadratic form is not unique.

Theorem 10.8 (Diagonalization of Symmetric Matrix).
Let $A \in \mathrm{Sym}_n(\mathbb{R})$. Then A is congruent to a diagonal matrix.

Proof.
Direct result from the definition (10.3) and theorem (10.6) ■

Definition 10.6 .
Let $A \in \mathrm{Sym}_n(\mathbb{R})$, and D be a diagonal matrix congruent to A. Let t be the number of positive diagonal entries of D, and m be the number of negative diagonal entries of D. Then

1. *t is called the* ***index*** *of A,*
2. *$t - m$ is called the* ***signature*** *of A.*

Theorem 10.9 (Sylvester's Law of Inertia).
Let $A \in \mathrm{Sym}_n(\mathbb{R})$. Suppose A is congruent to two diagonal matrices D_1 and D_2. Then D_1 and D_2 share the same index and signature.

Proof.
Let $\beta_1 = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$, $\beta_2 = \{\mathbf{v}'_1, \ldots, \mathbf{v}'_n\}$ such that

$$\psi_{\beta_1}(\tau) = D_1 = \mathrm{diag}(\alpha_1, \ldots, \alpha_n)$$

and

$$\psi_{\beta_2}(\tau) = D_2 = \mathrm{diag}(\alpha'_1, \ldots, \alpha'_n).$$

Define t_1, m_1 as the number of positive/negative entries in D_1, and t_2, m_2 for D_2.

Note that

$$\operatorname{rank}(D_1) = t_1 + m_1 = \operatorname{rank}(A) = t_2 + m_2 = \operatorname{rank}(D_2).$$

It suffices to show $t_1 = t_2$.
Suppose, WLOG, $t_1 > t_2$. Assume the first t_1 and t_2 elements are positive.
Consider $V_1 = \operatorname{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_{t_1}\}$ and $V_2' = \operatorname{span}\{\mathbf{v}'_{t_2+1}, \ldots, \mathbf{v}'_n\}$.
Claim. $V_1 \cap V_2' = \{\mathbf{0}\}$. Suppose not. Take $\mathbf{v} \in V_1 \cap V_2'$ such that $\mathbf{v} \neq \mathbf{0}$. Since $\mathbf{v} \in V_1$,

$$q(\mathbf{v}) = \sum_{i=1}^{t_1} \alpha_i x_i^2 > 0,$$

because $\alpha_i > 0$ and $\mathbf{v} \neq \mathbf{0}$. Since $\mathbf{v} \in V_2'$,

$$q(\mathbf{v}) = \sum_{i=t_2+1}^{n} \alpha_i' y_i^2 \leq 0,$$

because $\alpha_i' \leq 0$. This is absurd.
On the other hand, the dimension formula gives

$$\begin{aligned}
\dim(V_1 \cap V_2') &= \dim(V_1) + \dim(V_2') - \dim(V_1 + V_2') \\
&= t_1 + (n - t_2) - \dim(V_1 + V_2') \\
&\geq t_1 + n - t_2 - n \\
&> 0.
\end{aligned}$$

This contradicts the claim. So we must have $t_1 = t_2$. ∎

Theorem 10.10 .
Let $A \in \operatorname{Sym}_n(\mathbb{R})$ and P be any invertible matrix. The number of positive eigenvalues of A and $P^\top AP$ are equal.

Proof.
By the spectral theorem, there exists an orthogonal matrix Q such that

$$Q^\top AQ = D_1 = \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$$

Thus A is congruent to D_1. The number of positive eigenvalues is the index t. Let $C = P^\top AP$, then $C^\top = C$ is symmetric. By the spectral theorem, there exists an orthogonal matrix $\widetilde{Q}$ such that

$$\widetilde{Q}^\top C\widetilde{Q} = D_2 = \operatorname{diag}(\mu_1, \ldots, \mu_n).$$

So C is congruent to D_2. But C is congruent to A, which means A is congruent to D_2. By Sylvester's Law of Inertia, D_1 and D_2 must have the same index. Thus, the number of positive eigenvalues is preserved. ■

Corollary .
Let $A \in \mathrm{Sym}_n(\mathbb{R})$ and P be any invertible matrix. The number of negative eigenvalues of A and $P^\top AP$ are equal.

Corollary .

1. *$A \in \mathrm{Sym}_n(\mathbb{R})$ is PD iff the index of A is n.*
2. *$A \in \mathrm{Sym}_n(\mathbb{R})$ is ND iff the signature of A is $-n$.*

Theorem 10.11 .
Let $A, B \in \mathrm{Sym}_n(\mathbb{R})$. TFAE:

(1) A and B are congruent.

(2) A and B share the same index and signature.

Proof.
(1) $\implies$ (2). Suppose $A = P^\top BP$. By the previous theorem and corollary, A and B have the same number of positive and negative eigenvalues, so they share the same index and signature.

(2) $\implies$ (1). Suppose

$$A \sim D_1 = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$$

and

$$B \sim D_2 = \mathrm{diag}(\mu_1, \dots, \mu_n).$$

Since they have the same index t and negative eigenvalues m, we can assume the first t entries are positive, the next m are negative, and the rest are zero for both D_1 and D_2.
Consider the diagonal matrix

$$Q = \mathrm{diag}\left(\sqrt{\frac{\mu_1}{\lambda_1}}, \dots, \sqrt{\frac{\mu_t}{\lambda_t}}, \sqrt{\frac{\mu_{t+1}}{\lambda_{t+1}}}, \dots, 1, \dots\right).$$

It is clear that Q is invertible and

$$D_2 = Q^\top D_1 Q.$$

Write

$$D_1 = P^\top AP, \qquad D_2 = \widetilde{P}^\top B\widetilde{P}.$$

Then

$$\widetilde{P}^\top B\widetilde{P} = Q^\top (P^\top AP)Q.$$

Solving for B,

$$\begin{aligned} B &= (\widetilde{P}^\top)^{-1} Q^\top P^\top APQ\widetilde{P}^{-1} \\ &= (PQ\widetilde{P}^{-1})^\top A(PQ\widetilde{P}^{-1}). \end{aligned}$$

Thus, A and B are congruent. ■

Corollary .
If A is PD/ND/PSD/NSD, then any matrix congruent to A is also PD/ND/PSD/NSD.

Proof.
For PD/ND case, we can prove them with an alternative method.
Let B be congruent to A, so there is an invertible P such that $B = P^\top AP$.
Let $q_A(\mathbf{v}) = \mathbf{v}^\top A\mathbf{v}$, for B, $q_B(\mathbf{v}) = \mathbf{v}^\top (P^\top AP)\mathbf{v} = (P\mathbf{v})^\top A(P\mathbf{v})$.
Let $\mathbf{w} = P\mathbf{v}$, since A is PD, $q_A(\mathbf{w}) > 0$ for all $\mathbf{w} \neq \mathbf{0}$. Since P is invertible, $q_B(\mathbf{v}) = q_A(\mathbf{w}) > 0 \ \forall\, \mathbf{v} \neq \mathbf{0}$, meaning B is also PD. ■

10.2 Definiteness

Definition 10.7 (Definiteness).
Let A be a symmetric matrix. Then A is called:

1. ***positive-definite (PD)*** *if $\lambda > 0$ for all eigenvalues λ of A.*
2. ***positive-semidefinite (PSD)*** *if $\lambda \geq 0$ for all eigenvalues λ of A.*
3. ***negative-definite (ND)*** *if $\lambda < 0$ for all eigenvalues λ of A.*
4. ***negative-semidefinite (NSD)*** *if $\lambda \leq 0$ for all eigenvalues λ of A.*
5. ***indefinite*** *if it is neither PSD nor NSD (i.e., there exists at least one strictly positive eigenvalue and one strictly negative eigenvalue).*

Remark .
In particular, PD and ND matrices are special kinds of PSD and NSD matrices respectively. This definition can be extended to Hermitian matrices because all the eigenvalues of a Hermitian matrix are real.

Theorem 10.12 .
Let A be an $n \times n$ symmetric matrix and $q_A(\mathbf{v}) = \mathbf{v}^t A \mathbf{v}$ be its associated quadratic form. Note that clearly $q_A(\mathbf{0}) = 0$.

1. *A is PD if and only if $q_A(\mathbf{v}) > 0$ for any $\mathbf{v} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$.*
2. *A is PSD if and only if $q_A(\mathbf{v}) \geq 0$ for any $\mathbf{v} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$.*
3. *A is ND if and only if $q_A(\mathbf{v}) < 0$ for any $\mathbf{v} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$.*
4. *A is NSD if and only if $q_A(\mathbf{v}) \leq 0$ for any $\mathbf{v} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$.*
5. *A is indefinite if and only if there exist $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^n$ such that $q_A(\mathbf{v}_1) \leq 0$ and $q_A(\mathbf{v}_2) > 0$.*

Proof.
As A is symmetric, by the Real Spectral Theorem, there exists an orthonormal basis (ONB) $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ of $\mathbb{R}^n$ consisting of eigenvectors of A, such that the quadratic form can be written as:

$$q_A(\mathbf{v}) = \lambda_1 \langle \mathbf{v}, \mathbf{u}_1 \rangle^2 + \lambda_2 \langle \mathbf{v}, \mathbf{u}_2 \rangle^2 + \cdots + \lambda_n \langle \mathbf{v}, \mathbf{u}_n \rangle^2$$

where λ_i is the eigenvalue of A corresponding to $\mathbf{u}_i$. The other parts are similar, so we prove only the first statement (PD case).
($\Rightarrow$) Suppose A is PD. By definition, all eigenvalues satisfy $\lambda_1, \lambda_2, \ldots, \lambda_n > 0$. For any $\mathbf{v} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$, since $\{\mathbf{u}_i\}$ forms a basis, we must have $\langle \mathbf{v}, \mathbf{u}_i \rangle \neq 0$ for some i. Since $\lambda_i > 0$ and $\langle \mathbf{v}, \mathbf{u}_j \rangle^2 \geq 0$ for all j, it follows that

$$q_A(\mathbf{v}) \geq \lambda_i \langle \mathbf{v}, \mathbf{u}_i \rangle^2 > 0.$$

($\Leftarrow$) Suppose $q_A(\mathbf{v}) > 0$ for any $\mathbf{v} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$. In particular, choosing $\mathbf{v} = \mathbf{u}_i$ for each $i \in \{1, \ldots, n\}$, we have $\langle \mathbf{u}_i, \mathbf{u}_j \rangle = \delta_{ij}$. Thus,

$$\begin{aligned} q_A(\mathbf{u}_i) &= \lambda_i \langle \mathbf{u}_i, \mathbf{u}_i \rangle^2 \\ &= \lambda_i > 0. \end{aligned}$$

Since all $\lambda_i > 0$, A is PD by definition. ∎

Definition 10.8 (Leading Principal Minors).
Let A be an $n \times n$ matrix. The leading principal minors (LPM) are the determinants of the upper-left submatrices of A:

- *LPM_1: the determinant of the upper-left 1×1 corner of A,*

- *LPM_2: the determinant of the upper-left 2×2 corner of A,*
- *...*
- *LPM_n: the determinant of A itself (i.e., $\det(A)$).*

Definition 10.9 (Principal Minors).
Given an $n \times n$ matrix A, its principal minor of order k (PM_k) is the determinant of any $k \times k$ submatrix obtained by removing $n-k$ rows and the corresponding $n-k$ columns of A of the same indices.

Theorem 10.13 .
Let A be an $n \times n$ square matrix (not necessarily symmetric). For each $k \in \{1, 2, \dots, n\}$, let S_k be the sum of all principal minors of A of order k. By convention, $S_0 = 1$. Then the characteristic polynomial of A is given by:

$$p_A(x) = (-1)^n \sum_{i=0}^{n} (-1)^i S_i x^{n-i} = \pm \left(x^n - S_1 x^{n-1} + S_2 x^{n-2} - \cdots + (-1)^n S_n \right).$$

Proof.
We prove the formula by induction on n.
Let $A \in M_{n \times n}(\mathbb{C})$. Denote by A_{n-1} the upper-left $(n-1) \times (n-1)$ submatrix of A. Also, for $0 \leq i \leq n$, let

$$S_i(A) = \text{the sum of all principal minors of } A \text{ of order } i,$$

with the convention $S_0(A) = 1$.
We want to show that

$$\det(A - xI_n) = (-1)^n \sum_{i=0}^{n} (-1)^i S_i(A) x^{n-i}.$$

The case $n = 1$ is clear. Now assume the formula holds for $(n-1) \times (n-1)$ matrices. We prove it for an $n \times n$ matrix A.
Write

$$A - xI_n = M_1 + M_2,$$

where we split the last column as

$$\begin{pmatrix} a_{1n} \\ \vdots \\ a_{n-1,n} \\ a_{nn} - x \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ -x \end{pmatrix} + \begin{pmatrix} a_{1n} \\ \vdots \\ a_{n-1,n} \\ a_{nn} \end{pmatrix}.$$

By linearity of determinant in the last column,

$$\det(A - xI_n) = \det(M_1) + \det(M_2).$$

First, since the last column of M_1 has only one nonzero entry $-x$, expanding along the last column gives

$$\det(M_1) = -x \det(A_{n-1} - xI_{n-1}).$$

By the induction hypothesis,

$$\begin{aligned}
\det(M_1) &= -x \left[(-1)^{n-1} \sum_{i=0}^{n-1} (-1)^i S_i(A_{n-1}) x^{n-1-i} \right] \\
&= (-1)^n \sum_{i=0}^{n-1} (-1)^i S_i(A_{n-1}) x^{n-i}.
\end{aligned}$$

Next, compute $\det(M_2)$. When expanding $\det(M_2)$, choosing $n - i$ copies of $-x$ from the first $n-1$ diagonal entries leaves a principal minor of order i that must contain the n-th row and the n-th column.
Let $T_i^{(n)}$ be the sum of all principal minors of A of order i containing the n-th row and column. Then

$$\det(M_2) = \sum_{i=1}^{n} (-1)^{n-i} T_i^{(n)} x^{n-i}.$$

Therefore,

$$\begin{aligned}
\det(A - xI_n) &= \det(M_1) + \det(M_2) \\
&= (-1)^n \sum_{i=0}^{n-1} (-1)^i S_i(A_{n-1}) x^{n-i} + \sum_{i=1}^{n} (-1)^{n-i} T_i^{(n)} x^{n-i}.
\end{aligned}$$

Now observe that every principal minor of A of order i either does not use the n-th row and column, or does use the n-th row and column. Hence

$$S_i(A) = S_i(A_{n-1}) + T_i^{(n)} \qquad \text{for } 1 \leq i \leq n-1.$$

Also,

$$S_0(A) = 1, \qquad S_n(A) = T_n^{(n)} = \det(A).$$

Thus we may combine the two sums and obtain

$$\det(A - xI_n) = \sum_{i=0}^{n} (-1)^{n-i} S_i(A) x^{n-i}.$$

Equivalently,

$$\det(A - xI_n) = (-1)^n \sum_{i=0}^{n} (-1)^i S_i(A) x^{n-i}.$$

This proves the formula. ■

Theorem 10.14 (Sylvester's Criterion I).
Let A be an $n \times n$ symmetric matrix.

1. *A is PD if and only if $LPM_k > 0$ for every $k \in \{1, 2, \ldots, n\}$.*
2. *A is ND if and only if $(-1)^k \cdot LPM_k > 0$ for every $k \in \{1, 2, \ldots, n\}$.*

Proof.
We prove the positive-definite statement by induction on n.
($\Rightarrow$) Assume that the $n \times n$ symmetric matrix A is positive-definite. We show all its leading principal minors are positive.

- **Base case ($n = 1$):** $A = (\alpha_{11})$. Since A is PD, its sole eigenvalue is $\alpha_{11} > 0$. Thus $LPM_1 = \alpha_{11} > 0$, which is trivial.

- **Inductive step:** Suppose the statement holds for any $k \times k$ symmetric matrix. Consider A to be a $(k+1) \times (k+1)$ symmetric matrix. Write A as a block matrix:
$$A = \begin{pmatrix} A_k & \mathbf{y} \\ \mathbf{y}^T & \alpha_{k+1,k+1} \end{pmatrix}$$
where A_k is the $k \times k$ upper-left submatrix. Since A is PD, we have $\mathbf{x}^T A \mathbf{x} > 0$ for any non-zero $\mathbf{x} \in \mathbb{R}^{k+1}$. In particular, choose $\mathbf{x} = \begin{pmatrix} \mathbf{x}_k \\ 0 \end{pmatrix}$ where $\mathbf{x}_k \in \mathbb{R}^k \setminus \{\mathbf{0}\}$. Then:
$$\mathbf{x}^T A \mathbf{x} = \begin{pmatrix} \mathbf{x}_k^T & 0 \end{pmatrix} \begin{pmatrix} A_k & \mathbf{y} \\ \mathbf{y}^T & \alpha_{k+1,k+1} \end{pmatrix} \begin{pmatrix} \mathbf{x}_k \\ 0 \end{pmatrix} = \mathbf{x}_k^T A_k \mathbf{x}_k > 0.$$
This implies that A_k is also a positive-definite matrix. By the inductive hypothesis, all leading principal minors of A_k are strictly positive, which accounts for $LPM_1, LPM_2, \ldots, LPM_k$. The final leading principal minor is $LPM_{k+1} = \det(A)$. Since A is PD, all its eigenvalues are strictly positive, so $\det(A) = \prod_{i=1}^{k+1} \lambda_i > 0$. Hence, all leading principal minors of A are strictly positive.

($\Leftarrow$) Conversely, suppose all the leading principal minors of an $n \times n$ symmetric matrix A are positive. We show that A is positive-definite by induction on n.

- **Base case ($n = 1$):** $A = (\alpha_{11})$ with $LPM_1 = \alpha_{11} > 0$. Its eigenvalue is positive, so it is PD.

- **Inductive step:** Suppose this is true for any $k \times k$ symmetric matrix. Consider A to be a $(k+1) \times (k+1)$ symmetric matrix whose LPMs are all strictly positive. Write:

$$A = \begin{pmatrix} A_k & \vec{a} \\ \vec{a}^T & \beta \end{pmatrix}$$

where $A_k \in M_k(\mathbb{R})$ is symmetric, $\vec{a} \in \mathbb{R}^k$, and $\beta \in \mathbb{R}$. Since all LPMs of A are positive, all LPMs of A_k are also positive. By the induction hypothesis, A_k is positive-definite. Let $\{\vec{c}_1, \vec{c}_2, \ldots, \vec{c}_k\}$ be the columns of A_k. Since A_k is PD, $\det(A_k) \neq 0$, meaning these columns form a basis of $\mathbb{R}^k$. Thus, we can express $\vec{a}$ as a linear combination of the columns of A_k: $\vec{a} = \alpha_1\vec{c}_1 + \cdots + \alpha_k\vec{c}_k = A_k\vec{\alpha}$ where $\vec{\alpha} = (\alpha_1, \ldots, \alpha_k)^T$. We construct an elementary block matrix to eliminate $\vec{a}$ and $\vec{a}^T$:

$$P = \begin{pmatrix} I_k & -\vec{\alpha} \\ \vec{0}^T & 1 \end{pmatrix}.$$

Then compute the congruent transformation:

$$P^TAP = \begin{pmatrix} I_k & \vec{0} \\ -\vec{\alpha}^T & 1 \end{pmatrix} \begin{pmatrix} A_k & \vec{a} \\ \vec{a}^T & \beta \end{pmatrix} \begin{pmatrix} I_k & -\vec{\alpha} \\ \vec{0}^T & 1 \end{pmatrix} = \begin{pmatrix} A_k & \vec{0} \\ \vec{0}^T & d \end{pmatrix}$$

where $d = \beta - \vec{\alpha}^T A_k \vec{\alpha} \in \mathbb{R}$. Taking determinants on both sides:

$$\begin{aligned} \det(P^TAP) &= \det(P^T)\det(A)\det(P) \\ &= \det(A). \end{aligned}$$

On the other hand, from the block diagonal form, $\det(P^TAP) = \det(A_k) \cdot d$. Therefore, $\det(A) = \det(A_k) \cdot d \implies d = \frac{\det(A)}{\det(A_k)} = \frac{LPM_{k+1}}{LPM_k}$. Since both $LPM_{k+1} > 0$ and $LPM_k > 0$, we have $d > 0$.
The eigenvalues of the block diagonal matrix P^TAP are given by $\mathrm{Spec}(P^TAP) = \mathrm{Spec}(A_k) \cup \{d\}$. Since A_k is PD, all its eigenvalues are positive, and since $d > 0$, all eigenvalues of P^TAP are positive. Thus, P^TAP is positive-definite. Since P is invertible, this implies that A itself must be positive-definite.

For the negative-definite case, note that A is ND if and only if $-A$ is PD. Applying the above result to $-A$, we require $LPM_k(-A) > 0$ for all k. Since $-A_k$ scales every row by -1, $\det(-A_k) = (-1)^k \det(A_k) = (-1)^k LPM_k(A)$. Thus $(-1)^k \cdot LPM_k(A) > 0$ for all k. ■

Theorem 10.15 (Sylvester's Criterion II).
Let A be a symmetric matrix.

1. *A is PSD if and only if for every k and every principal minor PM_k of order k, $PM_k \geq 0$.*
2. *A is NSD if and only if for every k and every principal minor PM_k of order k, $(-1)^k \cdot PM_k \geq 0$.*
3. *A is indefinite if and only if the principal minors of A do not fit into the conditions in either (1) or (2).*

Proof.
($\Rightarrow$) The argument is similar to the proof in theorem (10.14).
($\Leftarrow$) Conversely, suppose A is a symmetric matrix whose principal minors of any order are non-negative. Our goal is to prove that A is PSD.
Claim: For any $t > 0$, the matrix $A + tI$ is positive-definite.
Proof of claim: Consider the leading principal minors of $A + tI$. Let A_k be the $k \times k$ upper-left submatrix of A. Then:

$$LPM_k(A + tI) = \det(A_k + tI) = p_{A_k}(-t).$$

Using the characteristic polynomial formula via principal minors for A_k:

$$p_{A_k}(-t) = (-1)^k \sum_{i=0}^{k} (-1)^i S_i(A_k)(-t)^{k-i} = \sum_{i=0}^{k} S_i(A_k) t^{k-i}$$

where $S_i(A_k)$ is the sum of all principal minors of order i of A_k. Since every principal minor of A_k is also a principal minor of A, and all principal minors of A are assumed to be non-negative, we have $S_i(A_k) \geq 0$ for all $i \geq 1$. For $i = 0$, $S_0(A_k) = 1$. Since $t > 0$, every term $S_i(A_k)t^{k-i}$ is non-negative, and the $i = 0$ term is $t^k > 0$. Thus, the total sum is strictly positive:

$$LPM_k(A + tI) > 0 \quad \text{for every } k \in \{1, \ldots, n\}.$$

By Sylvester's Criterion I, $A + tI$ must be positive-definite for any $t > 0$.
Since $A + tI$ is PD, for any non-zero vector $\mathbf{u} \in \mathbb{R}^n$, we have:

$$\mathbf{u}^T(A + tI)\mathbf{u} > 0 \implies \mathbf{u}^T A\mathbf{u} + t\|\mathbf{u}\|^2 > 0.$$

Taking the limit as $t \to 0^+$, we obtain:

$$\mathbf{u}^T A\mathbf{u} \geq 0$$

for any $\mathbf{u} \in \mathbb{R}^n$. Therefore, A is positive-semidefinite. ∎

Proposition 6 (General Factorization of PSD Matrices).
Let $B \in \mathrm{Sym}_n(\mathbb{R})$ and let $m \in \mathbb{N}$. Then there exists $A \in M_{m\times n}(\mathbb{R})$ such that $B = A^t A$ if and only if B is PSD and $\mathrm{rank}(B) \le m$.

Proof.
($\Rightarrow$) Suppose $B = A^t A$ for some $A \in M_{m\times n}(\mathbb{R})$. For any $\mathbf{v} \in \mathbb{R}^n$,

$$\mathbf{v}^t B\mathbf{v} = \mathbf{v}^t A^t A\mathbf{v} = \|A\mathbf{v}\|^2 \ge 0.$$

Hence B is PSD. Moreover, $\ker(A^t A) = \ker(A)$.
By rank-nullity, $\mathrm{rank}(A^t A) = \mathrm{rank}(A) \le m$. Thus $\mathrm{rank}(B) \le m$.

($\Leftarrow$) Suppose B is PSD and $\mathrm{rank}(B) = k \le m$.
If $k = 0$, then $B = \mathbf{O}$, and we may choose $A = \mathbf{O}_{m\times n}$. So assume $k \ge 1$.
By the Spectral Theorem, there exist orthonormal eigenvectors $\mathbf{p}_1, \dots, \mathbf{p}_k$ corresponding to the positive eigenvalues $\lambda_1, \dots, \lambda_k > 0$ such that

$$B = \sum_{i=1}^{k} \lambda_i \mathbf{p}_i \mathbf{p}_i^t.$$

Let $P_k = (\mathbf{p}_1 \cdots \mathbf{p}_k)$ and $\Sigma = \mathrm{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_k})$. Then $B = P_k \Sigma^2 P_k^t$.
Since $m \ge k$, define

$$A = \left(\begin{array}{c} \Sigma P_k^t \\ \hline \mathbf{O}_{(m-k)\times n} \end{array}\right) \in M_{m\times n}(\mathbb{R}).$$

Then

$$A^t A = \left(\begin{array}{c|c} P_k\Sigma & \mathbf{O}_{n\times(m-k)} \end{array}\right) \left(\begin{array}{c} \Sigma P_k^t \\ \hline \mathbf{O}_{(m-k)\times n} \end{array}\right) = P_k \Sigma^2 P_k^t = B.$$

Thus there exists $A \in M_{m\times n}(\mathbb{R})$ such that $B = A^t A$. ∎

Corollary .
Let $B \in \mathrm{Sym}_n(\mathbb{R})$ be PSD and let $\mathrm{rank}(B) = k$. Then there exists $A \in M_{k\times n}(\mathbb{R})$ such that

$$B = A^t A.$$

Moreover, k is the smallest possible number of rows of such an A.

Proof.
Existence follows from the previous proposition by taking $m = k$.

For minimality, suppose $B = A^t A$ for some $A \in M_{m\times n}(\mathbb{R})$. Then $\operatorname{rank}(B) = \operatorname{rank}(A^t A) = \operatorname{rank}(A) \leq m$. Hence $m \geq \operatorname{rank}(B) = k$. Therefore k is the smallest possible number of rows. ∎

Corollary .
Let $B \in \mathrm{Sym}_n(\mathbb{R})$. Then

$$B \text{ is PSD} \iff B = A^t A \quad \text{for some } A \in M_n(\mathbb{R}).$$

Proof.
This follows from the general factorization by taking $m = n$. Indeed, if B is PSD, then $\operatorname{rank}(B) \leq n$, so there exists $A \in M_n(\mathbb{R})$ such that $B = A^t A$.
Conversely, if $B = A^t A$, then for any $\mathbf{v} \in \mathbb{R}^n$,

$$\mathbf{v}^t B\mathbf{v} = \mathbf{v}^t A^t A\mathbf{v} = \|A\mathbf{v}\|^2 \geq 0.$$

Hence B is PSD. ∎

Corollary (Symmetric Square Root).
Let $B \in \mathrm{Sym}_n(\mathbb{R})$. Then

$$B \text{ is PSD} \iff B = X^2 \quad \text{for some } X \in \mathrm{Sym}_n(\mathbb{R}).$$

Proof.
($\Rightarrow$) Suppose B is PSD. By the Spectral Theorem, there exists $P \in O(n)$ such that

$$B = P \begin{pmatrix} \lambda_1 & & \mathbf{O} \\ & \ddots & \\ \mathbf{O} & & \lambda_n \end{pmatrix} P^t,$$

where $\lambda_i \geq 0$ for all i. Let

$$D = \begin{pmatrix} \lambda_1 & & \mathbf{O} \\ & \ddots & \\ \mathbf{O} & & \lambda_n \end{pmatrix}, \qquad \sqrt{D} = \begin{pmatrix} \sqrt{\lambda_1} & & \mathbf{O} \\ & \ddots & \\ \mathbf{O} & & \sqrt{\lambda_n} \end{pmatrix}.$$

Choose $X = P\sqrt{D}P^t$. Then $X \in \mathrm{Sym}_n(\mathbb{R})$, and

$$X^2 = (P\sqrt{D}P^t)(P\sqrt{D}P^t) = PDP^t = B.$$

($\Leftarrow$) Suppose $B = X^2$ for some $X \in \mathrm{Sym}_n(\mathbb{R})$. Since $X^t = X$, we have $B = X^2 = X^t X$. Hence B is PSD by the previous corollary. ∎

Remark .

The most general factorization is the following: if $B \in \mathrm{Sym}_n(\mathbb{R})$ is PSD and $\mathrm{rank}(B) = k$, then $B = A^t A$ for some $A \in M_{m\times n}(\mathbb{R})$ if and only if $m \geq k$.

The minimal choice is $m = k$. For any $m > k$, we can obtain a factorization in $M_{m\times n}(\mathbb{R})$ by adding zero rows.

The symmetric square root is a special square factorization. If B is PSD, then $B = X^2$ for some symmetric matrix X. Since $X^t = X$, this also gives $B = X^t X$ by taking $A = X$.

Geometry of Quadratic Forms

Let $A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ be symmetric and set $q_A(\mathbf{v}) = \mathbf{v}^t A\mathbf{v}$. We study the level set

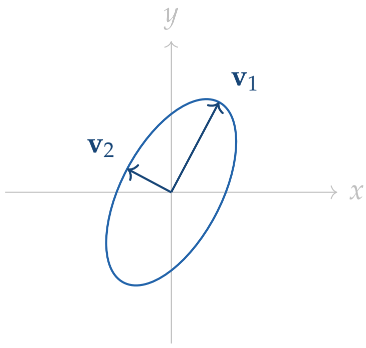
$$\Sigma_1(A) = \{\mathbf{v} \in \mathbb{R}^2 : q_A(\mathbf{v}) = 1\}.$$

The cross term $2bxy$ is geometrically a rotation of coordinates. By Spectral Theorem, there is an ONB $\{\mathbf{u}_1, \mathbf{u}_2\}$ such that, if $\mathbf{v} = s\mathbf{u}_1 + t\mathbf{u}_2$, then

$$q_A(\mathbf{v}) = \lambda_1 s^2 + \lambda_2 t^2.$$

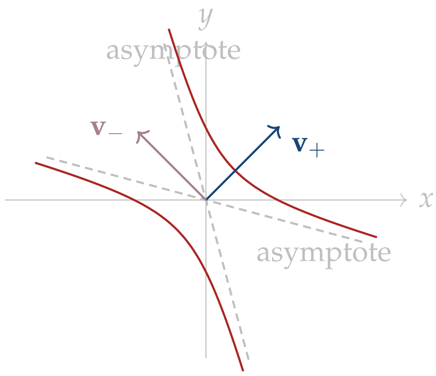
Thus the shape of $\Sigma_1(A)$ is determined by the signs of λ_1, λ_2.

1. **PD.** If $\lambda_1, \lambda_2 > 0$, then $\lambda_1 s^2 + \lambda_2 t^2 = 1$ is an ellipse. Its semi-axis lengths are $1/\sqrt{\lambda_1}$ and $1/\sqrt{\lambda_2}$, so the longer axis corresponds to the smaller eigenvalue.



   For example, take $A = \begin{pmatrix} 4 & -\frac{3}{2} \\ -\frac{3}{2} & 2 \end{pmatrix}$. Then $\Sigma_1(A)$ is $4x^2 - 3xy + 2y^2 = 1$. Since the eigenvalues of A are $(6+\sqrt{13})/2$ and $(6-\sqrt{13})/2$, this is an ellipse. The cross term $-3xy$ rotates the axes of the ellipse.

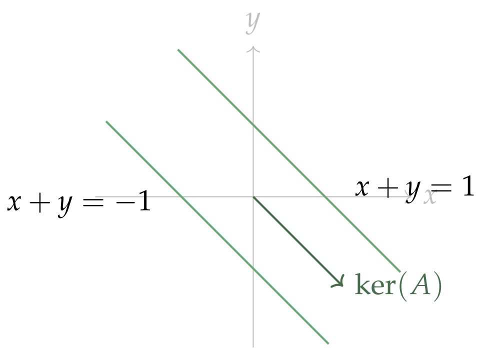
2. **Indefinite.** If $\lambda_1 > 0 > \lambda_2$, then $\lambda_1 s^2 + \lambda_2 t^2 = 1$ is a hyperbola. The positive eigendirection is the direction in which the level 1 opens. The equation $q_A(\mathbf{v}) = 0$ gives the two asymptotic directions.



For example, if $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$, then $\Sigma_1(A)$ is $x^2 + 4xy + y^2 = 1$. In the eigenbasis $\mathbf{v}_+ = (1,1)/\sqrt{2}$ and $\mathbf{v}_- = (1,-1)/\sqrt{2}$, this becomes $3s^2 - t^2 = 1$. The dashed lines come from $3s^2 - t^2 = 0$, so they are the asymptotes.

Think first about directions. For a direction $\mathbf{w} = s\mathbf{v}_+ + t\mathbf{v}_-$, the sign of $q_A(\mathbf{w}) = 3s^2 - t^2$ tells us whether this direction can hit the level $q_A = 1$ after scaling. If $3s^2 - t^2 > 0$, the $\mathbf{v}_+$ direction accounts more, and $c\mathbf{w}$ can be chosen so that $q_A(c\mathbf{w}) = 1$. If $3s^2 - t^2 \leq 0$, then no positive scaling can make $q_A = 1$. Near an asymptote, $3s^2 - t^2$ is positive but very small, so the needed scaling $c = 1/\sqrt{q_A(\mathbf{w})}$ is very large; this is why the branch stretches far out. Near the opening direction $\mathbf{v}_+$, the positive part clearly dominates, so only a small scaling is needed, and the branch stays close to the vertex/focus region.

3. **PSD.** If $\lambda_1 > 0$ and $\lambda_2 = 0$, then $\lambda_1 s^2 = 1$, so $s = \pm 1/\sqrt{\lambda_1}$ and t is free. Therefore $\Sigma_1(A)$ is two parallel lines, both parallel to the null direction $\ker(A)$.



For example, if $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, then $q_A(x,y) = (x+y)^2$, so $\Sigma_1(A)$ is $(x+y)^2 = 1$, i.e. $x + y = 1$ and $x + y = -1$. These lines are parallel to $\ker(A) = \mathrm{span}\{(1,-1)\}$.

4. **ND/NSD.** If $\lambda_1, \lambda_2 \leq 0$, then $q_A(\mathbf{v}) \leq 0$ for every $\mathbf{v} \in \mathbb{R}^2$. Hence the positive level set $\Sigma_1(A)$ is empty.

Application: Second Derivative Tests

Theorem 10.16 .
Let $f(x_1, x_2, \ldots, x_n)$ be a function with continuous third-order derivatives, and let $\mathbf{a} = (a_1, a_2, \ldots, a_n)$ be a critical point of f, i.e., $\frac{\partial f}{\partial x_i}(\mathbf{a}) = 0$ for each $i = 1, 2, \ldots, n$. Consider the $n \times n$ Hessian matrix at $\mathbf{a}$:

$$H_f(\mathbf{a}) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{a}) \right)_{i,j}.$$

1. *If $H_f(\mathbf{a})$ is PD, then $\mathbf{a}$ is a local minimum point for f.*
2. *If $H_f(\mathbf{a})$ is ND, then $\mathbf{a}$ is a local maximum point for f.*
3. *If $H_f(\mathbf{a})$ is indefinite (has at least one strictly positive and one strictly negative eigenvalue), then $\mathbf{a}$ is a saddle point.*
4. *In any other scenario (i.e., when $H_f(\mathbf{a})$ is PSD or NSD but not definite), the test is inconclusive.*

Proof.
We prove statement (1). By shifting the coordinate system and the function value suitably, we can assume without loss of generality that $\mathbf{a} = \mathbf{0}$ and $f(\mathbf{0}) = 0$. By multivariable Taylor's Theorem, writing $\mathbf{x} = (x_1, x_2, \ldots, x_n)$, we expand around $\mathbf{0}$:

$$f(\mathbf{x}) = f(\mathbf{0}) + \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(\mathbf{0}) x_i + \frac{1}{2} \sum_{i,j=1}^{n} \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{0}) x_i x_j + S(\mathbf{x})$$

where $S(\mathbf{x})$ is the remainder term satisfying $\lim_{\mathbf{x} \to \mathbf{0}} \frac{S(\mathbf{x})}{\|\mathbf{x}\|^2} = 0$. Since $\mathbf{0}$ is a critical point, the first-order sum vanishes. Thus:

$$f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T H_f(\mathbf{0}) \mathbf{x} + S(\mathbf{x}).$$

By the Real Spectral Theorem, since $H_f(\mathbf{0})$ is symmetric, there exists an ONB $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ of $\mathbb{R}^n$ consisting of eigenvectors of $H_f(\mathbf{0})$. We can express the quadratic form in terms of the eigenvalues $\lambda_i \in Spec(H_f(\mathbf{0}))$:

$$f(\mathbf{x}) = \frac{1}{2} \sum_{i=1}^{n} \lambda_i \langle \mathbf{x}, \mathbf{v}_i \rangle^2 + S(\mathbf{x}).$$

Assume $H_f(\mathbf{0})$ is positive-definite. Then all eigenvalues satisfy $\lambda_i > 0$. Take any $\varepsilon > 0$ chosen small enough such that $0 < \varepsilon < \frac{\lambda_i}{2}$ for all $i \in \{1, \ldots, n\}$. Since $\lim_{\mathbf{x}\to\mathbf{0}} \frac{S(\mathbf{x})}{\|\mathbf{x}\|^2} = 0$, there exists a $\delta > 0$ such that:

$$\begin{aligned}\|\mathbf{x}\| < \delta &\implies \left|\frac{S(\mathbf{x})}{\|\mathbf{x}\|^2}\right| < \varepsilon \\ &\implies |S(\mathbf{x})| < \varepsilon\|\mathbf{x}\|^2.\end{aligned}$$

This implies $S(\mathbf{x}) > -\varepsilon\|\mathbf{x}\|^2$. Substituting this into our expression for $f(\mathbf{x})$ yields:

$$f(\mathbf{x}) > \frac{1}{2}\sum_{i=1}^{n} \lambda_i \langle \mathbf{x}, \mathbf{v}_i\rangle^2 - \varepsilon\|\mathbf{x}\|^2.$$

Let us write $\mathbf{x} = \sum_{i=1}^{n} \alpha_i \mathbf{v}_i$, where $\alpha_i = \langle \mathbf{x}, \mathbf{v}_i\rangle$. By Pythagorean theorem, $\|\mathbf{x}\|^2 = \sum_{i=1}^{n} \alpha_i^2$. Substituting these representations gives:

$$\begin{aligned}f(\mathbf{x}) &> \frac{1}{2}\sum_{i=1}^{n} \lambda_i\alpha_i^2 - \varepsilon\sum_{i=1}^{n}\alpha_i^2 \\ &= \sum_{i=1}^{n}\left(\frac{1}{2}\lambda_i - \varepsilon\right)\alpha_i^2.\end{aligned}$$

Since we picked $\varepsilon < \frac{\lambda_i}{2}$, the coefficient $\left(\frac{1}{2}\lambda_i - \varepsilon\right)$ is strictly positive for every i. Therefore, for any $\mathbf{x} \neq \mathbf{0}$ within the neighborhood $\|\mathbf{x}\| < \delta$, at least one $\alpha_i \neq 0$, which ensures:

$$f(\mathbf{x}) > 0 = f(\mathbf{0}).$$

This proves that $\mathbf{a} = \mathbf{0}$ is a strict local minimum point for f. ■

Example 10.1 (2-Variable Function).
Let $f(x, y)$ be a C^3-function and fix (a, b) in the domain of f. Then one can write

$$\begin{aligned}f(x,y) &= f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b) \\ &\quad + \frac{1}{2}\left(f_{xx}(a,b)(x-a)^2 + 2f_{xy}(a,b)(x-a)(y-b) + f_{yy}(a,b)(y-b)^2\right) + R(x,y)\end{aligned}$$

where $R(x, y)$ is a remainder term that satisfies

$$\lim_{(x,y)\to(a,b)} \frac{R(x,y)}{(x-a)^2 + (y-b)^2} = 0.$$

Suppose $(0, 0)$ is a critical point, i.e., $f_x(0, 0) = 0$ and $f_y(0, 0) = 0$. Then the local behavior can be approximated by the quadratic form:

$$f(x,y) - f(0,0) \approx \frac{1}{2}\left(Ax^2 + 2Bxy + Cy^2\right)$$

where $A = f_{xx}(0,0)$, $B = f_{xy}(0,0)$, and $C = f_{yy}(0,0)$. This quadratic form is associated with the Hessian matrix:

$$\begin{pmatrix} A & B \\ B & C \end{pmatrix}.$$

The signs of the eigenvalues λ_1, λ_2 of the Hessian matrix determine the nature of the critical point:

- If $\lambda_1, \lambda_2 > 0$, then $f(x,y) > f(0,0)$ near $(0,0)$, so $(0,0)$ is a local minimum.
- If $\lambda_1, \lambda_2 < 0$, then $f(x,y) < f(0,0)$ near $(0,0)$, so $(0,0)$ is a local maximum.
- If $\lambda_1\lambda_2 < 0$, then $(0,0)$ is a saddle point.

Convexity and Global Minimum

The second derivative test is local: it tells us what happens near a critical point. Convexity gives a global version of this idea.

Definition 10.10 (Convex Set).
*A subset $U \subseteq \mathbb{R}^n$ is called **convex** if for every $\mathbf{x}, \mathbf{y} \in U$ and every $t \in [0,1]$, we have:*

$$(1-t)\mathbf{x} + t\mathbf{y} \in U.$$

Geometrically, this means that the straight line segment joining any two points of U is entirely contained within U.

Definition 10.11 (Convex Function).
Let $U \subseteq \mathbb{R}^n$ be a convex set.

1. *A function $f : U \to \mathbb{R}$ is called **convex** if for every $\mathbf{x}, \mathbf{y} \in U$ and every $t \in [0,1]$:*

$$f((1-t)\mathbf{x} + t\mathbf{y}) \le (1-t)f(\mathbf{x}) + tf(\mathbf{y}).$$

2. *If the inequality is strict whenever $\mathbf{x} \neq \mathbf{y}$ and $0 < t < 1$, then f is called **strictly convex**.*

Theorem 10.17 (Second-Order Criterion for Convexity).
Let $U \subseteq \mathbb{R}^n$ be open and convex, and let $f : U \to \mathbb{R}$ be a C^2-function. Then:

1. *f is convex if and only if $H_f(\mathbf{x})$ is PSD for all $\mathbf{x} \in U$.*
2. *If $H_f(\mathbf{x})$ is positive-definite for every $\mathbf{x} \in U$, then f is strictly convex.*

Proof.
We can connect multivariable convexity to single-variable calculus using line restrictions. Let $\mathbf{a} \in U$ and choose a direction vector $\mathbf{b} \in \mathbb{R}^n$. Define the single-variable function $g(t) = f(\mathbf{a} + t\mathbf{b})$.
Applying the chain rule, the first and second derivatives of $g(t)$ are:

$$\begin{aligned} g'(t) &= \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(\mathbf{a} + t\mathbf{b}) b_i \\ &= \nabla f(\mathbf{a} + t\mathbf{b}) \cdot \mathbf{b}. \end{aligned}$$

$$\begin{aligned} g''(t) &= \sum_{j=1}^{n} \sum_{i=1}^{n} \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{a} + t\mathbf{b}) b_i b_j \\ &= \mathbf{b}^T H_f(\mathbf{a} + t\mathbf{b}) \mathbf{b}. \end{aligned}$$

($\Rightarrow$) Suppose f is convex. Then the single-variable restriction $g(t)$ must be a convex function on its interval of definition. From single-variable calculus, a C^2 function is convex if and only if its second derivative is non-negative everywhere, meaning $g''(t) \geq 0$. Evaluating at $t = 0$ gives $g''(0) = \mathbf{b}^T H_f(\mathbf{a})\mathbf{b} \geq 0$. Since this holds for any vector $\mathbf{b} \in \mathbb{R}^n$ and any point $\mathbf{a} \in U$, the Hessian matrix $H_f(\mathbf{x})$ is positive-semidefinite for all $\mathbf{x} \in U$.
($\Leftarrow$) Conversely, suppose $H_f(\mathbf{x})$ is PSD for all $\mathbf{x} \in U$. Take any two points $\mathbf{x}, \mathbf{y} \in U$. Since U is a convex set, the segment $t\mathbf{x} + (1-t)\mathbf{y}$ stays in U for all $t \in [0, 1]$. Define $g(t) = f(\mathbf{y} + t(\mathbf{x} - \mathbf{y}))$. Its second derivative is given by:

$$g''(t) = (\mathbf{x} - \mathbf{y})^T H_f(t\mathbf{x} + (1-t)\mathbf{y})(\mathbf{x} - \mathbf{y}).$$

Since H_f is PSD everywhere on U, we know that $g''(t) \geq 0$ for all $t \in [0, 1]$. By single-variable calculus, $g(t)$ is a convex function on $[0, 1]$, which satisfies:

$$\begin{aligned} g(t) &= g(t \cdot 1 + (1-t) \cdot 0) \\ &\leq t g(1) + (1-t) g(0). \end{aligned}$$

Substituting the definition of $g(t)$ back into the inequality:

$$f(t\mathbf{x} + (1-t)\mathbf{y}) \leq t f(\mathbf{x}) + (1-t) f(\mathbf{y}).$$

This matches the definition of a convex function, proving that f is convex on U. ∎

Theorem 10.18 (Convex Functions and Global Minima).
Let $U \subseteq \mathbb{R}^n$ be a convex set, and let $f : U \to \mathbb{R}$ be a differentiable and convex function. If $\mathbf{a} \in U$ satisfies $\nabla f(\mathbf{a}) = \mathbf{0}$, then $\mathbf{a}$ is a global minimum of f on U.

Remark .
Furthermore, if f is strictly convex, then this global minimum is unique.

Proof.
Let $\mathbf{a}$ be a critical point of f, so $\nabla f(\mathbf{a}) = \mathbf{0}$. We want to show that $f(\mathbf{y}) \geq f(\mathbf{a})$ for any arbitrary point $\mathbf{y} \in U$. Let us construct the auxiliary line segment function $g(t) = f((1-t)\mathbf{a} + t\mathbf{y}) = f(\mathbf{a} + t(\mathbf{y} - \mathbf{a}))$ defined for $t \in [0,1]$. Since U is a convex set, this path lies entirely inside U.
Because f is given to be a convex function, its line restriction $g(t)$ is a convex function of a single variable on $[0,1]$. A differentiable single-variable function is convex if and only if its derivative $g'(t)$ is a monotonically increasing function. Therefore, we have:

$$g'(t) \geq g'(0) \quad \text{for all } t \in [0,1].$$

Integrating this derivative inequality from $t = 0$ to $t = 1$:

$$\int_0^1 g'(t)\,dt \geq \int_0^1 g'(0)\,dt \implies g(1) - g(0) \geq g'(0) \cdot (1-0) = g'(0).$$

Let us compute the value of $g'(0)$ explicitly via the multivariable chain rule:

$$g'(t) = \nabla f(\mathbf{a} + t(\mathbf{y} - \mathbf{a})) \cdot (\mathbf{y} - \mathbf{a}) \implies g'(0) = \nabla f(\mathbf{a}) \cdot (\mathbf{y} - \mathbf{a}).$$

Since $\mathbf{a}$ is a critical point, $\nabla f(\mathbf{a}) = \mathbf{0}$, which yields $g'(0) = \mathbf{0} \cdot (\mathbf{y} - \mathbf{a}) = 0$. Substituting this back into the integrated expression:

$$g(1) - g(0) \geq 0 \implies g(1) \geq g(0).$$

Since $g(1) = f(\mathbf{y})$ and $g(0) = f(\mathbf{a})$, we conclude that:

$$f(\mathbf{y}) \geq f(\mathbf{a})$$

for all $\mathbf{y} \in U$. This establishes that $\mathbf{a}$ is a global minimum of f on U. ∎

10.3 Min–Max Theorem & Weyl's Inequality

Proposition 7 .
Suppose $A \in M_n(\mathbb{C})$ is a positive-definite Hermitian matrix whose eigenvalues are given by $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$. Let the Rayleigh quotient defined by

$$f(\mathbf{x}) = \frac{\mathbf{x}^* A \mathbf{x}}{\mathbf{x}^* \mathbf{x}} \quad \forall \mathbf{x} \neq \mathbf{0}.$$

Then

$$\lambda_1 = \max_{\substack{\mathbf{x} \in \mathbb{C}^n \\ \mathbf{x} \neq \mathbf{0}}} f(\mathbf{x}), \qquad \lambda_n = \min_{\substack{\mathbf{x} \in \mathbb{C}^n \\ \mathbf{x} \neq \mathbf{0}}} f(\mathbf{x}).$$

Proof.
By spectral theorem, there exists an ONB $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ of eigenvectors of A. Write $\mathbf{x} = \sum_{i=1}^{n} \langle \mathbf{x}, \mathbf{v}_i \rangle \mathbf{v}_i = \sum_{i=1}^{n} \alpha_i \mathbf{v}_i$, then

$$\begin{aligned}
f(\mathbf{x}) &= \frac{\left(\sum_i \alpha_i \mathbf{v}_i\right)^* A \left(\sum_j \alpha_j \mathbf{v}_j\right)}{\|\mathbf{x}\|^2} \\
&= \frac{\left(\sum_i \alpha_i \mathbf{v}_i\right)^* \left(\sum_j \alpha_j A \mathbf{v}_j\right)}{\sum_i |\alpha_i|^2} \\
&= \frac{\left\langle \sum_i \alpha_i \mathbf{v}_i, \sum_j \alpha_j \lambda_j \mathbf{v}_j \right\rangle}{\sum_i |\alpha_i|^2} \\
&= \frac{\sum_i \sum_j \alpha_i \overline{\alpha_j} \lambda_j \langle \mathbf{v}_i, \mathbf{v}_j \rangle}{\sum_i |\alpha_i|^2}.
\end{aligned}$$

Since $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is an ONB, only the terms with $i = j$ survive. Hence

$$f(\mathbf{x}) = \frac{\sum_{i=1}^{n} |\alpha_i|^2 \lambda_i}{\sum_{i=1}^{n} |\alpha_i|^2}.$$

Thus the Rayleigh quotient is a weighted average of $\lambda_1, \lambda_2, \ldots, \lambda_n$.

In particular,

$$\lambda_n \leq f(\mathbf{x}) \leq \lambda_1.$$

Moreover,

$$\lambda_1 = f(\mathbf{v}_1), \qquad \lambda_n = f(\mathbf{v}_n).$$

Therefore,

$$\max_{\mathbf{x}\neq\mathbf{0}} f(\mathbf{x}) = \lambda_1, \qquad \min_{\mathbf{x}\neq\mathbf{0}} f(\mathbf{x}) = \lambda_n.$$

■

Theorem 10.19 (Courant–Fischer Min–Max).
Let A be Hermitian on a complex inner product space $(V, \langle -, - \rangle)$ with $\dim V = n$. Suppose the eigenvalues of A are ordered as

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n.$$

Then, for each $k = 1, 2, \ldots, n$,

$$\lambda_k = \min_{\substack{E\subseteq V\\ \dim E = n-k+1}} \max_{\substack{\mathbf{x}\in E\\ \mathbf{x}\neq\mathbf{0}}} \frac{\langle \mathbf{x}, A\mathbf{x}\rangle}{\langle \mathbf{x}, \mathbf{x}\rangle} = \max_{\substack{F\subseteq V\\ \dim F = k}} \min_{\substack{\mathbf{x}\in F\\ \mathbf{x}\neq\mathbf{0}}} \frac{\langle \mathbf{x}, A\mathbf{x}\rangle}{\langle \mathbf{x}, \mathbf{x}\rangle}.$$

Proof.
For $\mathbf{x} \neq \mathbf{0}$, write

$$f_A(\mathbf{x}) = \frac{\langle \mathbf{x}, A\mathbf{x}\rangle}{\langle \mathbf{x}, \mathbf{x}\rangle}.$$

Since $f_A(c\mathbf{x}) = f_A(\mathbf{x})$ for $c \neq 0$, it suffices to consider unit vectors.
By the Spectral Theorem, there exists an ONB $\{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ of eigenvectors of A such that $A\mathbf{u}_i = \lambda_i \mathbf{u}_i$, where $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$.

Proof of the first equality.
We prove

$$\lambda_k = \min_{\substack{E\subseteq V\\ \dim E = n-k+1}} \max_{\substack{\mathbf{x}\in E\\ \|\mathbf{x}\|=1}} \langle \mathbf{x}, A\mathbf{x}\rangle.$$

Case 1. Let $E_0 = \operatorname{span}\{\mathbf{u}_k, \mathbf{u}_{k+1}, \ldots, \mathbf{u}_n\}$. Then $\dim E_0 = n - k + 1$.
For any $\mathbf{x} \in E_0$ with $\|\mathbf{x}\| = 1$, write $\mathbf{x} = \sum_{i=k}^{n} \alpha_i \mathbf{u}_i$. Then $\sum_{i=k}^{n} |\alpha_i|^2 = 1$. Hence

$$\langle \mathbf{x}, A\mathbf{x}\rangle = \sum_{i=k}^{n} |\alpha_i|^2 \lambda_i \leq \sum_{i=k}^{n} |\alpha_i|^2 \lambda_k = \lambda_k.$$

On the other hand, taking $\mathbf{x} = \mathbf{u}_k$ gives $\langle \mathbf{u}_k, A\mathbf{u}_k\rangle = \lambda_k$. Therefore,

$$\max_{\substack{\mathbf{x}\in E_0\\ \|\mathbf{x}\|=1}} \langle \mathbf{x}, A\mathbf{x}\rangle = \lambda_k.$$

Thus,

$$\min_{\substack{E \subseteq V \\ \dim E = n-k+1}} \max_{\substack{\mathbf{x} \in E \\ \|\mathbf{x}\|=1}} \langle \mathbf{x}, A\mathbf{x} \rangle \leq \lambda_k.$$

Case 2. Let $E \subseteq V$ be any subspace with $\dim E = n - k + 1$.
Set $V_k = \operatorname{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$, so $\dim V_k = k$. We claim that $V_k \cap E \neq \{\mathbf{0}\}$. Indeed, if $V_k \cap E = \{\mathbf{0}\}$, then

$$\dim(V_k + E) = \dim V_k + \dim E - \dim(V_k \cap E) = k + (n - k + 1) - 0 = n + 1,$$

which is impossible since $V_k + E \subseteq V$. Hence, there exists $\mathbf{u} \in E$ such that $\mathbf{u} = \sum_{i=1}^{k} \beta_i \mathbf{u}_i$, where $\sum_{i=1}^{k} |\beta_i|^2 = 1$. Then

$$\langle \mathbf{u}, A\mathbf{u} \rangle = \sum_{i=1}^{k} |\beta_i|^2 \lambda_i \geq \sum_{i=1}^{k} |\beta_i|^2 \lambda_k = \lambda_k.$$

So we have

$$\max_{\substack{\mathbf{x} \in E \\ \|\mathbf{x}\|=1}} \langle \mathbf{x}, A\mathbf{x} \rangle \geq \lambda_k.$$

Because E was arbitrary,

$$\min_{\substack{E \subseteq V \\ \dim E = n-k+1}} \max_{\substack{\mathbf{x} \in E \\ \|\mathbf{x}\|=1}} \langle \mathbf{x}, A\mathbf{x} \rangle \geq \lambda_k.$$

Combining the two cases, we obtain

$$\lambda_k = \min_{\substack{E \subseteq V \\ \dim E = n-k+1}} \max_{\substack{\mathbf{x} \in E \\ \|\mathbf{x}\|=1}} \langle \mathbf{x}, A\mathbf{x} \rangle.$$

Proof of the second equality.
Apply the first equality to $-A$. Since the eigenvalues of $-A$ are $-\lambda_n \geq -\lambda_{n-1} \geq \cdots \geq -\lambda_1$, we have $\lambda_{n-k+1}(-A) = -\lambda_k(A)$. Therefore,

$$-\lambda_k(A) = \min_{\substack{F \subseteq V \\ \dim F = k}} \max_{\substack{\mathbf{x} \in F \\ \|\mathbf{x}\|=1}} \langle \mathbf{x}, (-A)\mathbf{x} \rangle.$$

That is,

$$-\lambda_k(A) = \min_{\substack{F \subseteq V \\ \dim F = k}} \max_{\substack{\mathbf{x} \in F \\ \|\mathbf{x}\|=1}} \left(- \langle \mathbf{x}, A\mathbf{x} \rangle\right).$$

Taking negative signs on both sides gives

$$\begin{aligned} \lambda_k(A) &= - \min_{\substack{F \subseteq V \\ \dim F = k}} \max_{\substack{\mathbf{x} \in F \\ \|\mathbf{x}\|=1}} \left(- \langle \mathbf{x}, A\mathbf{x} \rangle\right) \\ &= \max_{\substack{F \subseteq V \\ \dim F = k}} \min_{\substack{\mathbf{x} \in F \\ \|\mathbf{x}\|=1}} \langle \mathbf{x}, A\mathbf{x} \rangle. \end{aligned}$$

■

Remark .
Since
$$\frac{\langle c\mathbf{x}, A(c\mathbf{x})\rangle}{\langle c\mathbf{x}, c\mathbf{x}\rangle} = \frac{\langle \mathbf{x}, A\mathbf{x}\rangle}{\langle \mathbf{x}, \mathbf{x}\rangle} \qquad (c \neq 0),$$
the Rayleigh quotient only depends on the direction of $\mathbf{x}$*, not on its length. Hence we may restrict to unit vectors.*

When $k = 1$*, the first formula degenerates to*

$$\lambda_1 = \max_{\substack{\mathbf{x}\in V\\ \mathbf{x}\neq \mathbf{0}}} \frac{\langle \mathbf{x}, A\mathbf{x}\rangle}{\langle \mathbf{x}, \mathbf{x}\rangle},$$

because the only n-dimensional subspace of V is V itself.

When $k = n$*, the first formula degenerates to*

$$\lambda_n = \min_{\dim E=1} \max_{\substack{\mathbf{x}\in E\\ \mathbf{x}\neq \mathbf{0}}} \frac{\langle \mathbf{x}, A\mathbf{x}\rangle}{\langle \mathbf{x}, \mathbf{x}\rangle}.$$

But a one-dimensional subspace is of the form $E = \operatorname{span}\{\mathbf{v}\}$*. Since the quotient only depends on direction, the inner maximum over* $\operatorname{span}\{\mathbf{v}\}$ *is just the value at any nonzero vector in that line. Thus,*
$$\lambda_n = \min_{\substack{\mathbf{v}\in V\\ \mathbf{v}\neq \mathbf{0}}} \frac{\langle \mathbf{v}, A\mathbf{v}\rangle}{\langle \mathbf{v}, \mathbf{v}\rangle}.$$
Similarly, the second formula gives

$$\lambda_1 = \max_{\dim F=1} \min_{\substack{\mathbf{x}\in F\\ \mathbf{x}\neq \mathbf{0}}} \frac{\langle \mathbf{x}, A\mathbf{x}\rangle}{\langle \mathbf{x}, \mathbf{x}\rangle},$$

which is the same as maximizing over all directions.

Corollary .
Let $A \in \operatorname{Sym}_n(\mathbb{R})$ *and define* $f_A(\mathbf{x}) = \mathbf{x}^t A\mathbf{x}$*. Assume the eigenvalues of A are ordered as* $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$*. Then, for* $1 \leq k \leq n$*,*

$$\lambda_k = \min_{\substack{E\subseteq\mathbb{R}^n\\ \dim E=n-k+1}} \max_{\substack{\mathbf{x}\in E\\ \|\mathbf{x}\|=1}} f_A(\mathbf{x}) = \max_{\substack{E\subseteq\mathbb{R}^n\\ \dim F=k}} \min_{\substack{\mathbf{x}\in E\\ \|\mathbf{x}\|=1}} f_A(\mathbf{x}).$$

Proposition 8 .
For Hermitian matrices $A, H \in M_n(\mathbb{C})$*,*

$$\lambda_1(A+H) - \lambda_1(A) \leq \lambda_1(H).$$

Proof.
Using the Rayleigh quotient characterization of the largest eigenvalue,

$$\begin{aligned}
\lambda_1(A+H) &= \max_{\|\mathbf{x}\|=1} \langle \mathbf{x}, (A+H)\mathbf{x} \rangle \\
&= \max_{\|\mathbf{x}\|=1} \left(\langle \mathbf{x}, A\mathbf{x} \rangle + \langle \mathbf{x}, H\mathbf{x} \rangle \right) \\
&\leq \max_{\|\mathbf{x}\|=1} \langle \mathbf{x}, A\mathbf{x} \rangle + \max_{\|\mathbf{x}\|=1} \langle \mathbf{x}, H\mathbf{x} \rangle \\
&= \lambda_1(A) + \lambda_1(H).
\end{aligned}$$

Hence

$$\lambda_1(A+H) - \lambda_1(A) \leq \lambda_1(H).$$

■

Proposition 9 .
For Hermitian matrices $A, H \in M_n(\mathbb{C})$,

$$\lambda_n(A+H) - \lambda_n(A) \leq \lambda_1(H).$$

Proof.
For $\|\mathbf{x}\| = 1$, write $f_A(\mathbf{x}) = \langle \mathbf{x}, A\mathbf{x} \rangle$ and $f_H(\mathbf{x}) = \langle \mathbf{x}, H\mathbf{x} \rangle$. By the Rayleigh quotient characterization,

$$\lambda_n(A+H) = \min_{\|\mathbf{x}\|=1} \left(f_A(\mathbf{x}) + f_H(\mathbf{x}) \right).$$

Since $f_H(\mathbf{x}) \leq \lambda_1(H)$ for every $\|\mathbf{x}\| = 1$, we have

$$\begin{aligned}
\lambda_n(A+H) &= \min_{\|\mathbf{x}\|=1} \left(f_A(\mathbf{x}) + f_H(\mathbf{x}) \right) \\
&\leq \min_{\|\mathbf{x}\|=1} \left(f_A(\mathbf{x}) + \lambda_1(H) \right) \\
&= \min_{\|\mathbf{x}\|=1} f_A(\mathbf{x}) + \lambda_1(H) \\
&= \lambda_n(A) + \lambda_1(H).
\end{aligned}$$

Therefore,

$$\lambda_n(A+H) - \lambda_n(A) \leq \lambda_1(H).$$

■

Proposition 10 .

For Hermitian matrices $A, H \in M_n(\mathbb{C})$,

$$\lambda_1(A+H) - \lambda_1(A) \geq \lambda_n(H).$$

Proof.
Apply the previous proposition to $-A$ and $-H$:

$$\lambda_n((-A)+(-H)) - \lambda_n(-A) \leq \lambda_1(-H).$$

Since

$$\lambda_n(-(A+H)) = -\lambda_1(A+H), \quad \lambda_n(-A) = -\lambda_1(A), \quad \lambda_1(-H) = -\lambda_n(H),$$

we obtain

$$-\lambda_1(A+H) + \lambda_1(A) \leq -\lambda_n(H).$$

Thus

$$\lambda_1(A+H) - \lambda_1(A) \geq \lambda_n(H).$$

Combining this with Proposition 8, we get

$$\lambda_n(H) \leq \lambda_1(A+H) - \lambda_1(A) \leq \lambda_1(H).$$

∎

Theorem 10.20 (Weyl Inequality: Simple Version).
Let $A, H \in M_n(\mathbb{C})$ be Hermitian. For each $i = 1, 2, \ldots, n$,

$$\lambda_i(A+H) - \lambda_i(A) \leq \lambda_1(H).$$

Proof.
By Courant–Fischer,

$$\lambda_i(A+H) = \max_{\substack{E \subseteq \mathbb{C}^n \\ \dim E = i}} \min_{\substack{\mathbf{x} \in E \\ \|\mathbf{x}\|=1}} \langle \mathbf{x}, (A+H)\mathbf{x} \rangle .$$

We want to show

$$\max_{E} \min_{\substack{\mathbf{x} \in E \\ \|\mathbf{x}\|=1}} \langle \mathbf{x}, (A+H)\mathbf{x} \rangle \leq \lambda_i(A) + \lambda_1(H).$$

It is enough to show that for every subspace $E \subseteq \mathbb{C}^n$ with $\dim E = i$,

$$\min_{\substack{\mathbf{x} \in E \\ \|\mathbf{x}\|=1}} \langle \mathbf{x}, (A+H)\mathbf{x} \rangle \leq \lambda_i(A) + \lambda_1(H).$$

Choose a subspace $W_A \subseteq \mathbb{C}^n$ with $\dim W_A = n - i + 1$ such that

$$\max_{\substack{\mathbf{y} \in W_A \\ \|\mathbf{y}\|=1}} \langle \mathbf{y}, A\mathbf{y} \rangle = \lambda_i(A).$$

For any subspace E with $\dim E = i$, we have $\dim E + \dim W_A = i + (n - i + 1) = n + 1$. Hence $E \cap W_A \neq \{\mathbf{0}\}$.
Choose a unit vector $\mathbf{x} \in E \cap W_A$. Then

$$\langle \mathbf{x}, A\mathbf{x} \rangle \leq \lambda_i(A), \qquad \langle \mathbf{x}, H\mathbf{x} \rangle \leq \lambda_1(H).$$

Hence

$$\langle \mathbf{x}, (A+H)\mathbf{x} \rangle = \langle \mathbf{x}, A\mathbf{x} \rangle + \langle \mathbf{x}, H\mathbf{x} \rangle \leq \lambda_i(A) + \lambda_1(H).$$

Since this is true for some unit vector $\mathbf{x} \in E$,

$$\min_{\substack{\mathbf{z} \in E \\ \|\mathbf{z}\|=1}} \langle \mathbf{z}, (A+H)\mathbf{z} \rangle \leq \lambda_i(A) + \lambda_1(H).$$

Taking the maximum over all such E gives the desired inequality. ∎

Corollary .
For Hermitian matrices $A, H \in M_n(\mathbb{C})$,

$$\lambda_i(A+H) \geq \lambda_i(A) + \lambda_n(H).$$

Consequently,

$$\lambda_n(H) \leq \lambda_i(A+H) - \lambda_i(A) \leq \lambda_1(H).$$

Proof.
Use

$$\lambda_i(A+H) = -\lambda_{n-i+1}(-(A+H)).$$

By the previous theorem applied to $-A$ and $-H$,

$$\lambda_{n-i+1}((-A) + (-H)) \leq \lambda_{n-i+1}(-A) + \lambda_1(-H).$$

Multiplying by -1 gives

$$\begin{aligned} \lambda_i(A+H) &\geq -\lambda_{n-i+1}(-A) - \lambda_1(-H) \\ &= \lambda_i(A) + \lambda_n(H). \end{aligned}$$

Combining this with the upper bound gives

$$\lambda_n(H) \leq \lambda_i(A+H) - \lambda_i(A) \leq \lambda_1(H).$$

∎

Theorem 10.21 (Continuity of eigenvalues).
For Hermitian matrices $A, B \in M_n(\mathbb{C})$,

$$|\lambda_i(A) - \lambda_i(B)| \leq \|A - B\|_F, \qquad i = 1, 2, \ldots, n.$$

Hence the mapping from a Hermitian matrix to its i-th eigenvalue is 1*-Lipschitz continuous with respect to the Frobenius norm, and hence continuous.*

Proof.
Apply the previous corollary with base matrix B and perturbation $H = A - B$. Then $B + H = A$, so

$$\lambda_n(A - B) \leq \lambda_i(A) - \lambda_i(B) \leq \lambda_1(A - B).$$

Therefore,

$$|\lambda_i(A) - \lambda_i(B)| \leq \max\{|\lambda_1(A - B)|, |\lambda_n(A - B)|\}.$$

Using $\left|\lambda_j(C)\right| \leq \|C\|_F$ for every j and taking $C = A - B$, we get

$$\max\{|\lambda_1(A - B)|, |\lambda_n(A - B)|\} \leq \|A - B\|_F.$$

Therefore,

$$|\lambda_i(A) - \lambda_i(B)| \leq \|A - B\|_F.$$

∎

Theorem 10.22 .
For any $C \in M_n(\mathbb{C})$ and every eigenvalue $\lambda_j(C)$ of C, $\left|\lambda_j(C)\right| \leq \|C\|_F$.

Proof.
Let $\rho(C) = \max_j \left|\lambda_j(C)\right|$ be the spectral radius of C. It suffices to show that $\rho(C) \leq \|C\|_F$.
By JCF, there exists an invertible matrix P such that $C = PJP^{-1}$, where J is a Jordan matrix. Hence $C^m = PJ^mP^{-1}$ for every $m \geq 1$.
We first show that

$$\lim_{m\to\infty} \|C^m\|_F^{1/m} = \rho(C).$$

It is enough to prove this for J. By submultiplicativity of the Frobenius norm,

$$\|C^m\|_F = \left\|PJ^mP^{-1}\right\|_F \leq \|P\|_F \|J^m\|_F \left\|P^{-1}\right\|_F.$$

Let $K = \|P\|_F \|P^{-1}\|_F$. Then $\|C^m\|_F \leq K \|J^m\|_F$.

On the other hand,

$$\|J^m\|_F = \left\|P^{-1}C^mP\right\|_F \leq \left\|P^{-1}\right\|_F \|C^m\|_F \|P\|_F = K \|C^m\|_F.$$

Therefore,

$$\frac{1}{K} \|J^m\|_F \leq \|C^m\|_F \leq K \|J^m\|_F.$$

Taking the m-th root gives

$$K^{-1/m} \|J^m\|_F^{1/m} \leq \|C^m\|_F^{1/m} \leq K^{1/m} \|J^m\|_F^{1/m}.$$

Since $K > 0$ is independent of m, we have $K^{1/m} = e^{(\log K)/m} \to 1$ and $K^{-1/m} = e^{-(\log K)/m} \to 1$. Thus C^m and J^m have the same growth rate after taking the m-th root.

Now consider one Jordan block $J_r(\lambda) = \lambda I + N$, where $N^r = 0$.

If $\lambda = 0$, then $J_r(0) = N$ is nilpotent, so $J_r(0)^m = 0$ for all $m \geq r$. Hence

$$\lim_{m\to\infty} \|J_r(0)^m\|_F^{1/m} = 0 = |\lambda|.$$

Now assume $\lambda \neq 0$. By the binomial theorem,

$$J_r(\lambda)^m = (\lambda I + N)^m = \sum_{q=0}^{r-1} \binom{m}{q} \lambda^{m-q} N^q.$$

The sum stops at $r-1$ because $N^r = 0$. Since $q \leq r-1$ is bounded, $\binom{m}{q}$ grows at most polynomially in m. Hence there exists a constant $C_0 > 0$ such that

$$\|J_r(\lambda)^m\|_F \leq C_0 m^{r-1} |\lambda|^m$$

for all sufficiently large m. Taking the m-th root gives

$$\|J_r(\lambda)^m\|_F^{1/m} \leq C_0^{1/m} (m^{r-1})^{1/m} |\lambda|.$$

Since $C_0^{1/m} \to 1$ and $(m^{r-1})^{1/m} = e^{((r-1)\log m)/m} \to 1$, we obtain

$$\limsup_{m\to\infty} \|J_r(\lambda)^m\|_F^{1/m} \leq |\lambda|.$$

On the other hand, the diagonal entries of $J_r(\lambda)^m$ are all λ^m. Since the Frobenius norm is at least the absolute value of any entry, we have $\|J_r(\lambda)^m\|_F \geq |\lambda|^m$. Taking the m-th root gives $\|J_r(\lambda)^m\|_F^{1/m} \geq |\lambda|$, and hence

$$\liminf_{m\to\infty} \|J_r(\lambda)^m\|_F^{1/m} \geq |\lambda|.$$

Combining the upper and lower estimates,

$$|\lambda| \leq \liminf_{m\to\infty} \|J_r(\lambda)^m\|_F^{1/m} \leq \limsup_{m\to\infty} \|J_r(\lambda)^m\|_F^{1/m} \leq |\lambda|\,.$$

Therefore,

$$\lim_{m\to\infty} \|J_r(\lambda)^m\|_F^{1/m} = |\lambda|\,.$$

Now write J as a block diagonal matrix

$$J = \begin{pmatrix} J_{r_1}(\lambda_1) & & \mathbf{O} \\ & \ddots & \\ \mathbf{O} & & J_{r_s}(\lambda_s) \end{pmatrix}.$$

Then

$$\|J^m\|_F^2 = \sum_{\ell=1}^{s} \|J_{r_\ell}(\lambda_\ell)^m\|_F^2\,.$$

For each block, we have shown that $\|J_{r_\ell}(\lambda_\ell)^m\|_F^{1/m} \to |\lambda_\ell|$.
Let $M = \max_{1\leq\ell\leq s} |\lambda_\ell|$. Then $M = \rho(C)$. For the lower bound, choose ℓ_0 such that $\left|\lambda_{\ell_0}\right| = M$. Since $\|J^m\|_F \geq \left\|J_{r_{\ell_0}}(\lambda_{\ell_0})^m\right\|_F$, we get

$$\liminf_{m\to\infty} \|J^m\|_F^{1/m} \geq M.$$

For the upper bound, fix $\varepsilon > 0$. For each block, we have

$$\lim_{m\to\infty} \|J_{r_\ell}(\lambda_\ell)^m\|_F^{1/m} = |\lambda_\ell| \leq M.$$

Therefore, for each ℓ, there exists N_ℓ such that whenever $m \geq N_\ell$,

$$\|J_{r_\ell}(\lambda_\ell)^m\|_F^{1/m} \leq M + \varepsilon.$$

Equivalently,

$$\|J_{r_\ell}(\lambda_\ell)^m\|_F \leq (M+\varepsilon)^m.$$

Since there are only finitely many blocks, we can choose $N = \max\{N_1, \ldots, N_s\}$. Then for all $m \geq N$ and all $\ell = 1, \ldots, s$,

$$\|J_{r_\ell}(\lambda_\ell)^m\|_F \leq (M+\varepsilon)^m.$$

Since J is block diagonal,

$$\|J^m\|_F^2 = \sum_{\ell=1}^{s} \|J_{r_\ell}(\lambda_\ell)^m\|_F^2\,.$$

Thus, for all $m \geq N$,

$$\|J^m\|_F^2 \leq \sum_{\ell=1}^{s} (M+\varepsilon)^{2m} = s(M+\varepsilon)^{2m}.$$

Taking square roots gives

$$\|J^m\|_F \leq \sqrt{s}(M+\varepsilon)^m.$$

Taking the m-th root gives

$$\|J^m\|_F^{1/m} \leq s^{1/(2m)}(M+\varepsilon).$$

Since $s^{1/(2m)} \to 1$, we obtain

$$\limsup_{m\to\infty} \|J^m\|_F^{1/m} \leq M+\varepsilon.$$

Since $\varepsilon > 0$ is arbitrary,

$$\limsup_{m\to\infty} \|J^m\|_F^{1/m} \leq M.$$

Therefore,

$$\lim_{m\to\infty} \|J^m\|_F^{1/m} = M = \rho(C).$$

By the comparison between C^m and J^m above, we also get

$$\lim_{m\to\infty} \|C^m\|_F^{1/m} = \rho(C).$$

Finally, since $\|C^m\|_F \leq \|C\|_F^m$, we take the m-th root to give $\|C^m\|_F^{1/m} \leq \|C\|_F$. Letting $m \to \infty$, we obtain $\rho(C) \leq \|C\|_F$.
Since $\left|\lambda_j(C)\right| \leq \rho(C)$ for every j, we conclude that

$$\left|\lambda_j(C)\right| \leq \|C\|_F.$$

■

Remark .
Now we show that if C is Hermitian, then $\left|\lambda_j(C)\right| \leq \|C\|_F$ for every j.

Since C is Hermitian, by the Spectral Theorem, there exists a unitary matrix U such that $C = UDU^*$, where $D = \mathrm{diag}\big(\lambda_1(C), \lambda_2(C), \ldots, \lambda_n(C)\big)$. Hence

$$\|C\|_F^2 = \mathrm{tr}(C^*C) = \mathrm{tr}\big((UDU^*)^*(UDU^*)\big)$$

$$= \operatorname{tr}(UD^*DU^*) = \operatorname{tr}(U^*UD^*D) = \operatorname{tr}(D^*D) = \sum_{j=1}^{n} \left|\lambda_j(C)\right|^2.$$

Therefore, for every j,

$$\left|\lambda_j(C)\right|^2 \leq \sum_{j=1}^{n} |\lambda_\ell(C)|^2 = \|C\|_F^2.$$

Thus

$$\left|\lambda_j(C)\right| \leq \|C\|_F.$$

Theorem 10.23 (Weyl's Inequality for Singular Values).
For general matrices $A, B \in M_{m\times n}(\mathbb{C})$,

$$|\sigma_i(A) - \sigma_i(B)| \leq \sqrt{2}\,\|A - B\|_F.$$

Remark .
The mapping from a matrix to its singular values is therefore $\sqrt{2}$-Lipschitz, and hence continuous.

A stronger version is also true:

$$|\sigma_i(A) - \sigma_i(B)| \leq \|A - B\|_F,$$

but this is harder to prove.

Proof.
We have already shown that for Hermitian matrices,

$$|\lambda_i(A) - \lambda_i(B)| \leq \|A - B\|_F.$$

For general matrices $A, B \in M_{m\times n}(\mathbb{C})$, consider the Hermitian dilations

$$\widetilde{A} = \begin{pmatrix} \mathbf{O}_{m\times m} & A \\ A^* & \mathbf{O}_{n\times n} \end{pmatrix}, \qquad \widetilde{B} = \begin{pmatrix} \mathbf{O}_{m\times m} & B \\ B^* & \mathbf{O}_{n\times n} \end{pmatrix}.$$

Then $\widetilde{A}$ and $\widetilde{B}$ are Hermitian.
The goal is to compute the eigenvalues of $\widetilde{A}$. We have

$$\widetilde{A}^2 = \begin{pmatrix} \mathbf{O} & A \\ A^* & \mathbf{O} \end{pmatrix} \begin{pmatrix} \mathbf{O} & A \\ A^* & \mathbf{O} \end{pmatrix} = \begin{pmatrix} AA^* & \mathbf{O} \\ \mathbf{O} & A^*A \end{pmatrix}.$$

Hence the eigenvalues of $\widetilde{A}^2$ are the eigenvalues of AA^* together with the eigenvalues of A^*A.
Let $\operatorname{rank}(A) = k$. Then the nonzero eigenvalues of AA^* and A^*A are $\sigma_1(A)^2, \sigma_2(A)^2, \ldots, \sigma_k(A)^2$. Thus the nonzero eigenvalues of $\widetilde{A}^2$ are

$$\sigma_1(A)^2, \ldots, \sigma_k(A)^2, \sigma_1(A)^2, \ldots, \sigma_k(A)^2.$$

Therefore, every nonzero eigenvalue of $\widetilde{A}$ must be of the form $\pm\sigma_i(A)$.
Now we show that the signs appear in pairs. Let

$$S = \begin{pmatrix} I_m & \mathbf{O} \\ \mathbf{O} & -I_n \end{pmatrix}.$$

Then

$$S\widetilde{A}S = -\widetilde{A}.$$

Equivalently, $\widetilde{A}S = -S\widetilde{A}$. Hence, if $\widetilde{A}\mathbf{z} = \lambda\mathbf{z}$, then

$$\widetilde{A}(S\mathbf{z}) = -S\widetilde{A}\mathbf{z} = -\lambda S\mathbf{z}.$$

Since S is invertible, $S\mathbf{z} \neq \mathbf{0}$. Therefore, whenever λ is an eigenvalue of $\widetilde{A}$, $-\lambda$ is also an eigenvalue of $\widetilde{A}$ with the same multiplicity.
Therefore the eigenvalues of $\widetilde{A}$ are

$$\sigma_1(A), \ldots, \sigma_k(A), 0, \ldots, 0, -\sigma_k(A), \ldots, -\sigma_1(A).$$

Here the zero eigenvalues fill the remaining dimensions. In particular, if we pad the singular values by setting $\sigma_i(A) = 0$ for $i > \operatorname{rank}(A)$, then

$$\lambda_i(\widetilde{A}) = \sigma_i(A), \qquad i = 1, 2, \ldots, \min\{m, n\}.$$

Similarly,

$$\lambda_i(\widetilde{B}) = \sigma_i(B), \qquad i = 1, 2, \ldots, \min\{m, n\}.$$

Thus, by the continuity of eigenvalues for Hermitian matrices,

$$|\sigma_i(A) - \sigma_i(B)| = \left|\lambda_i(\widetilde{A}) - \lambda_i(\widetilde{B})\right| \leq \left\|\widetilde{A} - \widetilde{B}\right\|_F.$$

It remains to compute the Frobenius norm. Let $C = A - B$. Then

$$\widetilde{A} - \widetilde{B} = \begin{pmatrix} \mathbf{O} & C \\ C^* & \mathbf{O} \end{pmatrix}.$$

Since $\widetilde{A} - \widetilde{B}$ is Hermitian,

$$\begin{aligned} \left\|\widetilde{A} - \widetilde{B}\right\|_F^2 &= \operatorname{tr}\left((\widetilde{A} - \widetilde{B})^*(\widetilde{A} - \widetilde{B})\right) = \operatorname{tr}\left((\widetilde{A} - \widetilde{B})^2\right) \\ &= \operatorname{tr}\begin{pmatrix} CC^* & \mathbf{O} \\ \mathbf{O} & C^*C \end{pmatrix} \\ &= \operatorname{tr}(CC^*) + \operatorname{tr}(C^*C) = 2\operatorname{tr}(C^*C) = 2\|C\|_F^2 = 2\|A - B\|_F^2. \end{aligned}$$

Hence

$$\left\| \widetilde{A} - \widetilde{B} \right\|_F = \sqrt{2} \, \|A - B\|_F \, .$$

Therefore,

$$|\sigma_i(A) - \sigma_i(B)| \le \sqrt{2} \, \|A - B\|_F \, , \qquad i = 1, 2, \ldots, \min\{m, n\}.$$

■

11 Markov Chain and Perron-Frobenius

11.1 Perron-Frobenius Theorem

Definition 11.1 .
Consider a Markov chain of n finite states. Let p_{ij} be the probability to transit from state i to state j. Then $A = (p_{ij})^t$ is called the stochastic matrix of this Markov chain, where every entry $A_{ij} \geq 0$, and each column sums to 1.

Remark .
The official definition: A sequence of random variables $X_1, X_2, \dots$ is called a Markov chain if

$$\mathbb{P}(X_n = x_n \mid X_1 = x_1, \dots, X_{n-1} = x_{n-1}) = \mathbb{P}(X_n = x_n \mid X_{n-1} = x_{n-1}).$$

Theorem 11.1 .
Let $A \in M_n(\mathbb{C})$. Then $\lim\limits_{k\to\infty} A^k$ exists if and only if both of the following conditions are fulfilled:

1. *Every eigenvalue of A belongs to $S = \{z \in \mathbb{C} : |z| < 1 \text{ or } z = 1\}$;*
2. *If 1 is an eigenvalue of A, then its algebraic and geometric multiplicities are equal, i.e. $a_A(1) = g_A(1)$.*

Proof.
Proof of ($\Rightarrow$): Assume $\lim\limits_{m\to\infty} A^m$ exists. By JCF Theorem, there exists an invertible matrix P such that $A = PJP^{-1}$, where J is the Jordan canonical form of A. Hence $A^m = PJ^mP^{-1}$, so

$$J^m = P^{-1}A^mP.$$

Since $\lim\limits_{m\to\infty} A^m$ exists, $\lim\limits_{m\to\infty} J^m$ also exists. Thus it is enough to study each Jordan block of J.
Let $\lambda \in \mathrm{Spec}(A)$. Then some Jordan block $J_{\lambda,r}$ appears in J. The diagonal entries of $J_{\lambda,r}^m$ are all λ^m, so the convergence of J^m implies that λ^m converges as $m \to \infty$.
If $|\lambda| > 1$, then $|\lambda^m| \to \infty$, impossible. If $|\lambda| < 1$, then $\lambda^m \to 0$, which is allowed. Now suppose $|\lambda| = 1$ and $\lambda^m \to L$. Since $|\lambda^m| = 1$ for all m, we have $|L| = 1$, so $L \neq 0$. Also λ^{m+1} is just the shifted sequence of λ^m, so it has the same limit L. On the other hand, $\lambda^{m+1} = \lambda\lambda^m \to \lambda L$. Hence $\lambda L = L$.

Since $L \neq 0$, we get $\lambda = 1$. Therefore every eigenvalue of A lies in

$$S = \{z \in \mathbb{C} : |z| < 1 \text{ or } z = 1\}.$$

Next suppose $1 \in \operatorname{Spec}(A)$. We prove that $a_A(1) = g_A(1)$. Suppose not. Then there is a Jordan block for $\lambda = 1$ with index at least 2. That is, for some $r \geq 2$,

$$J_{1,r} = \begin{pmatrix} 1 & 1 & 0 & \cdots & 0 \\ 0 & 1 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix} = I_r + N,$$

where N is the nilpotent matrix with 1 on the superdiagonal and 0 elsewhere. By binomial expansion,

$$J_{1,r}^m = (I_r + N)^m = I_r + mN + \binom{m}{2} N^2 + \cdots + \binom{m}{r-1} N^{r-1}.$$

In particular, the $(1,2)$-entry of $J_{1,r}^m$ is m, which diverges as $m \to \infty$. This contradicts the convergence of J^m. Therefore every Jordan block corresponding to $\lambda = 1$ has index 1, so $a_A(1) = g_A(1)$.
Proof of ($\Leftarrow$): Assume A satisfies (1) and (2). By JCF Theorem,

$$A^m \sim J_{\lambda_1,k_1}^m \oplus J_{\lambda_2,k_2}^m \oplus \cdots \oplus J_{\lambda_l,k_l}^m.$$

It suffices to study the convergence of $J_{\lambda,k}^m$ as $m \to \infty$. By (1), $\lambda \in S$. There are two cases.
Case 1. $|\lambda| < 1$.

$$J_{\lambda,k} = \lambda I + N \implies J_{\lambda,k}^m = (\lambda I + N)^m.$$

Using binomial expansion, $a_{m,k} = \binom{m}{k}\lambda^{m-k}$ for the upper diagonal entries. By Ratio Test, as $m \to \infty$,

$$\left|\frac{a_{m+1,k}}{a_{m,k}}\right| = \left|\frac{m+1}{m+1-k}\lambda\right| \to |\lambda| < 1.$$

Hence $\lim\limits_{m\to\infty} a_{m,k} = 0$, meaning $\lim\limits_{m\to\infty} J_{\lambda,k}^m = \mathbf{O}$.
Case 2. $\lambda = 1$. By condition (2), $a_A(1) = g_A(1)$, meaning $A|_{E(1)}$ is diagonalizable. Every Jordan Block of $\lambda = 1$ has index 1. In this case, $J_{1,1}^m = (1)^m$, which clearly converges as $m \to \infty$. In all cases, $\lim\limits_{m\to\infty} A^m$ exists. ∎

Definition 11.2 .
Let $A \in M_n(\mathbb{C})$. Set, for each $1 \leq i, j \leq n$:

1. $\rho_i(A) \stackrel{\text{def}}{=} \sum_{j=1}^{n} |A_{ij}|$ *(the modulus length of entries in the i-th row)*
2. $\nu_j(A) \stackrel{\text{def}}{=} \sum_{i=1}^{n} |A_{ij}|$ *(the j-th column sum)*

In addition, we put $\rho(A) \stackrel{\text{def}}{=} \max_{1 \leq i \leq n} \{\rho_i(A)\}$ and $\nu(A) \stackrel{\text{def}}{=} \max_{1 \leq j \leq n} \{\nu_j(A)\}$.

Theorem 11.2 (Gershgorin's Disk).
For $A \in M_n(\mathbb{C})$, for each $1 \leq i \leq n$ define the i-th Gershgorin's disk as

$$C_i = \{z \in \mathbb{C} : |z - A_{ii}| \leq \rho_i(A) - |A_{ii}|\}.$$

Then for any eigenvalue λ of A, one has $\lambda \in C_1 \cup C_2 \cup \cdots \cup C_n$.

Proof.
Let $A = (A_{ij})_{i,j}$ and $\lambda \in \operatorname{Spec}(A)$, with $\mathbf{v} = (v_1, \ldots, v_n)^t$ being its eigenvector ($A\mathbf{v} = \lambda\mathbf{v}$). Let $1 \leq k \leq n$ be such that $|v_k| \geq |v_j|$ for all j. From $A\mathbf{v} = \lambda\mathbf{v}$, we get

$$|\lambda v_k - A_{kk} v_k| = \left| \sum_{j \neq k} A_{kj} v_j \right|.$$

Using the triangle inequality:

$$|\lambda - A_{kk}||v_k| \leq \sum_{j \neq k} |A_{kj}||v_j| \leq \sum_{j \neq k} |A_{kj}||v_k| = (\rho_k(A) - |A_{kk}|)|v_k|.$$

Since $|v_k| \neq 0$, after cancellation, $|\lambda - A_{kk}| \leq \rho_k(A) - |A_{kk}|$, implying $\lambda \in C_k$. ∎

Theorem 11.3 .
Let λ be an eigenvalue of $A \in M_n(\mathbb{C})$. Then $|\lambda| \leq \min\{\rho(A), \nu(A)\}$.

Proof.
By Gershgorin's Disk Theorem, $\lambda \in C_k$ for some k.

$$\begin{aligned} |\lambda| = |\lambda - A_{kk} + A_{kk}| &\leq |\lambda - A_{kk}| + |A_{kk}| \\ &\leq \rho_k(A) - |A_{kk}| + |A_{kk}| = \rho_k(A) \leq \rho(A). \end{aligned}$$

Since $A \sim A^t$, if $\lambda \in \operatorname{Spec}(A)$, then $\lambda \in \operatorname{Spec}(A^t)$. Thus $|\lambda| \leq \rho(A^t) = \nu(A)$, yielding $|\lambda| \leq \min\{\rho(A), \nu(A)\}$. ∎

Theorem 11.4 .
Let A be a stochastic matrix.

1. *If λ is an eigenvalue of A, then $|\lambda| \leq 1$.*

2. *1 is an eigenvalue of A.*

Proof.

1. For any $\lambda \in \mathrm{Spec}(A)$, we have $|\lambda| \leq \nu(A) = 1$ because the max column sum of a stochastic matrix is 1.

2. Each row of A^t sums to 1. Thus $A^t\mathbf{e} = 1\mathbf{e}$, where $\mathbf{e} = (1, 1, \ldots, 1)^t$. This implies $1 \in \mathrm{Spec}(A^t)$, and since $A \sim A^t$, $1 \in \mathrm{Spec}(A)$.

■

Definition 11.3 .
A matrix A is called positive if every entry of it is a strictly positive real number.

Theorem 11.5 (Perron-Frobenius, Positive Version).
Let A be a positive matrix. Suppose there exists an eigenvalue λ of A such that $|\lambda| = \rho(A)$, then:

1. *$\lambda = \rho(A)$ (i.e., λ is a positive real number)*

2. *$\ker(A - \lambda I) = \mathrm{span}\{\mathbf{e}\}$ where $\mathbf{e} = (1, 1, \ldots, 1)^t$ (i.e., $g_A(\lambda) = 1$)*

Remark .
This version does not assert the existence of an eigenvalue satisfying $|\lambda| = \rho(A)$. Rather, it tells us what can be concluded if such an eigenvalue exists.

Proof.
Suppose $\lambda \in \mathrm{Spec}(A)$ such that $|\lambda| = \rho(A)$. Let $\mathbf{v} = (v_1, \ldots, v_n)^t$ be any eigenvector of A corresponding to λ. Thus $A\mathbf{v} = \lambda\mathbf{v}$ and $\mathbf{v} \neq \mathbf{0}$. Choose k such that $|v_k| \geq |v_j|$ for every j. Then $|v_k| > 0$.

Looking at the k-th row of $A\mathbf{v} = \lambda\mathbf{v}$, we have $\lambda v_k = \sum_{j=1}^{n} A_{kj} v_j$. Hence

$$\begin{aligned}
|\lambda||v_k| = |\lambda v_k| &= \left|\sum_{j=1}^{n} A_{kj} v_j\right| \\
&\leq \sum_{j=1}^{n} |A_{kj} v_j| = \sum_{j=1}^{n} A_{kj} |v_j| \\
&\leq \sum_{j=1}^{n} A_{kj} |v_k| = \rho_k(A)|v_k| \\
&\leq \rho(A)|v_k| = |\lambda||v_k|.
\end{aligned}$$

The first and last terms are equal, so every inequality in this chain must be an equality.

First, from

$$\sum_{j=1}^{n} A_{kj}|v_j| = \sum_{j=1}^{n} A_{kj}|v_k|,$$

we get

$$\sum_{j=1}^{n} A_{kj}(|v_k| - |v_j|) = 0.$$

For every j, $A_{kj} > 0$ because A is positive, and $|v_k| - |v_j| \geq 0$ by the choice of k. Hence $|v_j| = |v_k|$ for all j.

Second, from

$$\left|\sum_{j=1}^{n} A_{kj} v_j\right| = \sum_{j=1}^{n} |A_{kj} v_j|,$$

we have equality in the triangle inequality. Recall that for nonzero complex numbers z_1, z_2, $|z_1 + z_2| = |z_1| + |z_2|$ if and only if $z_1 = r z_2$ for some $r > 0$. Similarly, for finitely many nonzero complex numbers, $A_{kj} v_j$ are positive multiples of each other for all j. So $A_{kj} v_j = c_j z$ for some $c_j > 0$, $z \in \mathbb{C}$ with $|z| = 1$. Since $A_{kj} > 0$, $v_j = \frac{c_j}{A_{kj}} z$. Moreover, $|v_j| = |v_k|$ for all j. Put $b = |v_k| > 0$. Then $\frac{c_j}{A_{kj}} = b$, so $v_j = bz$ for all j. Thus

$$\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} = \begin{pmatrix} bz \\ bz \\ \vdots \\ bz \end{pmatrix} = bz \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} = bz\mathbf{e}.$$

Thus every eigenvector corresponding to λ is a scalar multiple of $\mathbf{e}$, $\ker(A - \lambda I) = \operatorname{span}\{\mathbf{e}\}$. Since $\mathbf{v} = bz\mathbf{e}$ and $bz \neq 0$, from $A\mathbf{v} = \lambda\mathbf{v}$ we get

$$A(bz\mathbf{e}) = \lambda(bz\mathbf{e}).$$

Dividing both sides by bz, we obtain $A\mathbf{e} = \lambda\mathbf{e}$. Thus the i-th entry gives $\sum_{j=1}^{n} A_{ij} = \lambda$ for every i. Since A is positive, each row sum $\sum_{j=1}^{n} A_{ij}$ is a positive real number. Hence $\lambda > 0$. Together with $|\lambda| = \rho(A)$, we conclude that $\lambda = \rho(A)$. ■

Corollary .
Let A be a positive matrix. Suppose there exists an eigenvalue λ of A such that $|\lambda| = \nu(A)$, then:

1. *$\lambda = \nu(A)$ (i.e., λ is a positive real number)*
2. *$g_A(\lambda) = \dim_{\mathbb{C}} \ker(A - \lambda I) = 1$*

Remark .
Unlike the row-sum version, the column-sum version does not give the concrete description $\ker(A - \lambda I) = \operatorname{span}\{\mathbf{e}\}$. *Instead, it only tells us that the eigenspace has dimension* 1.

Proof.
Suppose $\lambda \in \operatorname{Spec}(A)$ such that $|\lambda| = \nu(A)$. Since A and A^t are similar, $\lambda \in \operatorname{Spec}(A^t)$. Also, A^t is positive and $\rho(A^t) = \nu(A)$. Therefore

$$|\lambda| = \nu(A) = \rho(A^t).$$

Applying the row-sum Perron-Frobenius positive version to A^t, we obtain $\lambda = \rho(A^t)$. Hence $\lambda = \nu(A)$, so λ is a positive real number. We also obtain

$$\ker(A^t - \lambda I) = \operatorname{span}\{\mathbf{e}\},$$

so $g_{A^t}(\lambda) = g_A(\lambda) = 1$. ■

Definition 11.4 .
A stochastic matrix A is called regular if there exists integer s such that A^s is a positive matrix.

Remark .
If A is positive and A is stochastic, then A^{s+1} is also positive.

Theorem 11.6 (Perron-Frobenius, Regular Stochastic Version).
Let A be a regular stochastic matrix and λ be an eigenvalue of A. Then:

1. $|\lambda| \leq 1$.
2. *If* $|\lambda| = 1$ *then* $\lambda = 1$ *and* $g_A(1) = 1$.
3. *There exists a 'positive eigenvector'* $\mathbf{v} = (v_1, \ldots, v_n)^t$ *of A with eigenvalue* 1 *such that* $v_i > 0$ *for all i and* $\sum_{i=1}^{n} v_i = 1$; *This* $\mathbf{v}$ *is called the stationary probability vector.*
4. $a_A(1) = 1$.
5. $L = \lim_{m\to\infty} A^m$ *exists.*
6. *The columns of L are identical and are equal to the eigenvector* $\mathbf{v}$ *in (3).*

Proof.
Proof of (1) and (2). Given A is a regular, A^s is a positive matrix for some s. Let $\lambda \in \text{Spec}(A)$. By Spectral Mapping Theorem, $\lambda^s \in \text{Spec}(A^s)$. Since A is a stochastic matrix, (so is A^s), by Gershgorin Disk Theorem, $|\lambda| \leq \nu(A) = 1$. Suppose $|\lambda| = 1$, then $|\lambda^s| = |1|^s = 1 = \nu(A^s)$. Apply Perron-Frobenius Theorem (positive ver., column) on A^s, $\lambda^s > 0 \implies \lambda^s = 1$. Repeat the argument for A^{s+1}, we obtain $\lambda^{s+1} = 1$. Hence, $\lambda = \dfrac{\lambda^{s+1}}{\lambda^s} = \dfrac{1}{1} = 1$.
Regarding geometric multiplicities: Note that $A\mathbf{v} = \lambda\mathbf{v} \implies A^s\mathbf{v} = \lambda^s\mathbf{v}$. Thus, $\ker(A - \lambda I) \subseteq \ker(A^s - \lambda^s I)$. For $\lambda = 1$, take dimension:

$$1 \leq g_A(1) \leq g_{A^s}(1) = 1 \qquad \text{(by PF. Theorem for positive matrices).}$$

Hence $g_A(1) = 1$.
Proof of (3). Let $\mathbf{v} = (v_1, v_2, \ldots, v_n)^t$ be an eigenvector of A of $\lambda = 1$. So $A\mathbf{v} = \mathbf{v}$. Entry-wise, we have $\sum_{j=1}^{n} A_{ij}v_j = v_i$ for each i. Then

$$|v_i| = \left|\sum_{j=1}^{n} A_{ij}v_j\right| \leq \sum_{j=1}^{n} A_{ij}|v_j|$$

by triangle inequality and $A_{ij} \geq 0$. Let $\delta_i = \left(\sum_{j=1}^{n} A_{ij}|v_j|\right) - |v_i| \geq 0$.
Now put $\boldsymbol{\delta} = (\delta_1, \delta_2, \ldots, \delta_n)^t$ and $\mathbf{v}^+ = (|v_1|, |v_2|, \ldots, |v_n|)^t$. Then $\boldsymbol{\delta} = A\mathbf{v}^+ - \mathbf{v}^+$. We want to show $\boldsymbol{\delta} = \mathbf{0}$. Let $\mathbf{e} = (1, 1, \ldots, 1)^t$. Consider the

inner product:

$$\langle A\mathbf{v}^+, \mathbf{e}\rangle = \langle \mathbf{v}^+, A^t\mathbf{e}\rangle = \langle \mathbf{v}^+, \mathbf{e}\rangle$$

since every column of A sums to 1, $A^t\mathbf{e} = \mathbf{e}$. On the other hand, $\langle A\mathbf{v}^+, \mathbf{e}\rangle = \langle \delta, \mathbf{e}\rangle + \langle \mathbf{v}^+, \mathbf{e}\rangle$, this implies $\langle \delta, \mathbf{e}\rangle = 0$, so $\sum_{i=1}^{n} \delta_i = 0$. Since $\delta_i \geq 0$, this forces $\delta_i = 0$ for all i. Hence, $A\mathbf{v}^+ = \mathbf{v}^+$.

As A is regular, A^s is positive for some s. So $A^s\mathbf{v}^+ = \mathbf{v}^+$. Entry-wise:

$$|v_i| = \sum_{j=1}^{n} (A^s)_{ij}|v_j| > 0$$

since $(A^s)_{ij} > 0$ and at least one $|v_j| > 0$ because $\mathbf{v} \neq \mathbf{0}$. Thus $|v_i| > 0$ for all i. We can rescale $\mathbf{v}^+$ by dividing by $\sum |v_i|$ to obtain a "positive eigenvector" whose entries sum to 1.

Proof of (4). By (2), we have $g_A(1) = 1 \implies$ There is only one Jordan block of eigenvalue 1. $P^{-1}AP = J_{1,a} \oplus K$, where K consists of Jordan blocks of other eigenvalues. So $P^{-1}A^mP = (J_{1,a})^m \oplus K^m$.

As A^m is stochastic, its entries are bounded between 0 and 1. So $P^{-1}A^mP$'s entries are bounded (independent of m). Suppose $a \geq 2$. Then

$$(J_{1,a})^m = \begin{pmatrix} 1 & m & \dots \\ 0 & 1 & m \\ \dots & \dots & \dots \end{pmatrix}$$

which is unbounded, contradicting the boundedness. This forces $a = 1$. Hence $a_A(1) = 1$.

Proof of (5). By (1) and (2), for any $\lambda \in \mathrm{Spec}(A)$, we have $\lambda \in S = \{z \in \mathbb{C} : |z| < 1 \text{ or } z = 1\}$. Secondly, for $\lambda = 1$, by (2) and (4), $a_A(1) = g_A(1) = 1$. By Theorem (11.1), $L = \lim_{m\to\infty} A^m$ converges.

Proof of (6). Write the columns of L as

$$L = \begin{pmatrix} | & | & & | \\ \mathbf{c}_1 & \mathbf{c}_2 & \cdots & \mathbf{c}_n \\ | & | & & | \end{pmatrix}.$$

Note: $A^{m+1} = A \cdot A^m \xrightarrow{m\to\infty} L = AL$. Thus

$$\begin{pmatrix} | & | & & | \\ \mathbf{c}_1 & \mathbf{c}_2 & \cdots & \mathbf{c}_n \\ | & | & & | \end{pmatrix} = \begin{pmatrix} | & | & & | \\ A\mathbf{c}_1 & A\mathbf{c}_2 & \cdots & A\mathbf{c}_n \\ | & | & & | \end{pmatrix}.$$

Hence $A\mathbf{c}_i = \mathbf{c}_i$, so $\mathbf{c}_i \in \ker(A - 1I) = \mathrm{span}\{\mathbf{v}\}$. As $\mathbf{c}_i$ have non-negative entries and sum to 1, we have $\mathbf{c}_i = \mathbf{v}$ for all i. ∎

Existence of Positive Eigenvalue and Eigenvector

Definition 11.5 .
Let $A \in M_n(\mathbb{R})$. We denote the spectral radius of A by

$$r(A) \overset{\text{def}}{=} \max\{|\lambda| : \lambda \text{ is an eigenvalue of } A\}.$$

i.e. the maximum modulus of all eigenvalues of A.

Theorem 11.7 (Brouwer's Fixed Point Theorem).
Let $K \subseteq \mathbb{R}^n$ be a non-empty, compact and convex set. If $T : K \to K$ is continuous, then there exists $\mathbf{x} \in K$ such that $T(\mathbf{x}) = \mathbf{x}$.

Proof.
Beyond the scope. ■

Theorem 11.8 .
Let $A = (A_{ij}) \in M_n(\mathbb{R})$ be a positive matrix, i.e. $A_{ij} > 0$ for all i, j. Then there exist $\lambda > 0$ and a vector $\mathbf{v} = (v_1, \dots, v_n)^t$ such that

$$A\mathbf{v} = \lambda\mathbf{v}, \qquad v_i > 0 \text{ for every } i.$$

Proof.
Consider the set (the collection of all "probability vectors")

$$\Delta = \left\{ \mathbf{x} = (x_1, \dots, x_n)^t \in \mathbb{R}^n : x_i \geq 0 \text{ for all } i, \sum_{i=1}^{n} x_i = 1 \right\}.$$

(Exercise: show that Δ is compact and convex). Let $A = (A_{ij})_{i,j}$ and $\mathbf{x} = (x_1, x_2, \dots, x_n)^t \in \Delta$ ($\mathbf{x} \neq \mathbf{0}$). Observe that

$$(A\mathbf{x})_i = \sum_{j=1}^{n} A_{ij} x_j > 0.$$

Define a function $T : \Delta \to \Delta$ by

$$T(\mathbf{x}) = \frac{A\mathbf{x}}{\sum_{i=1}^{n} (A\mathbf{x})_i}.$$

By Brouwer's Fixed Point Theorem, $\exists\, \mathbf{v} \in \Delta$ such that $T(\mathbf{v}) = \mathbf{v}$.
Let $\lambda = \sum_{i=1}^{n} (A\mathbf{v})_i$. By the observation, $\lambda > 0$.
Write $A\mathbf{v} = \lambda\mathbf{v}$, $\lambda > 0$ is an eigenvalue of A and $\mathbf{v} = \frac{1}{\lambda}(A\mathbf{v})$ is a positive eigenvector. ■

Theorem 11.9 .
Let $A \in M_n(\mathbb{R})$ be a positive matrix. Suppose $A\mathbf{v} = \lambda\mathbf{v}$ where $\lambda > 0$ and $\mathbf{v}$ has positive coordinates. Then $\lambda = r(A)$.

Proof.
Given $A\mathbf{v} = \lambda\mathbf{v}$ where $\lambda > 0$ and $\mathbf{v} \in (\mathbb{R}_{>0})^n$. We want to show $\lambda = r(A)$. Let $\mu \in \operatorname{Spec}(A)$ and $\mathbf{z} = (z_1, z_2, \ldots, z_n)^t$ be its eigenvector (i.e. $A\mathbf{z} = \mu\mathbf{z}$). Let $k \in \{1, 2, \ldots, n\}$ be such that

$$c = \frac{|z_k|}{v_k} \geq \frac{|z_i|}{v_i} \qquad \text{for all } i.$$

Since $A\mathbf{z} = \mu\mathbf{z}$, we have $\sum_{j=1}^{n} A_{kj} z_j = \mu z_k$. Hence

$$\begin{aligned} |\mu||z_k| &\leq \sum_{j=1}^{n} A_{kj}|z_j| \\ &\leq \sum_{j=1}^{n} A_{kj}(cv_j) = c\sum_{j=1}^{n} A_{kj} v_j. \end{aligned}$$

Since $A\mathbf{v} = \lambda\mathbf{v}$, we have

$$c\sum_{j=1}^{n} A_{kj} v_j = c(\lambda v_k) = \lambda \frac{|z_k|}{v_k} v_k = \lambda |z_k|.$$

In particular, $|\mu||z_k| \leq \lambda |z_k|$. After cancellation, $|\mu| \leq \lambda$. Hence, $\lambda = r(A)$. ∎

Theorem 11.10 .
Let $A \in M_n(\mathbb{R})$ be a positive matrix. Suppose $\lambda = r(A)$. Then $a_A(\lambda) = g_A(\lambda) = 1$.

Proof.
Let $A \in M_n(\mathbb{R})$ be a positive matrix.
Since A^t is also positive, Theorem (11.8) gives a positive eigenpair

$$A^t\mathbf{w} = \eta\mathbf{w}, \qquad \eta > 0, \qquad w_i > 0 \text{ for every } i.$$

By Theorem (11.9), this positive eigenvalue must be the spectral radius of A^t. Hence

$$\eta = r(A^t) = r(A) = \lambda.$$

Therefore there exists a positive vector $\mathbf{w} = (w_1, w_2, \dots, w_n)^t$ such that

$$A^t\mathbf{w} = \lambda\mathbf{w}, \qquad w_i > 0 \text{ for every } i.$$

Consider

$$D = \begin{pmatrix} w_1 & 0 & \cdots & 0 \\ 0 & w_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & w_n \end{pmatrix} \in GL_n(\mathbb{R}).$$

Let $P = \frac{1}{\lambda}DAD^{-1}$.

We claim that P is positive stochastic. Indeed,

$$P_{ij} = \frac{1}{\lambda}w_i A_{ij}\frac{1}{w_j} > 0$$

for all i, j. Moreover, for every fixed j,

$$\begin{aligned} \sum_{i=1}^{n} P_{ij} &= \sum_{i=1}^{n} \frac{1}{\lambda}\frac{w_i A_{ij}}{w_j} = \frac{1}{\lambda w_j}\sum_{i=1}^{n} A_{ij}w_i \\ &= \frac{1}{\lambda w_j}(A^t\mathbf{w})_j = \frac{1}{\lambda w_j}\lambda w_j = 1. \end{aligned}$$

Thus P is positive stochastic. In particular, P is regular stochastic. By Perron-Frobenius regular stochastic version, $a_P(1) = g_P(1) = 1$.

On the other hand, P is similar to $\frac{1}{\lambda}A$, we have $a_{\frac{1}{\lambda}A}(1) = g_{\frac{1}{\lambda}A}(1) = 1$.

For the geometric multiplicity,

$$\ker\left(\frac{1}{\lambda}A - I\right) = \ker(A - \lambda I),$$

so $g_A(\lambda) = 1$. For the algebraic multiplicity, suppose

$$p_A(x) = (x-\lambda)^\ell (x-\mu_1)(x-\mu_2)\cdots(x-\mu_{n-\ell}).$$

Then

$$\begin{aligned} p_{\frac{1}{\lambda}A}(x) &= \det\left(\frac{1}{\lambda}A - xI\right) = \det\left(\frac{1}{\lambda}(A - \lambda x I)\right) \\ &= \left(\frac{1}{\lambda}\right)^n p_A(\lambda x) \\ &= (x-1)^\ell\left(x - \frac{\mu_1}{\lambda}\right)\left(x - \frac{\mu_2}{\lambda}\right)\cdots\left(x - \frac{\mu_{n-\ell}}{\lambda}\right). \end{aligned}$$

Hence $a_{\frac{1}{\lambda}A}(1) = a_A(\lambda)$. Since $a_{\frac{1}{\lambda}A}(1) = 1$, we get $a_A(\lambda) = 1$. Therefore

$$a_A(\lambda) = g_A(\lambda) = 1.$$

∎

Theorem 11.11 (Perron-Frobenius Existence Theorem).
Let $A = (a_{ij}) \in M_n(\mathbb{R})$ be a nonzero matrix such that all its entries are non-negative (i.e. $a_{ij} \geq 0$ for all i, j). Then:

1. *$r(A)$ is an eigenvalue of A;*
2. *there exists an eigenvector $\mathbf{v} = (v_1, \dots, v_n)^t$ with $v_i \geq 0$ for all i such that $A\mathbf{v} = r(A)\mathbf{v}$.*

Proof.
Recall the simplex

$$\Delta = \left\{ \mathbf{x} = (x_1, \dots, x_n)^t \in \mathbb{R}^n : x_i \geq 0 \text{ for all } i,\ \sum_{i=1}^{n} x_i = 1 \right\}.$$

Every vector in Δ is nonzero and has non-negative entries. Also, Δ is closed and bounded in $\mathbb{R}^n$, hence compact.
Let J be the all-one matrix. For each $\epsilon > 0$, define

$$A_\epsilon = A + \epsilon J.$$

Since $A_{ij} \geq 0$ and $\epsilon > 0$, every entry of A_ϵ is strictly positive. Hence A_ϵ is a positive matrix.
By Theorem (11.8) and Theorem (11.9), $r(A_\epsilon)$ is an eigenvalue of A_ϵ with a positive eigenvector. Thus there exists $\mathbf{u}_\epsilon = (u_{\epsilon,1}, \dots, u_{\epsilon,n})^t$ such that

$$A_\epsilon \mathbf{u}_\epsilon = r(A_\epsilon)\mathbf{u}_\epsilon, \qquad u_{\epsilon,i} > 0 \text{ for every } i.$$

Normalize this eigenvector by setting

$$\mathbf{v}_\epsilon = \frac{\mathbf{u}_\epsilon}{\sum_{i=1}^{n} u_{\epsilon,i}}.$$

Then $\mathbf{v}_\epsilon \in \Delta$, and the eigenvector equation is unchanged:

$$A_\epsilon \mathbf{v}_\epsilon = r(A_\epsilon)\mathbf{v}_\epsilon.$$

Now take a sequence $\epsilon_m = \dfrac{1}{m}$. For each m, choose $\mathbf{v}_{\epsilon_m} \in \Delta$ as above. Since Δ is compact, by Bolzano-Weierstrass there exists a subsequence $\{\mathbf{v}_{\epsilon_{m_\ell}}\}_{\ell=1}^{\infty}$ and a vector $\mathbf{v} \in \Delta$ such that

$$\mathbf{v}_{\epsilon_{m_\ell}} \longrightarrow \mathbf{v} \qquad \text{as } \ell \to \infty.$$

Since $\epsilon_{m_\ell} \to 0$, we also have

$$A_{\epsilon_{m_\ell}} = A + \epsilon_{m_\ell} J \longrightarrow A.$$

Moreover, the spectral radius is continuous as a function of the entries of a matrix. Hence

$$r(A_{\epsilon_{m_\ell}}) \longrightarrow r(A).$$

For every ℓ, we have

$$A_{\epsilon_{m_\ell}} \mathbf{v}_{\epsilon_{m_\ell}} = r(A_{\epsilon_{m_\ell}}) \mathbf{v}_{\epsilon_{m_\ell}}.$$

Taking $\ell \to \infty$ on both sides, matrix multiplication and scalar multiplication are continuous, so

$$A\mathbf{v} = r(A)\mathbf{v}.$$

Finally, since $\mathbf{v} \in \Delta$, we have $\mathbf{v} \neq \mathbf{0}$ and $v_i \geq 0$ for every i. Therefore $r(A)$ is an eigenvalue of A, and $\mathbf{v}$ is a non-negative eigenvector corresponding to $r(A)$. ■

11.2 Google PageRank algorithm

PageRank, developed by Larry Page and Sergey Brin in 1996, is an algorithm used by Google Search to rank webpages in their search engine results. It assigns a ranking to each page on the web based on the structure of the hyperlinks between pages. The basic idea is that a page is important if it is linked to by other important pages.

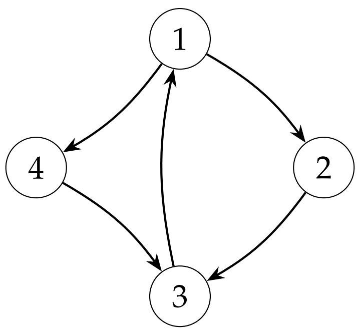


FIGURE 1: A small web as a directed graph. Each arrow represents a hyperlink from one page to another.

In the column-stochastic convention, the j-th column describes what happens when the surfer is currently on page j. If page j links to d_j pages, then each linked page receives probability $\frac{1}{d_j}$ in column j.

A dangling page is a page with no outgoing links. In the raw link matrix, this gives a zero column, so the matrix is not stochastic. The usual fix is to replace each zero column by $\frac{1}{n}\mathbf{e}$, meaning that if the surfer lands on a dangling page, the next page is chosen uniformly from all pages.

The random surfing idea has one more ingredient. At each step, with probability α, the surfer follows the hyperlink structure encoded by P; with probability $1-\alpha$, the surfer ignores the links and jumps uniformly to one of the n pages. This uniform jump is called teleportation. It prevents the surfer from getting trapped and makes every page reachable with positive probability.

Definition 11.6 .
Let P be the stochastic matrix associated with a given directed graph of n vertices. Fix a constant $0 < \alpha < 1$. The matrix

$$G = \alpha P + \frac{1-\alpha}{n}\mathbf{e}\mathbf{e}^t$$

is called the Google matrix associated with this graph. Here $\mathbf{e} = (1, 1, \ldots, 1)^t$, and α is called the 'damping factor'.

Theorem 11.12 .
For any $0 < \alpha < 1$, the Google matrix G is a positive stochastic matrix. In particular,

1. *$L = \lim_{k\to\infty} G^k$ exists.*
2. *There exists an eigenvector $\mathbf{v}_{PR} = (v_1, \ldots, v_n)^t$ of G with eigenvalue 1 such that each $v_i > 0$ and $\sum_{i=1}^{n} v_i = 1$.*
3. *The columns of L are identical and are equal to $\mathbf{v}_{PR}$.*

The vector $\mathbf{v}_{PR}$ is called the PageRank vector for the Google matrix G.

Proof.
Positivity: $G_{ij} = \alpha P_{ij} + \dfrac{1-\alpha}{n} \cdot 1 > 0.$
Stochastic:

$$\sum_{i=1}^{n} G_{ij} = \alpha \sum_{i=1}^{n} P_{ij} + \frac{1-\alpha}{n} \sum_{i=1}^{n} 1 = \alpha(1) + \frac{1-\alpha}{n}(n) = 1.$$

Hence, G is a positive stochastic matrix. By Perron-Frobenius Theorem for regular stochastic matrices, $L = \lim_{k\to\infty} G^k$ converges, and

$$L = \begin{pmatrix} | & | & & | \\ \mathbf{v}_{PR} & \mathbf{v}_{PR} & \cdots & \mathbf{v}_{PR} \\ | & | & & | \end{pmatrix}.$$

■

In reality, it is difficult to compute the PageRank vector $\mathbf{v}_{PR}$ explicitly. By the Perron-Frobenius theorem for regular stochastic matrices, for any probability vector $\mathbf{v}$, we have

$$\lim_{k\to\infty} G^k\mathbf{v} = \mathbf{v}_{PR}.$$

For convenience, we may take

$$\mathbf{v} = \mathbf{v}_n = \left(\frac{1}{n}, \frac{1}{n}, \ldots, \frac{1}{n}\right)^t.$$

So $G^k\mathbf{v}_n \approx \mathbf{v}_{PR}$ when k is large. In the following, we derive an upper bound for the difference $G^k\mathbf{v}_n - \mathbf{v}_{PR}$.

Definition 11.7 .
Let $\mathbf{x} \in \mathbb{R}^n$. The 1-norm of $\mathbf{x}$ is defined to be the sum of the absolute values of its coordinates. i.e. if $\mathbf{x} = (x_1, x_2, \ldots, x_n)^t$ then

$$\|\mathbf{x}\|_1 \stackrel{\text{def}}{=} |x_1| + |x_2| + \cdots + |x_n|.$$

Remark .

1. $\|\mathbf{x}\|_1 = 0$ *if and only if* $\mathbf{x} = \mathbf{0}$.

2. $\|\mathbf{x}+\mathbf{y}\|_1 \le \|\mathbf{x}\|_1 + \|\mathbf{y}\|_1$.

Lemma .
If P is stochastic, then for any $\mathbf{x} \in \mathbb{R}^n$,

$$\|P\mathbf{x}\|_1 \le \|\mathbf{x}\|_1. \tag{1}$$

Proof.
Let $\mathbf{x} = (x_1, x_2, \ldots, x_n)^t$. Since P is stochastic, $P_{ij} \ge 0$ and $\sum_{i=1}^{n} P_{ij} = 1$ for

every j. Hence

$$\begin{aligned}\|P\mathbf{x}\|_1 &= \sum_{i=1}^{n}\left|\sum_{j=1}^{n} P_{ij}x_j\right| \leq \sum_{i=1}^{n}\sum_{j=1}^{n} P_{ij}|x_j| \\ &= \sum_{j=1}^{n}\left(\sum_{i=1}^{n} P_{ij}\right)|x_j| = \sum_{j=1}^{n}|x_j| = \|\mathbf{x}\|_1.\end{aligned}$$

■

Theorem 11.13 .
Let G be a Google matrix and $\mathbf{v}_{PR}$ be its PageRank vector of a directed graph of n vertices. For any integer $k \geq 0$,

$$\|G^k\mathbf{v}_n - \mathbf{v}_{PR}\|_1 \leq \alpha^k\|\mathbf{v}_n - \mathbf{v}_{PR}\|_1. \tag{2}$$

where α is the damping factor.

Proof.
We use induction on k.
For $k = 0$, there is nothing to prove. Suppose the inequality is true for k. Consider the case $k+1$. Since $G\mathbf{v}_{PR} = \mathbf{v}_{PR}$,

$$G^{k+1}\mathbf{v}_n - \mathbf{v}_{PR} = G(G^k\mathbf{v}_n - \mathbf{v}_{PR}).$$

Write $G = \alpha P + (1-\alpha)E_n$, where

$$E_n = \frac{1}{n}\mathbf{e}\mathbf{e}^t.$$

For any probability vector $\mathbf{p}$,

$$E_n\mathbf{p} = \frac{1}{n}\mathbf{e}(\mathbf{e}^t\mathbf{p}) = \frac{1}{n}\mathbf{e} = \mathbf{v}_n.$$

Since $G^k\mathbf{v}_n$ and $\mathbf{v}_{PR}$ are both probability vectors,

$$E_nG^k\mathbf{v}_n = E_n\mathbf{v}_{PR} = \mathbf{v}_n.$$

Hence

$$\begin{aligned}G^{k+1}\mathbf{v}_n - \mathbf{v}_{PR} &= (\alpha P + (1-\alpha)E_n)\,G^k\mathbf{v}_n - (\alpha P + (1-\alpha)E_n)\,\mathbf{v}_{PR} \\ &= \alpha P(G^k\mathbf{v}_n - \mathbf{v}_{PR}) + (1-\alpha)E_nG^k\mathbf{v}_n - (1-\alpha)E_n\mathbf{v}_{PR} \\ &= \alpha P(G^k\mathbf{v}_n - \mathbf{v}_{PR}).\end{aligned}$$

Apply the 1-norm and use (1):

$$\begin{aligned}\|G^{k+1}\mathbf{v}_n - \mathbf{v}_{PR}\|_1 &= \alpha\|P(G^k\mathbf{v}_n - \mathbf{v}_{PR})\|_1 \\ &\leq \alpha\|G^k\mathbf{v}_n - \mathbf{v}_{PR}\|_1 \\ &\leq \alpha \cdot \alpha^k\|\mathbf{v}_n - \mathbf{v}_{PR}\|_1 \\ &= \alpha^{k+1}\|\mathbf{v}_n - \mathbf{v}_{PR}\|_1.\end{aligned}$$

By induction, the proof is complete. ■

Remark .

From (2),

$$\begin{aligned}\|G^k\mathbf{v}_n - \mathbf{v}_{PR}\|_1 &\leq \alpha^k\|\mathbf{v}_n - \mathbf{v}_{PR}\|_1 \\ &\leq \alpha^k(\|\mathbf{v}_n\|_1 + \|\mathbf{v}_{PR}\|_1) \\ &= \alpha^k(1+1) = 2\alpha^k.\end{aligned}$$

Therefore, for large k, $G^k\mathbf{v}_n$ approximates $\mathbf{v}_{PR}$, and the error is bounded by $2\alpha^k$.

12 Matrix Calculus

12.1 Differentiation

Definition 12.1 .
For $A, B \in M_{m\times n}(\mathbb{R})$, the Frobenius inner product is

$$\langle A, B\rangle_F = \mathrm{tr}(A^\top B).$$

The associated Frobenius norm is $\|A\|_F = \sqrt{\mathrm{tr}(A^\top A)}$.

Remark .

1. *When $n = 1$, the matrices A, B are column vectors, and the Frobenius inner product reduces to the usual dot product.*

2. *If $A = (A_{ij})$ and $B = (B_{ij})$, then*

$$\mathrm{tr}(A^\top B) = \sum_{j=1}^{n}\sum_{i=1}^{m} A_{ij}B_{ij}.$$

   In words, the Frobenius inner product is the ordinary dot product after writing the matrix entries as one long vector.

Definition 12.2 (Derivative).
Let $f : M_{m\times n}(\mathbb{R}) \to \mathbb{R}$ be differentiable. The derivative of f at X is the linear map

$$df_X : M_{m\times n}(\mathbb{R}) \to \mathbb{R}$$

characterized by

$$f(X + H) - f(X) = df_X(H) + r_X(H),$$

Here $df_X(H)$ captures the terms which are linear in H. The remainder $r_X(H)$ is of size $o(\|H\|_F)$, i.e.

$$\frac{|r_X(H)|}{\|H\|_F} \to 0 \text{ as } H \to 0.$$

Remark .
This is the matrix version of the definition of differentiability in multivariable calculus: near X, the function f should be well approximated by a linear map, with relative error tending to zero. In symbols, this means

$$\lim_{H\to 0} \frac{|f(X + H) - f(X) - df_X(H)|}{\|H\|_F} = 0.$$

Example 12.1 .

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x_1, x_2) = x_1^2 x_2$. Writing $\mathbf{x} = (x_1, x_2)^\top$ and $\mathbf{h} = (h_1, h_2)^\top$, we have

$$\begin{aligned} f(\mathbf{x}+\mathbf{h}) - f(\mathbf{x}) &= (x_1 + h_1)^2 (x_2 + h_2) - x_1^2 x_2 \\ &= 2x_1 x_2 h_1 + x_1^2 h_2 + 2x_1 h_1 h_2 + x_2 h_1^2 + h_1^2 h_2. \end{aligned}$$

The linear part in $\mathbf{h}$ is $2x_1x_2h_1 + x_1^2 h_2$, while the remaining terms are higher order. Hence

$$df_{\mathbf{x}}(\mathbf{h}) = 2x_1 x_2 h_1 + x_1^2 h_2 = \left\langle \begin{pmatrix} 2x_1x_2 \\ x_1^2 \end{pmatrix}, \begin{pmatrix} h_1 \\ h_2 \end{pmatrix} \right\rangle.$$

Therefore, $\nabla_{\mathbf{x}} f = (2x_1x_2, x_1^2)^\top$.

Definition 12.3 .

Let $R(H)$ be a scalar-valued function of a matrix variable $H \in M_{m\times n}(\mathbb{R})$.

We write $R(H) = O(\|H\|_F)$ as $H \to \mathbf{O}$ if there exist constants $C > 0$ and $\delta > 0$ such that

$$|R(H)| \le C\,\|H\|_F \qquad \text{whenever } 0 < \|H\|_F < \delta.$$

In words, $R(H) = O(\|H\|_F)$ means that $R(H)$ is at most of first order in the size of H.

We write $R(H) = o(\|H\|_F)$ as $H \to \mathbf{O}$ if

$$\frac{|R(H)|}{\|H\|_F} \to 0 \qquad \text{as } H \to \mathbf{O}.$$

In words, $R(H) = o(\|H\|_F)$ means that $R(H)$ is negligible compared with the size of H.

Remark .

The same notation is often used for a one-dimensional perturbation $t \in \mathbb{R}$. Since t may be regarded as a 1×1 matrix, its Frobenius norm is $\|t\|_F = |t|$. Thus $R(t) = O(t)$ means $|R(t)| \le C|t|$ for all sufficiently small nonzero t, and $R(t) = o(t)$ means

$$\frac{|R(t)|}{|t|} \to 0 \qquad \text{as } t \to 0.$$

This is the notation used when we restrict a matrix problem to a line such as $X + tH$.

Example 12.2 .

We have $\|H\|_F^2 = o(\|H\|_F)$ because

$$\frac{\|H\|_F^2}{\|H\|_F} = \|H\|_F \to 0 \qquad \text{as } H \to \mathbf{O}.$$

On the other hand, $7\,\|H\|_F = O(\|H\|_F)$ because $7\,\|H\|_F \le 7\,\|H\|_F$, but $7\,\|H\|_F \ne o(\|H\|_F)$ since

$$\frac{7\,\|H\|_F}{\|H\|_F} = 7 \not\to 0.$$

Definition 12.4 (Gradient).
Let $f : M_{m\times n}(\mathbb{R}) \to \mathbb{R}$ be differentiable. The unique matrix $G \in M_{m\times n}(\mathbb{R})$ such that

$$df_X(H) = \langle G, H\rangle_F = \operatorname{tr}(G^\top H) \quad \textit{for all } H \in M_{m\times n}(\mathbb{R}).$$

is called the gradient of f at X, and is denoted by $\nabla_X f$.

Since df_X is a linear functional on the inner product space $M_{m\times n}(\mathbb{R})$, the Riesz representation theorem gives a unique matrix $G \in M_{m\times n}(\mathbb{R})$ such that the definition (12.4) is valid. Thus the gradient is characterized by

$$df_X(H) = \operatorname{tr}\big((\nabla_X f)^\top H\big).$$

Informally, one may think of dX as representing a small change H in the matrix X. With this convention, the identity above is written as

$$df = \operatorname{tr}\big((\nabla_X f)^\top dX\big).$$

The above "equality" gives the differential notation.

Remark .

1. *For $f : \mathbb{R}^n \to \mathbb{R}$, the gradient is the column vector collecting all partial derivatives:*

$$\nabla_{\mathbf{x}} f = \left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n}\right)^\top, \qquad df_{\mathbf{x}}(\mathbf{h}) = (\nabla_{\mathbf{x}} f)^\top \mathbf{h}.$$

   Along a differentiable curve $\mathbf{x}(t)$, the chain rule gives

$$\frac{d}{dt} f(\mathbf{x}(t)) = df_{\mathbf{x}(t)}(\mathbf{x}'(t)) = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}\frac{dx_i}{dt}.$$

   Writing $df = \frac{d}{dt} f\, dt$ and $dx_i = \frac{dx_i}{dt}\, dt$ yields

$$df = (\nabla_{\mathbf{x}} f)^\top d\mathbf{x} = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}\, dx_i.$$

   Thus the differential formula is precisely the chain rule written without the parameter t.

2. *The derivative df_X is a linear map. When f is scalar-valued, it is a linear functional and the gradient is its Frobenius inner-product representative. Derivatives and differentials apply to scalar-, vector-, and matrix-valued functions, whereas a gradient is reserved for scalar-valued functions.*

Theorem 12.1 (Interior Extrema).
Let $U \subseteq M_{m\times n}(\mathbb{R})$ be open, and let $f : U \to \mathbb{R}$ be differentiable. If $X_0 \in U$ is a local maximum or a local minimum of f, then

$$\nabla_X f(X_0) = 0.$$

Proof.
Fix any direction $H \in M_{m\times n}(\mathbb{R})$. Define a one-variable function $\varphi(t) = f(X_0 + tH)$ for t near 0. Since X_0 is an interior local maximum or local minimum of f, the point $t = 0$ is a local maximum or local minimum of φ.
By Calculus 1, $\varphi'(0) = 0$.
By the definition of the derivative,

$$\varphi'(0) = df_{X_0}(H).$$

Thus $df_{X_0}(H) = 0$ for every direction H. Since

$$df_{X_0}(H) = \mathrm{tr}\big((\nabla_X f(X_0))^\top H\big),$$

we have

$$\mathrm{tr}\big((\nabla_X f(X_0))^\top H\big) = 0 \qquad \text{for all } H.$$

Taking $H = \nabla_X f(X_0)$ gives

$$\|\nabla_X f(X_0)\|_F^2 = 0.$$

Hence $\nabla_X f(X_0) = 0$. ∎

Let $A = (A_{ij})$ be an $m \times n$ matrix whose entries are scalar-valued differentiable functions. We call A a matrix-valued function.

Definition 12.5 (Matrix Differential).
Let $A = (A_{ij})$ be an $m \times n$ matrix-valued function. Its differential is the $m \times n$ matrix

$$dA \overset{\text{def}}{=} (dA_{ij})_{i,j} = \begin{pmatrix} dA_{11} & dA_{12} & \cdots & dA_{1n} \\ dA_{21} & dA_{22} & \cdots & dA_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ dA_{m1} & dA_{m2} & \cdots & dA_{mn} \end{pmatrix}.$$

Example 12.3 .
Let

$$A(x,y) = \begin{pmatrix} x^2y & e^x + y \\ \sin x & x - y^2 \end{pmatrix}.$$

Then

$$dA = \begin{pmatrix} 2xy\,dx + x^2\,dy & e^x\,dx + dy \\ \cos x\,dx & dx - 2y\,dy \end{pmatrix}.$$

Theorem 12.2 (Basic Differential Rules).

Let A and B be matrices of functions with compatible sizes, and let $\alpha \in \mathbb{R}$ be a constant scalar. Whenever the expressions are defined,

1. $d(A+B) = dA + dB$;
2. $d(\alpha A) = \alpha\, dA$;
3. $d(AB) = (dA)B + A(dB)$;
4. $d(A^\top) = (dA)^\top$;
5. $d\,\mathrm{tr}(A) = \mathrm{tr}(dA)$.

Proof.
The first two rules follow entry by entry from the linearity of ordinary differentials. For the product rule, the (i,j)-entry of AB is

$$(AB)_{ij} = \sum_k A_{ik} B_{kj}.$$

Taking the differential entry by entry,

$$d(AB)_{ij} = \sum_k dA_{ik}\, B_{kj} + \sum_k A_{ik}\, dB_{kj}.$$

This is exactly the (i,j)-entry of $(dA)B + A(dB)$.
For transpose, the (i,j)-entry of $d(A^\top)$ is $d(A^\top_{ij}) = d(A_{ji})$, which is the (i,j)-entry of $(dA)^\top$. Finally,

$$d\,\mathrm{tr}(A) = d\left(\sum_i A_{ii}\right) = \sum_i dA_{ii} = \mathrm{tr}(dA).$$

∎

Recall for a scalar-valued function f, the gradient is characterized by

$$df = \mathrm{tr}\big((\nabla_X f)^\top dX\big).$$

This gives a concrete strategy for computing matrix gradients:

1. Express the scalar function using traces whenever convenient.
2. Take the differential using the rules above.
3. Move terms cyclically inside traces until the result has the form $\mathrm{tr}(G^\top dX)$.

4. Conclude that $\nabla_X f = G$.

The most frequently used trace identities are

$$\operatorname{tr}(PQ) = \operatorname{tr}(QP), \quad \operatorname{tr}(PQR) = \operatorname{tr}(QRP) = \operatorname{tr}(RPQ), \quad \operatorname{tr}(P) = \operatorname{tr}(P^\top).$$

Only cyclic permutations are valid in general; the order of factors may not be arbitrarily rearranged.

Example 12.4 .
Let $f(X) = \|X\|_F^2$. *Since* $\|X\|_F^2 = \operatorname{tr}(X^\top X)$,

$$\begin{aligned} df &= d\operatorname{tr}(X^\top X) = \operatorname{tr}(d(X^\top X)) \\ &= \operatorname{tr}((dX)^\top X) + \operatorname{tr}(X^\top dX). \end{aligned}$$

Using $\operatorname{tr}((dX)^\top X) = \operatorname{tr}(((dX)^\top X)^\top) = \operatorname{tr}(X^\top dX)$, *we get*

$$df = 2\operatorname{tr}(X^\top dX) = \operatorname{tr}((2X)^\top dX).$$

Therefore

$$\nabla_X f = 2X.$$

Example 12.5 .
Let $f : M_n(\mathbb{R}) \to \mathbb{R}$ *be defined by* $f(X) = \operatorname{tr}(X^2)$. *Then*

$$\begin{aligned} df &= d\operatorname{tr}(X^2) = \operatorname{tr}(d(X \cdot X)) \\ &= \operatorname{tr}((dX)X) + \operatorname{tr}(X(dX)) \\ &= \operatorname{tr}(XdX) + \operatorname{tr}(XdX) \\ &= \operatorname{tr}(2X\,dX). \end{aligned}$$

Since $df = \operatorname{tr}(G^\top dX)$, *we have* $G^\top = 2X$. *Therefore*

$$\nabla_X f = 2X^\top.$$

Remark .
Consider $F : M_n(\mathbb{R}) \to M_n(\mathbb{R})$ *defined by* $F(X) = X^2$. *Its differential is*

$$dF = d(X^2) = d(X \cdot X) = (dX)X + X(dX).$$

Thus the derivative at X *is the linear map*

$$dF_X(H) = HX + XH.$$

Since F *is not scalar-valued, the notion of gradient does not apply here. But its derivative and differential are still well-defined.*

Theorem 12.3 (Chain Rule).
Let $f : \mathcal{X} \to \mathcal{Y}$ and $g : \mathcal{Y} \to \mathcal{Z}$ be differentiable maps between finite-dimensional matrix spaces. Then

$$d(g \circ f)_X = dg_{f(X)} \circ df_X.$$

Equivalently, for every direction $H \in \mathcal{X}$,

$$d(g \circ f)_X(H) = dg_{f(X)}\big(df_X(H)\big).$$

There is a particularly useful gradient form of this rule. If $L : \mathcal{X} \to \mathcal{Y}$ is linear, its adjoint $L^* : \mathcal{Y} \to \mathcal{X}$ is characterized by

$$\langle L(H), K\rangle_{\mathcal{Y}} = \langle H, L^*(K)\rangle_{\mathcal{X}} \qquad \text{for all } H \in \mathcal{X},\ K \in \mathcal{Y}.$$

Here H is an arbitrary input direction, while K is an arbitrary test direction in the output space. The two inner products are dictated by the spaces in which the vectors live: $L(H)$ and K belong to $\mathcal{Y}$, whereas H and $L^*(K)$ belong to $\mathcal{X}$. Thus L^* depends on the chosen inner products on both $\mathcal{X}$ and $\mathcal{Y}$; under the usual Euclidean or Frobenius inner products, its matrix representation is the transpose of the matrix representing L.

Theorem 12.4 .
Let $f : \mathcal{X} \to \mathcal{Y}$ and $g : \mathcal{Y} \to \mathbb{R}$ be differentiable. Then

$$\nabla_X(g \circ f) = df_X^*\big(\nabla_Y g(f(X))\big),$$

where df_X^* is the adjoint of the linear map df_X.

Proof.
For every input direction $H \in \mathcal{X}$, the ordinary chain rule gives

$$d(g \circ f)_X(H) = dg_{f(X)}\big(df_X(H)\big).$$

Now $df_X(H) \in \mathcal{Y}$ is a direction at $f(X)$. Since $g : \mathcal{Y} \to \mathbb{R}$ is scalar-valued, the definition of its gradient states that, for every $Y, K \in \mathcal{Y}$,

$$dg_Y(K) = \langle \nabla_Y g(Y), K\rangle_{\mathcal{Y}} = \langle K, \nabla_Y g(Y)\rangle_{\mathcal{Y}},$$

where the second equality uses symmetry of the real Frobenius inner product. Taking $Y = f(X)$ and $K = df_X(H)$ therefore gives

$$d(g \circ f)_X(H) = \langle df_X(H), \nabla_Y g(f(X))\rangle_{\mathcal{Y}}.$$

Both entries lie in $\mathcal{Y}$. To identify the gradient with respect to X, however, the differential must have the form $\langle H, G\rangle_{\mathcal{X}}$ for some $G \in \mathcal{X}$. The adjoint is defined precisely to move df_X across the inner product:

$$\langle df_X(H), \nabla_Y g(f(X))\rangle_{\mathcal{Y}} = \langle H, df_X^*(\nabla_Y g(f(X)))\rangle_{\mathcal{X}}.$$

Hence $G = df_X^*(\nabla_Y g(f(X)))$, which proves the result. In this sense, the adjoint pulls the output gradient in $\mathcal{Y}$ back to an input gradient in $\mathcal{X}$. ∎

Remark (Jacobian form).
Let $f : \mathbb{R}^n \to \mathbb{R}^m$ and $g : \mathbb{R}^m \to \mathbb{R}$. Writing $f = (f_1, \dots, f_m)^\top$, the Jacobian of f is

$$J_f(\mathbf{x}) = \left(\frac{\partial f_i}{\partial x_j}(\mathbf{x})\right)_{i,j} \in M_{m\times n}(\mathbb{R}),$$

and it represents the derivative:

$$df_{\mathbf{x}}(\mathbf{h}) = J_f(\mathbf{x})\mathbf{h}.$$

With the usual inner products, the adjoint of this derivative is represented by the transpose Jacobian, since

$$\langle J_f(\mathbf{x})\mathbf{h}, \mathbf{k}\rangle = \left\langle \mathbf{h}, J_f(\mathbf{x})^\top \mathbf{k}\right\rangle.$$

Therefore the gradient chain rule becomes

$$\nabla_{\mathbf{x}}(g \circ f)(\mathbf{x}) = J_f(\mathbf{x})^\top \nabla_{\mathbf{y}} g(f(\mathbf{x})).$$

Thus the adjoint formula is the coordinate-free version of "multiply the output gradient by the transpose Jacobian."

Example 12.6 .
Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ and $g : \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$f(\mathbf{x}) = \begin{pmatrix} x_1^2 + x_2 \\ x_1 x_2 \end{pmatrix}, \qquad g(\mathbf{y}) = \frac{1}{2}(y_1^2 + y_2^2).$$

Then

$$J_f(\mathbf{x}) = \begin{pmatrix} 2x_1 & 1 \\ x_2 & x_1 \end{pmatrix}, \qquad \nabla_{\mathbf{y}} g(\mathbf{y}) = \mathbf{y}.$$

Hence

$$\begin{aligned} \nabla_{\mathbf{x}}(g \circ f)(\mathbf{x}) &= J_f(\mathbf{x})^\top f(\mathbf{x}) \\ &= \begin{pmatrix} 2x_1(x_1^2 + x_2) + x_1 x_2^2 \\ x_1^2 + x_2 + x_1^2 x_2 \end{pmatrix}. \end{aligned}$$

This agrees with direct differentiation of

$$(g \circ f)(\mathbf{x}) = \frac{1}{2}\left((x_1^2 + x_2)^2 + (x_1 x_2)^2\right).$$

Proposition 11 (Affine Matrix Least Squares).
Let

$$A \in M_{m\times p}(\mathbb{R}), \qquad X \in M_{p\times q}(\mathbb{R}), \qquad B \in M_{q\times n}(\mathbb{R}), \qquad C \in M_{m\times n}(\mathbb{R}).$$

Define

$$F(X) = \frac{1}{2}\left\| AXB - C \right\|_F^2.$$

Then

$$\nabla_X F = A^\top (AXB - C) B^\top.$$

Proof.
Let $Y = AXB - C$.
Then

$$F(X) = \frac{1}{2}\|Y\|_F^2 = \frac{1}{2}\operatorname{tr}(Y^\top Y).$$

Taking differential, we get

$$dY = d(AXB - C) = A(dX)B.$$

Since $d\left(\frac{1}{2}\|Y\|_F^2\right) = \langle Y, dY\rangle_F$, we have

$$\begin{aligned} dF &= \langle Y, dY\rangle_F \\ &= \langle AXB - C, A(dX)B\rangle_F \\ &= \operatorname{tr}\left((AXB - C)^\top A(dX)B\right). \end{aligned}$$

Now move the matrices cyclically inside the trace:

$$\begin{aligned} dF &= \operatorname{tr}\left(B(AXB - C)^\top A(dX)\right) \\ &= \operatorname{tr}\left(\left(A^\top (AXB - C) B^\top\right)^\top dX\right). \end{aligned}$$

By the definition of the gradient,

$$dF = \operatorname{tr}\left((\nabla_X F)^\top dX\right).$$

Hence

$$\nabla_X F = A^\top (AXB - C) B^\top.$$

■

Remark (Least-squares interpretations).
The rule $\nabla_X \frac{1}{2} \|AXB - C\|_F^2 = A^\top(AXB - C)B^\top$ reflects the two adjoints: left multiplication by A contributes $A^\top$, while right multiplication by B contributes $B^\top$. For $B = I$, it reduces to the usual matrix least-squares gradient $\nabla_X \frac{1}{2} \|AX - C\|_F^2 = A^\top(AX - C)$.

For the linear model $y = X_{\text{data}}\beta + \varepsilon$, take $A = X_{\text{data}}$, $X = \beta$, $B = I$, and $C = y$. Then

$$\nabla_\beta \frac{1}{2} \|X_{\text{data}}\beta - y\|^2 = X_{\text{data}}^\top(X_{\text{data}}\beta - y),$$

so the first-order condition is $X_{\text{data}}^\top X_{\text{data}}\hat{\beta} = X_{\text{data}}^\top y$.

If $X_{\text{data}}^\top X_{\text{data}}$ is invertible, equivalently if X_{data} has full column rank, then

$$\hat{\beta} = (X_{\text{data}}^\top X_{\text{data}})^{-1} X_{\text{data}}^\top y.$$

For multiple outcomes $Y = X_{\text{data}}\Gamma + E$, the same rule gives $\nabla_\Gamma \frac{1}{2} \|X_{\text{data}}\Gamma - Y\|_F^2 = X_{\text{data}}^\top(X_{\text{data}}\Gamma - Y)$; thus matrix least squares is OLS applied to all outcome columns at once.

The factor B allows a transformation on the right of a coefficient matrix. For example, in

$$Y = X_{\text{data}}\Theta R + E,$$

Θ is the free coefficient matrix to be estimated, whereas R is a known, fixed matrix that specifies how the columns of Θ are combined. The objective $\frac{1}{2} \|X_{\text{data}}\Theta R - Y\|_F^2$ is the full form with $A = X_{\text{data}}$, $X = \Theta$, $B = R$, and $C = Y$.

For an ordinary coefficient restriction such as $\beta_1 = \beta_2$, it is more natural to parameterize the coefficient vector as $\beta = R\theta$. For example, if $\beta \in \mathbb{R}^3$, set

$$R = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad \theta = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}.$$

Then

$$\beta = R\theta = \begin{bmatrix} \theta_1 \\ \theta_1 \\ \theta_2 \end{bmatrix},$$

so $\beta_1 = \beta_2$ holds automatically, while θ_1, θ_2 remain unrestricted. The restricted least-squares problem becomes $\frac{1}{2} \|X_{\text{data}}R\theta - y\|^2$, which has the same form with $A = X_{\text{data}}R$, $X = \theta$, $B = I$, and $C = y$.

Weighted least squares and GLS enter through a transformation on the left. Suppose

$$y = X_{\text{data}}\beta + \varepsilon, \qquad \mathbb{E}(\varepsilon) = 0, \qquad \operatorname{Cov}(\varepsilon) = \Omega \succ 0.$$

GLS weights the residual by its inverse covariance and minimizes

$$\begin{aligned} Q_{\mathrm{GLS}}(\beta) &= \frac{1}{2}(X_{\mathrm{data}}\beta - y)^\top \Omega^{-1}(X_{\mathrm{data}}\beta - y) \\ &= \frac{1}{2}\left\|\Omega^{-1/2}(X_{\mathrm{data}}\beta - y)\right\|^2. \end{aligned}$$

Thus GLS is ordinary least squares applied to the whitened data $\widetilde{X} = \Omega^{-1/2}X_{\mathrm{data}}$ *and* $\widetilde{y} = \Omega^{-1/2}y$. *In the notation of* $F(X)$, *this corresponds to* $X = \beta$, $A = \Omega^{-1/2}X_{\mathrm{data}}$, $B = I$, *and* $C = \Omega^{-1/2}y$. *Therefore*

$$\nabla_\beta Q_{\mathrm{GLS}}(\beta) = X_{\mathrm{data}}^\top \Omega^{-1}(X_{\mathrm{data}}\beta - y),$$

and the first-order condition is the GLS normal equation

$$X_{\mathrm{data}}^\top \Omega^{-1} X_{\mathrm{data}} \hat{\beta}_{\mathrm{GLS}} = X_{\mathrm{data}}^\top \Omega^{-1} y.$$

If $X_{\mathrm{data}}^\top \Omega^{-1} X_{\mathrm{data}}$ *is invertible, then*

$$\hat{\beta}_{\mathrm{GLS}} = (X_{\mathrm{data}}^\top \Omega^{-1} X_{\mathrm{data}})^{-1} X_{\mathrm{data}}^\top \Omega^{-1} y.$$

Weighted least squares is the special case in which Ω *is diagonal; general GLS also permits correlated errors through the off-diagonal entries of* Ω.

Proposition 12 (Derivative of Affine Mapping).
Let $A \in M_{m\times n}(\mathbb{R})$ *and* $\mathbf{b} \in \mathbb{R}^m$. *Define*

$$F : \mathbb{R}^n \to \mathbb{R}^m, \qquad F(\mathbf{x}) = A\mathbf{x} + \mathbf{b}.$$

Then F *is differentiable at every* $\mathbf{x} \in \mathbb{R}^n$, *and*

$$dF_{\mathbf{x}}(\mathbf{h}) = A\mathbf{h}.$$

Thus the matrix representing the derivative $dF_{\mathbf{x}}$ *is* A.

Proof.
Consider the candidate linear map $L : \mathbb{R}^n \to \mathbb{R}^m$ defined by $L(\mathbf{h}) = A\mathbf{h}$. For every nonzero $\mathbf{h}$,

$$\begin{aligned} &\frac{\|F(\mathbf{x}+\mathbf{h}) - F(\mathbf{x}) - L(\mathbf{h})\|}{\|\mathbf{h}\|} \\ =&\frac{\|A(\mathbf{x}+\mathbf{h}) + \mathbf{b} - (A\mathbf{x}+\mathbf{b}) - A\mathbf{h}\|}{\|\mathbf{h}\|} = 0. \end{aligned}$$

The quotient is identically zero, so its limit as $\mathbf{h} \to \mathbf{0}$ is zero. By the definition of the Fréchet derivative, $dF_{\mathbf{x}} = L$, whose matrix is A. ∎

Remark .
Because F is vector-valued, A is its derivative (or Jacobian), not its gradient. Componentwise, the rows of A are $(\nabla_{\mathbf{x}} F_i)^\top$.

Theorem 12.5 .
Let $A \in M_n(\mathbb{R})$ and define $f : \mathbb{R}^n \to \mathbb{R}$ by $f(\mathbf{x}) = \mathbf{x}^\top A\mathbf{x}$. Then

$$\nabla_{\mathbf{x}} f = (A + A^\top)\mathbf{x}.$$

In particular, if A is symmetric, then $\nabla_{\mathbf{x}} f = 2A\mathbf{x}$.

Proof.
The matrix A is not assumed to be symmetric. Since $\mathbf{x}^\top A\mathbf{x}$ is a scalar,

$$f(\mathbf{x}) = \operatorname{tr}(\mathbf{x}^\top A\mathbf{x}).$$

Taking the differential,

$$\begin{aligned} df &= \operatorname{tr}(d(\mathbf{x}^\top A\mathbf{x})) \\ &= \operatorname{tr}((d\mathbf{x})^\top A\mathbf{x}) + \operatorname{tr}(\mathbf{x}^\top A\, d\mathbf{x}). \end{aligned}$$

For the first term,

$$\operatorname{tr}((d\mathbf{x})^\top A\mathbf{x}) = \operatorname{tr}(((d\mathbf{x})^\top A\mathbf{x})^\top) = \operatorname{tr}(\mathbf{x}^\top A^\top d\mathbf{x}).$$

Hence

$$\begin{aligned} df &= \operatorname{tr}(\mathbf{x}^\top A^\top d\mathbf{x}) + \operatorname{tr}(\mathbf{x}^\top A\, d\mathbf{x}) \\ &= \operatorname{tr}\big(\mathbf{x}^\top (A^\top + A) d\mathbf{x}\big). \end{aligned}$$

Comparing this with $df = \operatorname{tr}((\nabla_{\mathbf{x}} f)^\top d\mathbf{x})$ gives

$$(\nabla_{\mathbf{x}} f)^\top = \mathbf{x}^\top (A^\top + A).$$

Therefore

$$\nabla_{\mathbf{x}} f = (A + A^\top)\mathbf{x}.$$

If $A = A^\top$, this becomes $\nabla_{\mathbf{x}} f = 2A\mathbf{x}$. ∎

Theorem 12.6 .
In each item below, all matrices and vectors other than the displayed variables are constant.

1. *If $\mathbf{x} \in \mathbb{R}^n$, $A \in M_{m\times n}(\mathbb{R})$, and $B \in M_{n\times m}(\mathbb{R})$, then*

$$\frac{\partial(A\mathbf{x})}{\partial \mathbf{x}^\top} = A, \qquad \frac{\partial(\mathbf{x}^\top B)}{\partial \mathbf{x}} = B.$$

2. *If* $\mathbf{x}, \mathbf{a} \in \mathbb{R}^n$*, then* $\dfrac{\partial}{\partial \mathbf{x}}(\mathbf{a}^\top \mathbf{x}) = \dfrac{\partial}{\partial \mathbf{x}}(\mathbf{x}^\top \mathbf{a}) = \mathbf{a}$.

3. *If* $\mathbf{x} \in \mathbb{R}^m$*,* $\mathbf{y} \in \mathbb{R}^n$*, and* $A \in M_{m\times n}(\mathbb{R})$*, define* $f(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top A\mathbf{y} \in \mathbb{R}$*. Then* $\frac{\partial f}{\partial \mathbf{x}} = A\mathbf{y}$ *and* $\frac{\partial f}{\partial \mathbf{y}} = A^\top \mathbf{x}$.

4. *If* $\mathbf{x} \in \mathbb{R}^n$ *and* $A \in M_n(\mathbb{R})$*, then* $\dfrac{\partial}{\partial \mathbf{x}}(\mathbf{x}^\top A\mathbf{x}) = (A + A^\top)\mathbf{x}$ *and* $\dfrac{\partial^2}{\partial \mathbf{x} \partial \mathbf{x}^\top}(\mathbf{x}^\top A\mathbf{x}) = A + A^\top$.

5. *If* $X, A \in M_{m\times n}(\mathbb{R})$*, then* $\dfrac{\partial}{\partial X}\operatorname{tr}(A^\top X) = A$.

6. *If* $X \in M_{m\times n}(\mathbb{R})$*, then* $\dfrac{\partial}{\partial X}\left(\dfrac{1}{2}\|X\|_F^2\right) = X$.

7. *If* $X \in M_{m\times n}(\mathbb{R})$*,* $A \in M_{p\times m}(\mathbb{R})$*,* $B \in M_{n\times q}(\mathbb{R})$*, and* $C \in M_{p\times q}(\mathbb{R})$*, then* $\dfrac{\partial}{\partial X}\left(\dfrac{1}{2}\|AXB - C\|_F^2\right) = A^\top(AXB - C)B^\top$.

Proof.
For (1), define

$$F(\mathbf{x}) = A\mathbf{x}, \qquad G(\mathbf{x}) = \mathbf{x}^\top B.$$

The preceding proposition gives $\partial F/\partial \mathbf{x}^\top = A$. Moreover, $G^\top(\mathbf{x}) = B^\top \mathbf{x}$, so

$$\frac{\partial G}{\partial \mathbf{x}} = \left(\frac{\partial G^\top}{\partial \mathbf{x}^\top}\right)^\top = (B^\top)^\top = B.$$

For (2), let $f(\mathbf{x}) = \mathbf{a}^\top \mathbf{x} = \mathbf{x}^\top \mathbf{a}$. Then

$$df = \operatorname{tr}(\mathbf{a}^\top d\mathbf{x}),$$

so $\frac{\partial f}{\partial \mathbf{x}} = (\mathbf{a}^\top)^\top = \mathbf{a}$.
For (3), the product rule gives

$$df = (d\mathbf{x})^\top A\mathbf{y} + \mathbf{x}^\top A\, d\mathbf{y} = (A\mathbf{y})^\top d\mathbf{x} + (A^\top \mathbf{x})^\top d\mathbf{y}.$$

Since $\mathbf{x}$ and $\mathbf{y}$ are both column variables, their total differential has the standard form

$$df = \left(\frac{\partial f}{\partial \mathbf{x}}\right)^\top d\mathbf{x} + \left(\frac{\partial f}{\partial \mathbf{y}}\right)^\top d\mathbf{y}.$$

Comparing the coefficients of $d\mathbf{x}$ and $d\mathbf{y}$ identifies both vector partial derivatives.
For (5), let $f(X) = \operatorname{tr}(A^\top X)$. Then

$$df = \operatorname{tr}(A^\top dX),$$

so $\frac{\partial f}{\partial X} = A$. The equivalent formula follows by replacing $A^\top$ with A.
Parts (4), (6), (7) were proved earlier. Note that every function in parts (2)–(7) is scalar-valued. Under our column convention, the vector or matrix obtained by collecting its partial derivatives is exactly its gradient; that is,

$$\frac{\partial f}{\partial \mathbf{x}} = \nabla_{\mathbf{x}} f, \qquad \frac{\partial f}{\partial \mathbf{y}} = \nabla_{\mathbf{y}} f, \qquad \frac{\partial f}{\partial X} = \nabla_X f.$$

■

Remark (Derivative-layout convention).
The transpose records the layout *of the derivative. For a scalar-valued function $f : \mathbb{R}^n \to \mathbb{R}$, we use the column-gradient convention*

$$df = \left(\frac{\partial f}{\partial \mathbf{x}}\right)^\top d\mathbf{x}, \qquad \frac{\partial f}{\partial \mathbf{x}} \in \mathbb{R}^n.$$

For a column-vector-valued function $F : \mathbb{R}^n \to \mathbb{R}^m$, its derivative must instead be an $m \times n$ matrix that acts on $d\mathbf{x}$:

$$dF = \underbrace{\frac{\partial F}{\partial \mathbf{x}^\top}}_{J_F \in M_{m\times n}(\mathbb{R})} d\mathbf{x}.$$

Thus, $\partial F / \partial \mathbf{x}^\top$ is the usual $m \times n$ Jacobian, whose rows correspond to the components F_i and whose columns correspond to the variables x_j. For a row-vector-valued function $G : \mathbb{R}^n \to \mathbb{R}^{1\times m}$, we instead define

$$dG = d\mathbf{x}^\top \underbrace{\frac{\partial G}{\partial \mathbf{x}}}_{\in M_{n\times m}(\mathbb{R})},$$

so that the output remains a row vector. Consequently,

$$d(A\mathbf{x}) = A\,d\mathbf{x} \implies \frac{\partial (A\mathbf{x})}{\partial \mathbf{x}^\top} = A, \qquad d(\mathbf{x}^\top B) = d\mathbf{x}^\top B \implies \frac{\partial (\mathbf{x}^\top B)}{\partial \mathbf{x}} = B.$$

This is the layout convention used throughout this section. Equivalently, one may first transpose the row-valued function and compute the ordinary Jacobian of $G^\top = B^\top \mathbf{x}$, then transpose the result back to the row-output layout:

$$\frac{\partial G}{\partial \mathbf{x}} = \left(\frac{\partial G^\top}{\partial \mathbf{x}^\top}\right)^\top = (B^\top)^\top = B.$$

Invertible matrices

The following formulas apply to functions whose domain is the set of invertible matrices

$$GL_n(\mathbb{R}) = \{X \in M_n(\mathbb{R}) : \det X \neq 0\},$$

called the general linear group over $\mathbb{R}$.

Theorem 12.7 (Inversion).
The inversion map $X \mapsto X^{-1}$ on $GL_n(\mathbb{R})$ satisfies

$$d(X^{-1}) = -X^{-1}(dX)X^{-1}.$$

Proof.
Start from the identity $X^{-1}X = I$. Differentiating both sides gives $d(X^{-1})\,X + X^{-1}dX = 0$, hence $d(X^{-1})\,X = -X^{-1}dX$. Multiplying on the right by X^{-1} gives

$$d(X^{-1}) = -X^{-1}(dX)X^{-1}.$$

The inversion map is matrix-valued, not scalar-valued, so we talk about its derivative or differential, not its gradient. ∎

Set $GL_n^+(\mathbb{R}) = \{X \in GL_n(\mathbb{R}) : \det X > 0\}$.

Theorem 12.8 (Log-determinant).
The function $f : GL_n^+(\mathbb{R}) \to \mathbb{R}$ defined by $f(X) = \log\det X$ has gradient

$$\nabla_X(\log\det X) = (X^{-1})^\top.$$

Equivalently,

$$d(\log\det X) = \operatorname{tr}(X^{-1}dX).$$

Proof.
Let $f(X) = \log\det X$. For a small matrix H,

$$\begin{aligned} f(X+H) - f(X) &= \log\det(X+H) - \log\det X \\ &= \log\det\bigl(X(I + X^{-1}H)\bigr) - \log\det X \\ &= \log\det X + \log\det(I + X^{-1}H) - \log\det X \\ &= \log\det(I + X^{-1}H). \end{aligned}$$

Now put $K = X^{-1}H$. We claim that

$$\det(I+K) = 1 + \operatorname{tr}(K) + O(\|K\|_F^2).$$

Let $\mathrm{Spec}(K) = \{\lambda_1, \dots, \lambda_n\}$, counted with algebraic multiplicity. Then

$$\begin{aligned}\det(I+K) &= \prod_{i=1}^{n}(1+\lambda_i)\\ &= 1 + \sum_{i=1}^{n}\lambda_i + \text{higher-order terms}\\ &= 1 + \mathrm{tr}(K) + O(\|K\|_F^2).\end{aligned}$$

The higher-order terms are sums of products of at least two eigenvalues, hence are $O(\|K\|_F^2)$ since $|\lambda_i| \le \|K\|_F \le \|X^{-1}\| \, \|H\|_F \to 0$ as $H \to 0$.
Next write $u = \mathrm{tr}(K) + O(\|K\|_F^2)$. The one-variable Taylor expansion of $g(u) = \log(1+u)$ at $u = 0$ gives $g(0) = 0$ and $g'(0) = 1$, so $\log(1+u) = u + O(u^2)$ as $u \to 0$. Hence

$$\begin{aligned}\log\det(I+K) &= \log\bigl(1 + \mathrm{tr}(K) + O(\|K\|_F^2)\bigr)\\ &= \mathrm{tr}(K) + O(\|K\|_F^2),\end{aligned}$$

where $O(\|K\|_F^2) = O(\|H\|_F^2) = o(\|H\|_F)$. Therefore

$$f(X+H) - f(X) = \mathrm{tr}(X^{-1}H) + o(\|H\|_F).$$

In differential notation, $df = \mathrm{tr}(X^{-1}dX)$, so $\nabla_X f = (X^{-1})^\top$. ■

Corollary (Determinant).
On $GL_n^+(\mathbb{R})$, $d(\det X) = \det(X)\,\mathrm{tr}(X^{-1}dX), \quad \nabla_X \det X = \det(X) X^{-\top}$.

Proof.
Chain Rule. ■

Hessian

The Hessian describes how the gradient changes. In one variable, $f'(x)$ measures slope, while $f''(x)$ measures how the slope changes. In several variables, the same idea remains true: the gradient gives the first-order change of the function, and the Hessian gives the first-order change of the gradient.

Vector Variable Let $U \subseteq \mathbb{R}^n$ be open and let $f : U \to \mathbb{R}$ be differentiable. For $\mathbf{x} \in U$, the derivative of f at $\mathbf{x}$ is the linear map $df_{\mathbf{x}} : \mathbb{R}^n \to \mathbb{R}$ characterized by

$$f(\mathbf{x}+\mathbf{h}) - f(\mathbf{x}) = df_{\mathbf{x}}(\mathbf{h}) + o(\|\mathbf{h}\|).$$

Since $df_{\mathbf{x}}$ is a linear functional, there exists a unique vector $\nabla f(\mathbf{x}) \in \mathbb{R}^n$ such that

$$df_{\mathbf{x}}(\mathbf{h}) = \langle \nabla f(\mathbf{x}), \mathbf{h} \rangle .$$

Thus the first-order approximation of f is

$$f(\mathbf{x} + \mathbf{h}) = f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{h} \rangle + o(\|\mathbf{h}\|).$$

Now assume f is C^2. Then the gradient $\nabla f : U \to \mathbb{R}^n$ is itself a differentiable vector-valued function. Its derivative at $\mathbf{x}$ is the Hessian matrix

$$H_f(\mathbf{x}) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}) \right)_{i,j} .$$

This means that if we perturb $\mathbf{x}$ in the direction $\mathbf{h}$ by replacing it with $\mathbf{x} + t\mathbf{h}$, then

$$\nabla f(\mathbf{x} + t\mathbf{h}) = \nabla f(\mathbf{x}) + tH_f(\mathbf{x})\mathbf{h} + o(t).$$

Proof.
Let $G(\mathbf{x}) = \nabla f(\mathbf{x})$. Since G is differentiable at $\mathbf{x}$, we have

$$G(\mathbf{x} + \mathbf{k}) = G(\mathbf{x}) + dG_{\mathbf{x}}(\mathbf{k}) + o(\|\mathbf{k}\|).$$

For $G = \nabla f$, the derivative $dG_{\mathbf{x}}$ is represented by the Hessian matrix. Hence $dG_{\mathbf{x}}(\mathbf{k}) = H_f(\mathbf{x})\mathbf{k}$, so

$$\nabla f(\mathbf{x} + \mathbf{k}) = \nabla f(\mathbf{x}) + H_f(\mathbf{x})\mathbf{k} + o(\|\mathbf{k}\|).$$

Now set $\mathbf{k} = t\mathbf{h}$. Since $\|\mathbf{k}\| = \|t\mathbf{h}\| = |t| \, \|\mathbf{h}\|$, for fixed $\mathbf{h}$ the remainder $o(\|t\mathbf{h}\|)$ can be written as $o(t)$. Therefore

$$\nabla f(\mathbf{x} + t\mathbf{h}) = \nabla f(\mathbf{x}) + tH_f(\mathbf{x})\mathbf{h} + o(t).$$

∎

Definition 12.6 (Vector Hessian).
Let $f : U \to \mathbb{R}$ be C^2. The Hessian of f at $\mathbf{x}$ is the linear map $\nabla^2 f(\mathbf{x}) : \mathbb{R}^n \to \mathbb{R}^n$ defined by

$$\nabla^2 f(\mathbf{x})[\mathbf{h}] = H_f(\mathbf{x})\mathbf{h}.$$

In coordinates, this is the usual Hessian matrix

$$H_f(\mathbf{x}) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}) \right)_{i,j} .$$

To understand the Hessian as curvature, fix a point $\mathbf{x}$ and a direction $\mathbf{h}$. Define the one-variable function $\phi(t) = f(\mathbf{x} + t\mathbf{h})$. Then ϕ records the behavior of f along the line through $\mathbf{x}$ in the direction $\mathbf{h}$.

By the chain rule, $\phi'(t) = \langle \nabla f(\mathbf{x} + t\mathbf{h}), \mathbf{h} \rangle$. Using the first-order approximation of the gradient, write

$$\nabla f(\mathbf{x} + t\mathbf{h}) = \nabla f(\mathbf{x}) + tH_f(\mathbf{x})\mathbf{h} + \mathbf{r}(t), \qquad \frac{\|\mathbf{r}(t)\|}{|t|} \to 0.$$

Then

$$\begin{aligned}\phi'(t) &= \left\langle \nabla f(\mathbf{x}) + tH_f(\mathbf{x})\mathbf{h} + \mathbf{r}(t), \mathbf{h} \right\rangle \\ &= \langle \nabla f(\mathbf{x}), \mathbf{h} \rangle + t \left\langle H_f(\mathbf{x})\mathbf{h}, \mathbf{h} \right\rangle + \langle \mathbf{r}(t), \mathbf{h} \rangle \\ &= \langle \nabla f(\mathbf{x}), \mathbf{h} \rangle + t \left\langle H_f(\mathbf{x})\mathbf{h}, \mathbf{h} \right\rangle + o(t),\end{aligned}$$

Indeed, by Cauchy–Schwarz,

$$\frac{|\langle \mathbf{r}(t), \mathbf{h} \rangle|}{|t|} \leq \frac{\|\mathbf{r}(t)\|}{|t|} \|\mathbf{h}\|.$$

Since $\mathbf{h}$ is fixed and $\|\mathbf{r}(t)\| / |t| \to 0$, we get $\langle \mathbf{r}(t), \mathbf{h} \rangle = o(t)$. Therefore

$$\phi''(0) = \left\langle H_f(\mathbf{x})\mathbf{h}, \mathbf{h} \right\rangle = \mathbf{h}^\top H_f(\mathbf{x})\mathbf{h}.$$

This number measures the curvature of f at $\mathbf{x}$ along the direction $\mathbf{h}$.

Remark .
The scalar

$$q_{\mathbf{x}}(\mathbf{h}) = \mathbf{h}^\top H_f(\mathbf{x})\mathbf{h}$$

is the quadratic form associated with the Hessian at $\mathbf{x}$*. Its sign is the second-order curvature of f along the direction* $\mathbf{h}$*: positive means upward, negative means downward, and zero means flat to second order.*

Proposition 13 (Second-Order Approximation).
Let $f : U \to \mathbb{R}$ be C^2. Then for small $\mathbf{h}$*,*

$$f(\mathbf{x} + \mathbf{h}) = f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{h} \rangle + \frac{1}{2}\mathbf{h}^\top H_f(\mathbf{x})\mathbf{h} + o(\|\mathbf{h}\|^2).$$

> *Proof.*
> Define $\phi(t) = f(\mathbf{x} + t\mathbf{h})$. By the one-variable Taylor expansion at $t = 0$,
>
> $$\phi(1) = \phi(0) + \phi'(0) + \frac{1}{2}\phi''(0) + o(\|\mathbf{h}\|^2).$$

Now $\phi(0) = f(\mathbf{x})$ and $\phi(1) = f(\mathbf{x}+\mathbf{h})$. Also,

$$\phi'(0) = \langle \nabla f(\mathbf{x}), \mathbf{h} \rangle, \qquad \phi''(0) = \mathbf{h}^\top H_f(\mathbf{x})\mathbf{h}.$$

Substituting these identities gives

$$f(\mathbf{x}+\mathbf{h}) = f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{h} \rangle + \frac{1}{2}\mathbf{h}^\top H_f(\mathbf{x})\mathbf{h} + o(\|\mathbf{h}\|^2).$$

■

Matrix Variable Now let $f : M_{m\times n}(\mathbb{R}) \to \mathbb{R}$ be a scalar-valued function of a matrix variable X.

For a small perturbation $H \in M_{m\times n}(\mathbb{R})$, differentiability means

$$f(X+H) - f(X) = df_X(H) + o(\|H\|_F).$$

Since df_X is a linear functional on the matrix space, there exists a unique matrix $\nabla_X f(X)$ such that

$$df_X(H) = \langle \nabla_X f(X), H \rangle_F = \mathbf{tr}\left((\nabla_X f(X))^\top H \right).$$

Therefore

$$f(X+H) = f(X) + \langle \nabla_X f(X), H \rangle_F + o(\|H\|_F).$$

Now assume f is twice differentiable. The gradient $\nabla_X f : M_{m\times n}(\mathbb{R}) \to M_{m\times n}(\mathbb{R})$ is a matrix-valued function. Therefore its derivative is a linear map from matrix directions to matrix directions.

Definition 12.7 (Matrix Hessian).
The Hessian of f at X is the linear operator

$$\nabla_X^2 f(X) : M_{m\times n}(\mathbb{R}) \to M_{m\times n}(\mathbb{R})$$

defined by

$$\nabla_X^2 f(X)[H] = d(\nabla_X f)_X(H).$$

In words, $\nabla_X^2 f(X)[H]$ is the first-order change of the gradient when X moves in the direction H.

Thus, if we replace X by $X + tH$, then

$$\nabla_X f(X+tH) = \nabla_X f(X) + t\nabla_X^2 f(X)[H] + o(t).$$

Proof.
Let $G(X) = \nabla_X f(X)$. Since G is differentiable at X, we have

$$G(X+K) = G(X) + dG_X(K) + R(K),$$

where $\|R(K)\|_F \,/\, \|K\|_F \to 0$ as $K \to 0$. By definition, $dG_X(K) = \nabla_X^2 f(X)[K]$. Now take $K = tH$. Since $\|K\|_F = \|tH\|_F = |t|\,\|H\|_F$, for fixed H the remainder satisfies $\|R(tH)\|_F \,/|t| \to 0$. Hence $R(tH) = o(t)$, and therefore

$$\nabla_X f(X+tH) = \nabla_X f(X) + t\nabla_X^2 f(X)[H] + o(t).$$

■

To interpret the matrix Hessian as curvature, fix a matrix direction H and define the one-variable function $\phi(t) = f(X+tH)$. Then $\phi'(t) = \langle \nabla_X f(X+tH), H\rangle_F$. Using the first-order approximation of the gradient,

$$\begin{aligned}\phi'(t) &= \left\langle \nabla_X f(X) + t\nabla_X^2 f(X)[H] + o(t), H \right\rangle_F \\ &= \langle \nabla_X f(X), H\rangle_F + t\left\langle \nabla_X^2 f(X)[H], H\right\rangle_F + o(t).\end{aligned}$$

Therefore

$$\phi''(0) = \left\langle \nabla_X^2 f(X)[H], H\right\rangle_F.$$

This is the matrix-variable analogue of $\mathbf{h}^\top H_f(\mathbf{x})\mathbf{h}$.

Proposition 14 (Second-Order Approximation for Matrix Variables).
Let $f : M_{m\times n}(\mathbb{R}) \to \mathbb{R}$ be twice differentiable. Then

$$f(X+H) = f(X) + \langle \nabla_X f(X), H\rangle_F + \frac{1}{2}\left\langle \nabla_X^2 f(X)[H], H\right\rangle_F + o(\|H\|_F^2).$$

Proof.
Define $\phi(t) = f(X+tH)$. By the one-variable Taylor expansion,

$$\phi(1) = \phi(0) + \phi'(0) + \frac{1}{2}\phi''(0) + o(\|H\|_F^2).$$

Now $\phi(0) = f(X)$ and $\phi(1) = f(X+H)$. Moreover,

$$\phi'(0) = \langle \nabla_X f(X), H\rangle_F, \qquad \phi''(0) = \left\langle \nabla_X^2 f(X)[H], H\right\rangle_F.$$

Substituting these into the Taylor expansion gives

$$f(X+H) = f(X) + \langle \nabla_X f(X), H\rangle_F + \frac{1}{2}\left\langle \nabla_X^2 f(X)[H], H\right\rangle_F + o(\|H\|_F^2).$$

■

Proposition 15 (Hessian of Affine Matrix Least Squares).
Let

$$f(X) = \frac{1}{2} \|AXB - C\|_F^2 .$$

Then

$$\nabla_X f(X) = A^\top (AXB - C) B^\top,$$

and

$$\nabla_X^2 f(X)[H] = A^\top AHBB^\top.$$

Moreover,

$$\left\langle \nabla_X^2 f(X)[H], H \right\rangle_F = \|AHB\|_F^2 \geq 0.$$

Therefore f is convex in X.

Proof.
Let $R(X) = AXB - C$. Then $f(X) = \frac{1}{2} \|R(X)\|_F^2$. We already know that

$$\nabla_X f(X) = A^\top (AXB - C) B^\top.$$

To find the Hessian, we differentiate the gradient.
Replace X by $X + tH$. Then

$$\begin{aligned}
\nabla_X f(X + tH) &= A^\top \big(A(X + tH)B - C\big) B^\top \\
&= A^\top (AXB - C) B^\top + tA^\top AHBB^\top \\
&= \nabla_X f(X) + tA^\top AHBB^\top.
\end{aligned}$$

Comparing this with

$$\nabla_X f(X + tH) = \nabla_X f(X) + t\nabla_X^2 f(X)[H] + o(t),$$

we obtain

$$\nabla_X^2 f(X)[H] = A^\top AHBB^\top.$$

Now compute the curvature in the direction H:

$$\begin{aligned}
\left\langle \nabla_X^2 f(X)[H], H \right\rangle_F &= \left\langle A^\top AHBB^\top, H \right\rangle_F \\
&= \operatorname{tr}\left((A^\top AHBB^\top)^\top H \right) \\
&= \operatorname{tr}\left(BB^\top H^\top A^\top AH \right) \\
&= \operatorname{tr}\left(B^\top H^\top A^\top AHB \right) \\
&= \operatorname{tr}\left((AHB)^\top (AHB) \right) \\
&= \|AHB\|_F^2 \geq 0.
\end{aligned}$$

Hence the curvature is nonnegative in every direction H. Therefore the least-squares objective is convex in X. ∎

Remark .
This proposition is important because it turns a least-squares problem into a curvature statement. The Hessian is the linear operator

$$L(H) = A^\top AHBB^\top.$$

Its quadratic form is

$$\langle L(H), H\rangle_F = \|AHB\|_F^2 \geq 0,$$

so the least-squares objective is convex. Hence solving $\nabla_X f(X) = 0$ *gives a global minimizer, not merely a local candidate. If* $AHB = \mathbf{O}$ *only when* $H = \mathbf{O}$*, then the curvature is positive in every nonzero direction, so the minimizer is unique.*

The main points of this section are:

1. The Hessian is the derivative of the gradient. For vectors,

$$\nabla f(\mathbf{x} + t\mathbf{h}) = \nabla f(\mathbf{x}) + tH_f(\mathbf{x})\mathbf{h} + o(t).$$

   For matrices,

$$\nabla_X f(X + tH) = \nabla_X f(X) + t\nabla_X^2 f(X)[H] + o(t).$$

2. Curvature is obtained by pairing the Hessian output with the original direction:

$$\mathbf{h}^\top H_f(\mathbf{x})\mathbf{h}, \qquad \left\langle \nabla_X^2 f(X)[H], H \right\rangle_F.$$

   Nonnegative curvature in every direction is the second-order meaning of convexity.

3. For $f(X) = \frac{1}{2}\|AXB - C\|_F^2$,

$$\nabla_X^2 f(X)[H] = A^\top AHBB^\top, \qquad \left\langle \nabla_X^2 f(X)[H], H \right\rangle_F = \|AHB\|_F^2 \geq 0.$$

   Hence least-squares objectives are convex, and solving $\nabla_X f = 0$ gives a global minimizer in the unconstrained case.

12.2 Applications

Deriving Estimators

Consider the linear regression model written in matrix form

$$\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

where $\mathbf{y} \in \mathbb{R}^n$, $X \in M_{n\times k}(\mathbb{R})$, $\boldsymbol{\beta} \in \mathbb{R}^k$, and $\boldsymbol{\varepsilon} \in \mathbb{R}^n$. The sum of squared errors is

$$SSE(\boldsymbol{\beta}) = \sum_{i=1}^{n}(y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2 = \|\mathbf{y} - X\boldsymbol{\beta}\|^2 .$$

Here $\mathbf{x}_i^\top$ is the i-th row of X.

Theorem 12.9 (OLS estimator).
If $\operatorname{rank}(X) = k$, *then*

$$\widehat{\beta}_{OLS} = (X^\top X)^{-1} X^\top \mathbf{y}.$$

Remark .
This is just a special case of proposition (11).

Proof.
Write

$$SSE(\boldsymbol{\beta}) = \|X\boldsymbol{\beta} - \mathbf{y}\|^2 = \operatorname{tr}((X\boldsymbol{\beta} - \mathbf{y})^\top (X\boldsymbol{\beta} - \mathbf{y})).$$

Since $d(X\boldsymbol{\beta}) = X\, d\boldsymbol{\beta}$,

$$\begin{aligned} dSSE(\boldsymbol{\beta}) &= \operatorname{tr}(d(X\boldsymbol{\beta})^\top (X\boldsymbol{\beta} - \mathbf{y})) + \operatorname{tr}((X\boldsymbol{\beta} - \mathbf{y})^\top d(X\boldsymbol{\beta})) \\ &= \operatorname{tr}((X\, d\boldsymbol{\beta})^\top (X\boldsymbol{\beta} - \mathbf{y})) + \operatorname{tr}((X\boldsymbol{\beta} - \mathbf{y})^\top X\, d\boldsymbol{\beta}) \\ &= 2\operatorname{tr}((X\boldsymbol{\beta} - \mathbf{y})^\top X\, d\boldsymbol{\beta}). \end{aligned}$$

Therefore

$$\nabla_{\boldsymbol{\beta}} SSE = \left(2(X\boldsymbol{\beta} - \mathbf{y})^\top X\right)^\top = 2X^\top (X\boldsymbol{\beta} - \mathbf{y}).$$

At the minimizer,

$$2X^\top (X\widehat{\boldsymbol{\beta}} - \mathbf{y}) = 0.$$

Thus

$$X^\top X\widehat{\boldsymbol{\beta}} = X^\top \mathbf{y}.$$

Since $\operatorname{rank}(X) = k$, the matrix $X^\top X$ is invertible. Hence

$$\widehat{\beta}_{OLS} = (X^\top X)^{-1} X^\top \mathbf{y}.$$

The matrix equation is the same as the summation version. Indeed,

$$X^\top X = \sum_{i=1}^n \mathbf{x}_i\mathbf{x}_i^\top, \qquad X^\top \mathbf{y} = \sum_{i=1}^n \mathbf{x}_i y_i.$$

Therefore,
the normal equation $X^\top X\widehat{\boldsymbol{\beta}} = X^\top \mathbf{y}$ is exactly $\left(\sum_{i=1}^n \mathbf{x}_i\mathbf{x}_i^\top\right)\widehat{\boldsymbol{\beta}} = \sum_{i=1}^n \mathbf{x}_i y_i.$ ■

Remark .
The notation $SSE(\boldsymbol{\beta})$ emphasizes the squared-error criterion as a function of an arbitrary coefficient vector:

$$SSE(\boldsymbol{\beta}) = \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2.$$

After an estimator $\widehat{\boldsymbol{\beta}}$ is chosen, the same quantity is often called the residual sum of squares:

$$RSS = RSS(\widehat{\boldsymbol{\beta}}) = SSE(\widehat{\boldsymbol{\beta}}) = \sum_{i=1}^n \widehat{e}_i^2.$$

Thus SSE emphasizes the objective function, while RSS emphasizes the fitted residuals. Many texts use the two names interchangeably.

Definition 12.8 (Likelihood and log-likelihood).
For fixed data $z_1, \ldots, z_n$ with density or probability mass function $f(z;\theta)$, define

$$L(\theta) = \prod_{i=1}^n f(z_i;\theta), \qquad \mathcal{L}(\theta) = \log L(\theta) = \sum_{i=1}^n \log f(z_i;\theta).$$

Since log *is increasing, maximizing $L(\theta)$ is equivalent to maximizing $\mathcal{L}(\theta)$.*

Theorem 12.10 (Normal Regression).
Condition on fixed regressors $\mathbf{x}_1, \ldots, \mathbf{x}_n \in \mathbb{R}^k$. Suppose

$$y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + e_i, \qquad e_i \overset{iid}{\sim} N(0, \sigma^2).$$

Equivalently, $y_i \overset{iid}{\sim} \mathbf{x}_i \sim N(\mathbf{x}_i^\top \boldsymbol{\beta}, \sigma^2)$. If $\sum_{i=1}^n \mathbf{x}_i\mathbf{x}_i^\top$ is invertible, then

$$\widehat{\boldsymbol{\beta}}_{MLE} = \widehat{\boldsymbol{\beta}}_{OLS} = \left(\sum_{i=1}^n \mathbf{x}_i\mathbf{x}_i^\top\right)^{-1}\left(\sum_{i=1}^n \mathbf{x}_i y_i\right), \qquad \widehat{\sigma}^2_{MLE} = \frac{1}{n}\sum_{i=1}^n \widehat{e}_i^2,$$

where

$$\widehat{e}_i = y_i - \mathbf{x}_i^\top \widehat{\boldsymbol{\beta}}_{MLE}.$$

Proof.
For one observation,

$$f(y_i \mid \mathbf{x}_i; \boldsymbol{\beta}, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2\right).$$

Thus

$$L(\boldsymbol{\beta}, \sigma^2) = \prod_{i=1}^{n} f(y_i \mid \mathbf{x}_i; \boldsymbol{\beta}, \sigma^2),$$

and the log-likelihood is, up to an additive constant C,

$$\mathcal{L}(\boldsymbol{\beta}, \sigma^2) = C - \frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2.$$

For fixed σ^2, put $r_i = \mathbf{x}_i^\top \boldsymbol{\beta} - y_i$. Then $(y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2 = r_i^2$ and $dr_i = \mathbf{x}_i^\top d\boldsymbol{\beta}$. Since r_i is a scalar,

$$\begin{aligned} d(r_i^2) &= d\ \mathrm{tr}(r_i^\top r_i) \\ &= \mathrm{tr}((dr_i)^\top r_i) + \mathrm{tr}(r_i^\top dr_i) \\ &= 2r_i \mathbf{x}_i^\top d\boldsymbol{\beta}. \end{aligned}$$

Therefore

$$\begin{aligned} d_{\boldsymbol{\beta}}\mathcal{L} &= -\frac{1}{2\sigma^2}\sum_{i=1}^{n} 2r_i \mathbf{x}_i^\top d\boldsymbol{\beta} \\ &= \mathrm{tr}\left(-\frac{1}{\sigma^2}\sum_{i=1}^{n} r_i \mathbf{x}_i^\top d\boldsymbol{\beta}\right). \end{aligned}$$

Comparing with $d_{\boldsymbol{\beta}}\mathcal{L} = (\nabla_{\boldsymbol{\beta}}\mathcal{L})^\top d\boldsymbol{\beta}$, we read off

$$(\nabla_{\boldsymbol{\beta}}\mathcal{L})^\top = -\frac{1}{\sigma^2}\sum_{i=1}^{n} r_i \mathbf{x}_i^\top.$$

Equivalently,

$$\nabla_{\boldsymbol{\beta}}\mathcal{L} = -\frac{1}{\sigma^2}\sum_{i=1}^{n} \mathbf{x}_i r_i = -\frac{1}{\sigma^2}\sum_{i=1}^{n} \mathbf{x}_i(\mathbf{x}_i^\top \boldsymbol{\beta} - y_i).$$

Hence the first-order condition is

$$-\frac{1}{\sigma^2}\sum_{i=1}^{n} \mathbf{x}_i(\mathbf{x}_i^\top \boldsymbol{\beta} - y_i) = 0.$$

Equivalently,

$$\left(\underbrace{\sum_{i=1}^{n} \mathbf{x}_i \mathbf{x}_i^\top}_{X^\top X}\right)\boldsymbol{\beta} = \underbrace{\sum_{i=1}^{n} \mathbf{x}_i y_i}_{X^\top \mathbf{y}}.$$

Hence

$$\widehat{\boldsymbol{\beta}}_{MLE} = \left(\sum_{i=1}^{n} \mathbf{x}_i \mathbf{x}_i^\top\right)^{-1} \left(\sum_{i=1}^{n} \mathbf{x}_i y_i\right).$$

This is exactly the OLS estimator.
Next fix $\boldsymbol{\beta}$ and put

$$RSS(\boldsymbol{\beta}) = \sum_{i=1}^{n} (y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2.$$

Ignoring constants,

$$\mathcal{L}(\boldsymbol{\beta}, \sigma^2) = -\frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} RSS(\boldsymbol{\beta}).$$

Treating σ^2 as the scalar variable,

$$\frac{\partial \mathcal{L}}{\partial \sigma^2} = -\frac{n}{2} \frac{1}{\sigma^2} + \frac{1}{2} \frac{1}{(\sigma^2)^2} RSS(\boldsymbol{\beta}).$$

The first-order condition is

$$-\frac{n}{2} \frac{1}{\sigma^2} + \frac{1}{2} \frac{1}{(\sigma^2)^2} RSS(\boldsymbol{\beta}) = 0.$$

Multiplying by $2(\sigma^2)^2$ gives

$$-n\sigma^2 + RSS(\boldsymbol{\beta}) = 0.$$

Thus, at $\widehat{\boldsymbol{\beta}}_{MLE}$,

$$\widehat{\sigma}^2_{MLE} = \frac{1}{n} RSS(\widehat{\boldsymbol{\beta}}_{MLE}) = \frac{1}{n} \sum_{i=1}^{n} \widehat{e}_i^2.$$

■

Theorem 12.11 (MLE Estimator).
Let $\mathbf{x}_1, \ldots, \mathbf{x}_n \overset{iid}{\sim} N(\boldsymbol{\mu}, \Sigma)$, where $\mathbf{x}_i \in \mathbb{R}^p$, $\boldsymbol{\mu} \in \mathbb{R}^p$, and $\Sigma \in M_p(\mathbb{R})$ is symmetric positive definite. Put

$$\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^{n} \mathbf{x}_i, \qquad S = \sum_{i=1}^{n} (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top.$$

If S is positive definite, then

$$\widehat{\boldsymbol{\mu}}_{MLE} = \bar{\mathbf{x}}, \qquad \widehat{\Sigma}_{MLE} = \frac{1}{n} S.$$

Proof.
For one observation,

$$p(\mathbf{x}_i; \boldsymbol{\mu}, \Sigma) = \frac{1}{(2\pi)^{p/2}(\det \Sigma)^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x}_i - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x}_i - \boldsymbol{\mu})\right).$$

Hence

$$L(\boldsymbol{\mu}, \Sigma) = \prod_{i=1}^{n} p(\mathbf{x}_i; \boldsymbol{\mu}, \Sigma),$$

and the log-likelihood is

$$\mathcal{L}(\boldsymbol{\mu}, \Sigma) = C - \frac{n}{2} \log \det \Sigma - \frac{1}{2} \sum_{i=1}^{n} (\mathbf{x}_i - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x}_i - \boldsymbol{\mu}).$$

Here C does not depend on $\boldsymbol{\mu}$ or Σ.
First fix Σ. Since Σ^{-1} is symmetric,

$$\begin{aligned} d\left[(\mathbf{x}_i - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x}_i - \boldsymbol{\mu})\right] &= -(d\boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x}_i - \boldsymbol{\mu}) - (\mathbf{x}_i - \boldsymbol{\mu})^\top \Sigma^{-1} d\boldsymbol{\mu} \\ &= -2(\mathbf{x}_i - \boldsymbol{\mu})^\top \Sigma^{-1} d\boldsymbol{\mu}. \end{aligned}$$

Therefore

$$\begin{aligned} d_{\boldsymbol{\mu}}\mathcal{L} &= \sum_{i=1}^{n} (\mathbf{x}_i - \boldsymbol{\mu})^\top \Sigma^{-1} d\boldsymbol{\mu} \\ &= \left(\Sigma^{-1} \sum_{i=1}^{n} (\mathbf{x}_i - \boldsymbol{\mu})\right)^\top d\boldsymbol{\mu}. \end{aligned}$$

Comparing with $d_{\boldsymbol{\mu}}\mathcal{L} = (\nabla_{\boldsymbol{\mu}}\mathcal{L})^\top d\boldsymbol{\mu}$, we get

$$\nabla_{\boldsymbol{\mu}}\mathcal{L} = \Sigma^{-1} \sum_{i=1}^{n} (\mathbf{x}_i - \boldsymbol{\mu}).$$

Since Σ is invertible, the first-order condition gives

$$\sum_{i=1}^{n} \mathbf{x}_i - n\boldsymbol{\mu} = 0, \qquad \widehat{\boldsymbol{\mu}}_{MLE} = \frac{1}{n} \sum_{i=1}^{n} \mathbf{x}_i.$$

Now plug in $\widehat{\boldsymbol{\mu}}_{MLE} = \bar{\mathbf{x}}$. For each i,

$$\begin{aligned} &(\mathbf{x}_i - \widehat{\boldsymbol{\mu}}_{MLE})^\top \Sigma^{-1} (\mathbf{x}_i - \widehat{\boldsymbol{\mu}}_{MLE}) \\ &\qquad = \operatorname{tr}\left[(\mathbf{x}_i - \widehat{\boldsymbol{\mu}}_{MLE})^\top \Sigma^{-1} (\mathbf{x}_i - \widehat{\boldsymbol{\mu}}_{MLE})\right] \\ &\qquad = \operatorname{tr}\left[\Sigma^{-1} (\mathbf{x}_i - \widehat{\boldsymbol{\mu}}_{MLE})(\mathbf{x}_i - \widehat{\boldsymbol{\mu}}_{MLE})^\top\right]. \end{aligned}$$

Hence, with

$$S = \sum_{i=1}^{n} (\mathbf{x}_i - \widehat{\boldsymbol{\mu}}_{MLE})(\mathbf{x}_i - \widehat{\boldsymbol{\mu}}_{MLE})^\top,$$

we have

$$\begin{aligned}\sum_{i=1}^{n} d\left[(\mathbf{x}_i - \widehat{\boldsymbol{\mu}}_{MLE})^\top \Sigma^{-1} (\mathbf{x}_i - \widehat{\boldsymbol{\mu}}_{MLE})\right] &= d\ \mathrm{tr}(\Sigma^{-1} S) \\ &= \mathrm{tr}\big(d(\Sigma^{-1}) S\big) \\ &= \mathrm{tr}\big(-\Sigma^{-1}(d\Sigma)\Sigma^{-1} S\big) \\ &= -\,\mathrm{tr}\big(\Sigma^{-1} S \Sigma^{-1} d\Sigma\big).\end{aligned}$$

where we used $d(\Sigma^{-1}) = -\Sigma^{-1}(d\Sigma)\Sigma^{-1}$. Together with $d(\log\det\Sigma) = \mathrm{tr}(\Sigma^{-1} d\Sigma)$, this gives

$$d\mathcal{L} = \mathrm{tr}\left[\left(-\frac{n}{2}\Sigma^{-1} + \frac{1}{2}\Sigma^{-1} S \Sigma^{-1}\right) d\Sigma\right].$$

Hence the first-order condition is

$$-\frac{n}{2}\Sigma^{-1} + \frac{1}{2}\Sigma^{-1} S \Sigma^{-1} = 0.$$

Multiplying by 2, then by Σ on the left and by Σ on the right, gives

$$-n\Sigma + S = 0.$$

Therefore

$$\widehat{\Sigma}_{MLE} = \frac{1}{n} S = \frac{1}{n}\sum_{i=1}^{n} (\mathbf{x}_i - \widehat{\boldsymbol{\mu}}_{MLE})(\mathbf{x}_i - \widehat{\boldsymbol{\mu}}_{MLE})^\top.$$

■

Remark .

Let $\mathbf{x}_1, \ldots, \mathbf{x}_n \in \mathbb{R}^p$, and let

$$S = \sum_{i=1}^{n} (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$$

be the centered scatter matrix. The covariance MLE is

$$\widehat{\Sigma} = \frac{1}{n} S.$$

If S is singular, then $\widehat{\Sigma}$ is also singular, so it is not positive definite. Hence there is no covariance MLE inside the positive definite cone.

In particular, since

$$\sum_{i=1}^{n} (\mathbf{x}_i - \bar{\mathbf{x}}) = 0,$$

the centered vectors $\mathbf{x}_i - \bar{\mathbf{x}}$ span a space of dimension at most $n-1$. Therefore

$$\mathrm{rank}(S) \le n - 1.$$

If $n \le p$, then $n-1 < p$, so S cannot have full rank p. Thus S cannot be positive definite. Hence a necessary condition for a positive definite covariance MLE in $\mathbb{R}^p$ is $n > p$: the sample size must be larger than the dimension, i.e. larger than the number of coordinates to be estimated together.

Low-rank approximation

Question .
Let $A \in M_{m\times n}(\mathbb{R})$ and $r \le \operatorname{rank}(A)$. Find $B \in M_{m\times n}(\mathbb{R})$ with $\operatorname{rank}(B) = r$ such that $\|A - B\|_F$ is minimized.

A rank-r matrix can be written in the form

$$B = XZ^\top, \qquad X \in M_{m\times r}(\mathbb{R}), \qquad Z \in M_{n\times r}(\mathbb{R}).$$

To see the meaning of this factorization, write $Z = (\mathbf{z}_1, \ldots, \mathbf{z}_r)$ with $\mathbf{z}_j \in \mathbb{R}^n$. Then

$$Z^\top = \begin{pmatrix} \mathbf{z}_1^\top \\ \vdots \\ \mathbf{z}_r^\top \end{pmatrix}.$$

If X_i denotes the i-th row of X, then the i-th row of B is $B_{i\cdot} = X_i Z^\top$, which is a linear combination of the rows $\mathbf{z}_1^\top, \ldots, \mathbf{z}_r^\top$. Therefore, the row space of B is spanned by $\{\mathbf{z}_1^\top, \ldots, \mathbf{z}_r^\top\}$.

There is redundancy in this description. If $Q \in M_r(\mathbb{R})$ is invertible, then

$$XZ^\top = (XQ)(ZQ^{-\top})^\top.$$

So X and Z are not uniquely determined by B. To remove part of this redundancy, impose the constraint

$$Z^\top Z = I_r.$$

This means the columns $\mathbf{z}_1, \ldots, \mathbf{z}_r$ of Z are orthonormal. Notice that Z itself is not an orthogonal matrix unless $r = n$, because Z is generally not square.

Thus we study the constrained optimization problem

$$\text{minimize } f(X, Z) = \frac{1}{2}\left\|A - XZ^\top\right\|_F^2 \qquad \text{subject to } Z^\top Z = I_r.$$

The factor $1/2$ is inserted only to simplify derivatives later.

Writing $Z = (\mathbf{z}_1, \ldots, \mathbf{z}_r)$, the constraint $Z^\top Z = I_r$ means

$$\mathbf{z}_i^\top \mathbf{z}_j = \delta_{ij} \qquad \text{for all } 1 \le i, j \le r.$$

The scalar constraints are the entries of $Z^\top Z - I_r$. We organize the corresponding Lagrange multipliers into an $r \times r$ matrix $\Lambda = (\lambda_{ij})$. The trace term in the Lagrangian is the Frobenius inner product form of all multiplier terms at once. Indeed, if $C = Z^\top Z - I_r$, then $\operatorname{tr}(\Lambda C) = \langle \Lambda^\top, C \rangle_F$. Since C is symmetric, we may choose Λ symmetric. Then

$$\operatorname{tr}\big(\Lambda(Z^\top Z - I_r)\big) = \Big\langle \Lambda, Z^\top Z - I_r \Big\rangle_F = \sum_{i=1}^{r}\sum_{j=1}^{r} \lambda_{ij}(\mathbf{z}_i^\top \mathbf{z}_j - \delta_{ij}).$$

Thus $\frac{1}{2}\operatorname{tr}\big(\Lambda(Z^\top Z - I_r)\big)$ packages all scalar constraints $\mathbf{z}_i^\top \mathbf{z}_j = \delta_{ij}$ and their Lagrange multipliers into one matrix expression. Define

$$\mathcal{L}(X, Z, \Lambda) = \frac{1}{2}\operatorname{tr}(RR^\top) - \frac{1}{2}\operatorname{tr}\big(\Lambda(Z^\top Z - I_r)\big),$$

where $R = A - XZ^\top$.

Step 1: First order condition in X

Theorem 12.12 (First order condition).
When $\mathcal{L}$ is minimized, one has $X = AZ$.

Proof.
Regard Z and Λ as constants. Since the constraint term does not depend on X and $R = A - XZ^\top$, we have $d_X R = -dX\, Z^\top$. By the chain rule and the differential of the squared Frobenius norm,

$$\begin{aligned}
d_X \mathcal{L} &= \frac{1}{2} d_X \operatorname{tr}(RR^\top) \\
&= d_X \left(\frac{1}{2}\|R\|_F^2\right) \\
&= d_R \left(\frac{1}{2}\|R\|_F^2\right)[d_X R] \\
&= \Big\langle R, -dX\, Z^\top \Big\rangle_F \\
&= \operatorname{tr}\Big(R^\top(-dX\, Z^\top)\Big) \\
&= -\operatorname{tr}(Z^\top R^\top dX) = -\operatorname{tr}((RZ)^\top dX).
\end{aligned}$$

Hence $\nabla_X \mathcal{L} = -RZ$. At a minimizer, $0 = RZ = (A - XZ^\top)Z = AZ - X$, where $Z^\top Z = I_r$ is used in the last equality. Thus $X = AZ$. ∎

Step 2: Reduction to a problem in Z

Theorem 12.13 .

When $X = AZ$, the objective function becomes

$$f(AZ, Z) = \frac{1}{2}\left(\operatorname{tr}(A^\top A) - \operatorname{tr}(Z^\top A^\top AZ)\right).$$

Therefore minimizing $f(AZ, Z)$ subject to $Z^\top Z = I_r$ is equivalent to maximizing

$$\operatorname{tr}(Z^\top A^\top AZ) \qquad \text{subject to } Z^\top Z = I_r.$$

Proof.

At a minimizer in X, we have $X = AZ$, so $XZ^\top = AZZ^\top$. Put $P = ZZ^\top$. Since $Z^\top Z = I_r$, $P^\top = P$ and $P^2 = Z(Z^\top Z)Z^\top = ZZ^\top = P$.

Now $f(AZ, Z) = \frac{1}{2}\left\|A - AZZ^\top\right\|_F^2 = \frac{1}{2}\|A(I-P)\|_F^2$. Using $P^\top = P$ and $(I-P)^2 = I - P$,

$$\begin{aligned}
f(AZ, Z) &= \frac{1}{2}\operatorname{tr}\big((A(I-P))^\top A(I-P)\big) \\
&= \frac{1}{2}\operatorname{tr}\big((I-P)A^\top A(I-P)\big) \\
&= \frac{1}{2}\operatorname{tr}\big(A^\top A(I-P)^2\big) \\
&= \frac{1}{2}\operatorname{tr}\big(A^\top A(I-P)\big) \\
&= \frac{1}{2}\left(\operatorname{tr}(A^\top A) - \operatorname{tr}(A^\top AP)\right).
\end{aligned}$$

Since $P = ZZ^\top$, cyclicity of trace gives $\operatorname{tr}(A^\top AP) = \operatorname{tr}(Z^\top A^\top AZ)$. Therefore

$$f(AZ, Z) = \frac{1}{2}\left(\operatorname{tr}(A^\top A) - \operatorname{tr}(Z^\top A^\top AZ)\right).$$

The first term $\operatorname{tr}(A^\top A)$ does not depend on Z. Hence minimizing $f(AZ, Z)$ is equivalent to maximizing $\operatorname{tr}(Z^\top A^\top AZ)$ subject to $Z^\top Z = I_r$. ∎

Theorem 12.14 .

Let $P \in M_n(\mathbb{R})$. If $P^\top = P$ and $P^2 = P$, then P is the orthogonal projection onto $\operatorname{Im}(P)$.

Proof.

Suppose $P^\top = P$ and $P^2 = P$.

First, P acts as the identity on $\operatorname{Im}(P)$. Indeed, if $\mathbf{y} \in \operatorname{Im}(P)$, then $\mathbf{y} = P\mathbf{x}$ for some $\mathbf{x}$. Hence $P\mathbf{y} = P^2\mathbf{x} = P\mathbf{x} = \mathbf{y}$.

Also, P kills every vector in $\operatorname{Ker}(P)$. If $\mathbf{y} \in \operatorname{Ker}(P)$, then $P\mathbf{y} = 0$ by definition.

It remains to show that these two subspaces are orthogonal. Let $\mathbf{u} \in \operatorname{Im}(P)$ and $\mathbf{v} \in \operatorname{Ker}(P)$. Then $\mathbf{u} = P\mathbf{x}$ for some $\mathbf{x}$, so

$$\langle \mathbf{u}, \mathbf{v} \rangle = \langle P\mathbf{x}, \mathbf{v} \rangle = \langle \mathbf{x}, P^\top \mathbf{v} \rangle = \langle \mathbf{x}, P\mathbf{v} \rangle = 0.$$

Thus $\operatorname{Im}(P) \perp \operatorname{Ker}(P)$.
Since $P^2 = P$, every $\mathbf{x} \in \mathbb{R}^n$ can be decomposed as $\mathbf{x} = P\mathbf{x} + (\mathbf{x} - P\mathbf{x})$, where $P\mathbf{x} \in \operatorname{Im}(P)$ and

$$P(\mathbf{x} - P\mathbf{x}) = P\mathbf{x} - P^2\mathbf{x} = 0.$$

Hence $\mathbf{x} - P\mathbf{x} \in \operatorname{Ker}(P)$, and therefore $\mathbb{R}^n = \operatorname{Im}(P) \oplus \operatorname{Ker}(P)$. By orthogonal decomposition, we have $\operatorname{Ker}(P) = \operatorname{Im}(P)^\perp$.
Therefore P preserves $\operatorname{Im}(P)$ and kills $\operatorname{Im}(P)^\perp$. Hence P is the orthogonal projection onto $\operatorname{Im}(P)$. ■

Remark .
Idempotent alone is not enough to guarantee orthogonal projection. For example, let

$$P = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}.$$

Then $P^2 = P$, so P is idempotent. However, $P^\top \neq P$, so it is not symmetric.

Indeed, $\operatorname{Im}(P) = \operatorname{span}\{(1,0)^t\}$ *and* $\operatorname{Ker}(P) = \operatorname{span}\{(1,-1)^t\}$. *These two subspaces are not orthogonal, since $\langle (1,0)^t, (1,-1)^t \rangle = 1$. Hence P is a projection, but not an orthogonal projection.*

Remark .
In our case, $P = ZZ^\top$. Since $Z^\top Z = I_r$, we have $P^\top = P$ and

$$P^2 = ZZ^\top ZZ^\top = Z(Z^\top Z)Z^\top = ZZ^\top = P.$$

Thus P is symmetric and idempotent. By the theorem above, P is the orthogonal projection onto $\operatorname{Im}(P)$.

Moreover, $\operatorname{Im}(P)$ *is exactly the column space of Z. Indeed, from $P = ZZ^\top$, it is clear that* $\operatorname{Im}(P)$ *is contained in the column space of Z. Conversely, if* $\mathbf{y}$ *belongs to the column space of Z, then* $\mathbf{y} = Z\mathbf{a}$ *for some* $\mathbf{a}$, *and*

$$P\mathbf{y} = ZZ^\top Z\mathbf{a} = Z\mathbf{a} = \mathbf{y}.$$

Thus $\mathbf{y} \in \operatorname{Im}(P)$. *Therefore* $\operatorname{Im}(P)$ *is the column space of Z.*

So $P = ZZ^\top$ is the orthogonal projection onto the column space of Z.

Step 3: Linear algebra input

Theorem 12.15 (Rayleigh).
Let B be symmetric and positive semidefinite, with eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0$. Then the maximum value of $\operatorname{tr}(Z^\top BZ)$ subject to $Z^\top Z = I_r$ is $\lambda_1 + \cdots + \lambda_r$.

The maximum is attained when the columns of Z are eigenvectors corresponding to the r largest eigenvalues of B.

Proof.
By the spectral theorem, there is an orthonormal basis $\mathbf{v}_1, \ldots, \mathbf{v}_n$ of eigenvectors of B such that $B\mathbf{v}_i = \lambda_i \mathbf{v}_i$. Write the columns of Z as $\mathbf{z}_1, \ldots, \mathbf{z}_r$. Since $\mathbf{v}_1, \ldots, \mathbf{v}_n$ is an orthonormal basis, write $\mathbf{z}_j = \sum_{i=1}^{n} \alpha_{ij} \mathbf{v}_i$. Then

$$
\begin{aligned}
\operatorname{tr}(Z^\top BZ) &= \sum_{j=1}^{r} \mathbf{z}_j^\top B \mathbf{z}_j \\
&= \sum_{j=1}^{r} \left\langle B\left(\sum_{i=1}^{n} \alpha_{ij}\mathbf{v}_i\right), \sum_{k=1}^{n} \alpha_{kj}\mathbf{v}_k \right\rangle \\
&= \sum_{j=1}^{r} \left\langle \sum_{i=1}^{n} \lambda_i \alpha_{ij}\mathbf{v}_i, \sum_{k=1}^{n} \alpha_{kj}\mathbf{v}_k \right\rangle \\
&= \sum_{j=1}^{r} \sum_{i=1}^{n} \lambda_i \alpha_{ij}^2.
\end{aligned}
$$

Define $\gamma_i = \sum_{j=1}^{r} \alpha_{ij}^2$. Then $\operatorname{tr}(Z^\top BZ) = \sum_{i=1}^{n} \lambda_i \gamma_i$. We need two observations about γ_i.

- Since the columns of Z are orthonormal,

$$
\begin{aligned}
\sum_{i=1}^{n} \gamma_i = \sum_{i=1}^{n}\sum_{j=1}^{r} \alpha_{ij}^2 &= \sum_{j=1}^{r}\sum_{i=1}^{n} \alpha_{ij}^2 \\
&= \sum_{j=1}^{r} \left\|\mathbf{z}_j\right\|^2 = r.
\end{aligned}
$$

- Let $V_r = \operatorname{span}\{\mathbf{z}_1, \ldots, \mathbf{z}_r\}$. Since $\mathbf{z}_1, \ldots, \mathbf{z}_r$ is an orthonormal basis of V_r,

$$
P_{V_r}(\mathbf{v}_i) = \sum_{j=1}^{r} \left\langle \mathbf{v}_i, \mathbf{z}_j \right\rangle \mathbf{z}_j.
$$

Therefore

$$\begin{aligned}\|P_{V_r}(\mathbf{v}_i)\|^2 &= \sum_{j=1}^{r} \left|\langle \mathbf{v}_i, \mathbf{z}_j \rangle\right|^2 \\ &= \sum_{j=1}^{r} \left|\langle \mathbf{z}_j, \mathbf{v}_i \rangle\right|^2 = \sum_{j=1}^{r} \alpha_{ij}^2 = \gamma_i.\end{aligned}$$

By the minimal-distance property of orthogonal projection,

$$0 \le \gamma_i = \|P_{V_r}(\mathbf{v}_i)\|^2 \le \|\mathbf{v}_i\|^2 = 1 \qquad \text{for every } 1 \le i \le n.$$

It remains to maximize $\sum_{i=1}^{n} \lambda_i \gamma_i$ subject to $0 \le \gamma_i \le 1$ and $\sum_{i=1}^{n} \gamma_i = r$. Since $\lambda_1 \ge \cdots \ge \lambda_n$, the maximum is achieved by placing all available weight on the largest eigenvalues: $\gamma_1 = \cdots = \gamma_r = 1$ and $\gamma_{r+1} = \cdots = \gamma_n = 0$. Thus the maximum value is $\lambda_1 + \cdots + \lambda_r$, attained by taking $Z = (\mathbf{v}_1, \ldots, \mathbf{v}_r)$. ■

Theorem 12.16 (Low-rank approximation theorem).
Let $A = U\Sigma V^\top$ be a singular value decomposition of A, with singular values $\sigma_1 \ge \sigma_2 \ge \cdots \ge 0$. Then the best rank-$r$ approximation to A in Frobenius norm is the truncated SVD

$$B_r = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top.$$

Equivalently, if $U_r = (\mathbf{u}_1, \ldots, \mathbf{u}_r)$, $V_r = (\mathbf{v}_1, \ldots, \mathbf{v}_r)$, and $\Sigma_r = \operatorname{diag}(\sigma_1, \ldots, \sigma_r)$, then

$$B_r = U_r \Sigma_r V_r^\top.$$

Proof.
From Step 2, the problem reduces to maximizing

$$\operatorname{tr}(Z^\top A^\top A Z) \qquad \text{subject to } Z^\top Z = I_r.$$

The matrix $A^\top A$ is symmetric and positive semidefinite. Let

$$A^\top A \mathbf{v}_i = \lambda_i \mathbf{v}_i, \qquad \lambda_1 \ge \lambda_2 \ge \cdots \ge \lambda_n \ge 0,$$

where $\mathbf{v}_1, \ldots, \mathbf{v}_n$ is an orthonormal basis. The eigenvalues of $A^\top A$ are the squares of the singular values of A, so $\lambda_i = \sigma_i^2$. By Rayleigh's theorem, the maximum of $\operatorname{tr}(Z^\top A^\top A Z)$ is

$$\lambda_1 + \cdots + \lambda_r = \sigma_1^2 + \cdots + \sigma_r^2,$$

and it is attained when $Z = (\mathbf{v}_1, \ldots, \mathbf{v}_r) = V_r$. From Step 1, the corresponding X is $X = AZ = AV_r$. Write $U = (U_r\ U_\perp)$. Since the columns of V are orthonormal,

$$V^\top V_r = \begin{pmatrix} I_r \\ \mathbf{O}_{(n-r)\times r} \end{pmatrix}.$$

Thus multiplication by $V^\top V_r$ selects the first r columns of Σ, and hence

$$\Sigma V^\top V_r = \begin{pmatrix} \Sigma_r \\ \mathbf{O}_{(m-r)\times r} \end{pmatrix}.$$

Therefore

$$\begin{aligned} B &= XZ^\top = AV_rV_r^\top \\ &= U\Sigma V^\top V_r V_r^\top \\ &= (U_r\ U_\perp) \begin{pmatrix} \Sigma_r \\ \mathbf{O}_{(m-r)\times r} \end{pmatrix} V_r^\top \\ &= U_r\Sigma_r V_r^\top = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top. \end{aligned}$$

Thus the best rank-r approximation is the truncated SVD. ∎